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**NORTHERNAIRE SITE
CADILLAC MICHIGAN**

**SUPPLEMENTAL
REMEDIAL INVESTIGATION**

**PREPARED FOR
MICHIGAN DEPARTMENT OF
NATURAL RESOURCES**

JANUARY 1987

EC-JORDAN CO.

EPA Region 5 Records Ctr.



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E.C. JORDAN CO.
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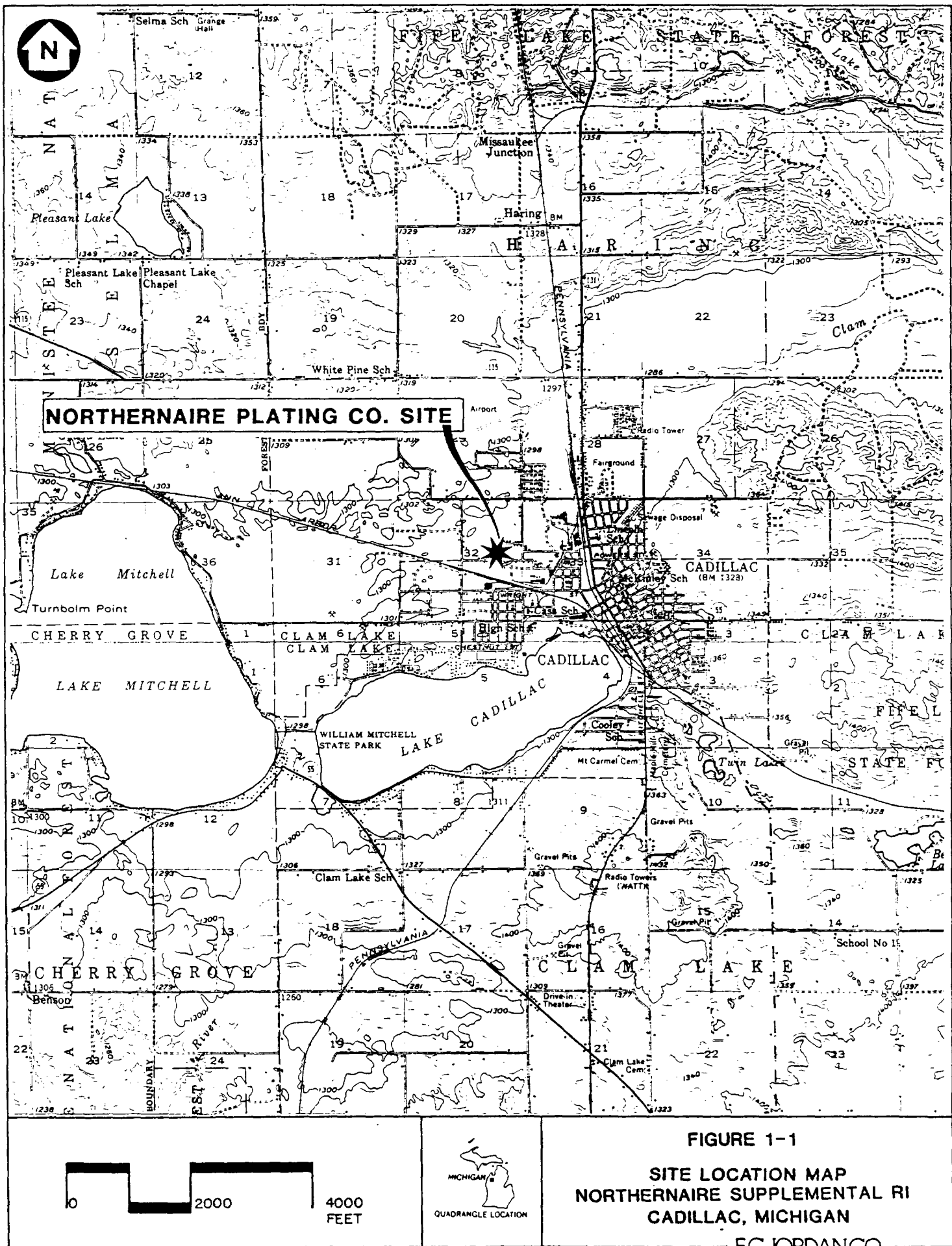
A. BORING LOGS AND MONITORING WELL DIAGRAMS

1.0 EXECUTIVE SUMMARY

Recent site explorations (November, 1985 to April, 1986) at the Northernnaire Plating facility (Northernnaire) in Cadillac, Michigan (see Figure 1-1) have been conducted by the E.C. Jordan Co. (Jordan) for the Michigan Department of Natural Resources (MDNR) to supplement the original Northernnaire Site, Phase II Remedial Investigation (RI) conducted by Jordan in 1984-1985 and issued in May 1985. The previous RI work had identified the presence of hex-chrome in the intermediate aquifer in wells MW-7 and K-15 (see Figure 2-1), but did not provide sufficient data with which to identify the extent and distribution of the contamination. This supplemental work was designed to assess hexavalent chromium (hex-chrome) contamination in the aquifers downgradient from this site. The installation of monitoring wells into the intermediate aquifer, the collection of groundwater samples for field screening and laboratory analysis, and the collection of groundwater elevation data in area wells have provided necessary information to complete the present phases of the Northernnaire RI.

Groundwater flow directions in the water table and intermediate aquifers were found to be consistent with those determined in earlier studies. Groundwater flow in the water table or shallow aquifer is in a northeasterly direction and flow in the intermediate aquifer is generally in a northwesterly direction.

In the original RI work (see "Northernnaire Site Phase II Remedial Investigation", May, 1985), hex-chrome and cadmium had not been found in the shallow aquifer. However, hex-chrome contamination at 1.6 parts per million



(ppm) and cadmium at 0.053 ppm were found in monitoring well MW-5 in the shallow aquifer during the February, 1986 sampling event. The concentrations of hex-chrome and cadmium had decreased to 0.54 ppm and 0.018 ppm, respectively, during the May, 1986 sampling event. These detections reaffirm the previous determinations that operations at Northernair led to hex-chrome contamination in the aquifers insomuch as MW-5 is directly downgradient of the facility with no other known intervening source. It is expected that further decreases in concentrations in MW-5 will occur over time because the plume is probably fairly localized and should be moving rapidly through the upper aquifer. It should also become diluted with increased groundwater mixing.

Geologic data from the borings drilled for monitoring well installations during a previous study for Kysor Industrial Corporation indicate that the upper clay aquitard is thin in the vicinity of wells K-6 and K-8. This fact, coupled with recent groundwater level data which suggest that substantial downward vertical gradients exist between the shallow and intermediate aquifers where the clay is thin, indicates that there is a mechanism by which hex-chrome can migrate from the upper to the intermediate aquifer.

Data from ten monitoring wells (MW-9 to MW-18) installed for the supplemental studies have defined the extent of hex-chrome contamination in the intermediate aquifer (see Figure 5-3). Presently the plume is beneath the undeveloped areas of the Industrial Park north of Northernair. The leading edge of the plume apparently does not yet extend beyond Holman and Saunders Streets to the west and north respectively. The plume in the intermediate aquifer is approximately 1100 feet long (N-S), 650 feet wide (E-W) and from 5 to 15 feet thick. The

maximum concentration of chromium determined in the plume was almost 2 ppm in well K-15 (February, 1986).

The hex-chrome plume in both aquifers is situated just above the confining clay layers of each unit. This is probably a result of relatively small but significant downward vertical gradients that are known to exist within the separate aquifers.

Based on available data, the present environmental impact of the hex-chrome plume is believed to be minimal inasmuch as no current receptors have been identified. With the elimination of the source of contamination at Northernnaire, the hex-chrome in both aquifers should continue to become diluted as it moves through the aquifer. However, the potential for migration of the plume from the intermediate to the deeper aquifer in which the city water supply wells are screened must be considered in light of existing hydrologic factors. Because horizontal flow is essentially uniform across the site, mass balance considerations for the groundwater flow indicate probable seepage of groundwater through the lower aquitard, at least in some areas. A more detailed assessment of the rate and quantity of seepage, and the exact impact it may have on groundwater quality in the deeper aquifer would necessitate a thorough investigation of the lower clay aquitard. This work was beyond the scope of the supplemental work since MDNR has rightly taken the precaution of prohibiting the penetration of the lower aquitard at this point in time due to the presence of other sources of contamination in the intermediate aquifer. However, the probability is small that Northernnaire derived contaminants could significantly affect the water quality of the municipal water supply.

2.0 INTRODUCTION

2.1 SCOPE OF WORK

This supplement was designed to further the remedial investigation (RI) of the Northernnaire Plating Superfund site in Cadillac, Michigan, as performed by E.C. Jordan Co. (Jordan) during the fall and winter of 1984 and 1985. The supplement to the RI has the primary objective of investigating the extent and concentration distribution of hexavalent chromium (hex-chrome) contamination in the aquifer beneath the industrial park north of the Northernnaire facility. A monitoring well installation program was conducted in the winter of 1985-1986 by Jordan in cooperation with the Michigan Department of Natural Resources (MDNR). This program was implemented in accordance with the supplemental Work Plan of the Remedial Investigation/Feasibility Study, dated October 1985 (revised December 1985) as approved by the MDNR and United States Environmental Protection Agency (USEPA).

Several work tasks were performed from November 1985 to April 1986 to assess the degree of hex-chrome contamination and to provide additional information about the site geology and hydrogeology. These tasks included:

- 1) determinations of the geology in the vicinity of the plume by performing deep auger borings;
- 2) monitoring well installations to determine the extent of contamination

- 3) field screening for groundwater quality;
- 4) collection of groundwater samples for analysis by the USEPA Contract Laboratory Program (CLP); and
- 5) gathering groundwater level data for confirmation of groundwater flow direction.

This report describes the field program, the data collected, and interpretations of the data, and presents conclusions and recommendations regarding the data and assessments of potential migration and impact. This report is intended as a supplement to Jordan's Remedial Investigation report of May 1985. As such, this supplemental report will only reference or summarize relevant sections in the earlier report. The reader desiring a more detailed history of the facility or of previous investigations is referred to the May 1985 RI report.

2.2 BACKGROUND AND PREVIOUS STUDIES

In 1978, complaints of poor quality groundwater from domestic wells of residents along Eighth Street adjacent to the Industrial Park, Cadillac, Michigan, led to the detection of hexavalent chromium in the groundwater. Those residents with affected wells were subsequently supplied with city water. The hex-chrome contamination was attributed by MDNR and the City to the operations at the nearby Northern Plating facility. The City rescinded Northern's

sewer discharge permit and plugged the discharge sewer line. Early in 1981 Northernair ceased operations and eventually filed for bankruptcy. Due to the hazardous nature of hex-chrome and other chemicals present at Northernair and the relative closeness of the Cadillac public water supply wellfield, this abandoned site was listed on the USEPA's National Priorities List. The MDNR and USEPA initiated immediate response actions to remove wastes and have subsequently implemented a remedial investigation program to determine the extent of the remaining site contamination. The RI will serve as a basis for the conduct of a feasibility study which will evaluate and recommend remedial action alternatives.

The initial RI studies clearly defined limited soil contamination in the immediate vicinity of the Northernair building. This problem became the object of a focused feasibility study for remediation. The selected remedial action is currently under design and is expected to be implemented during 1986.

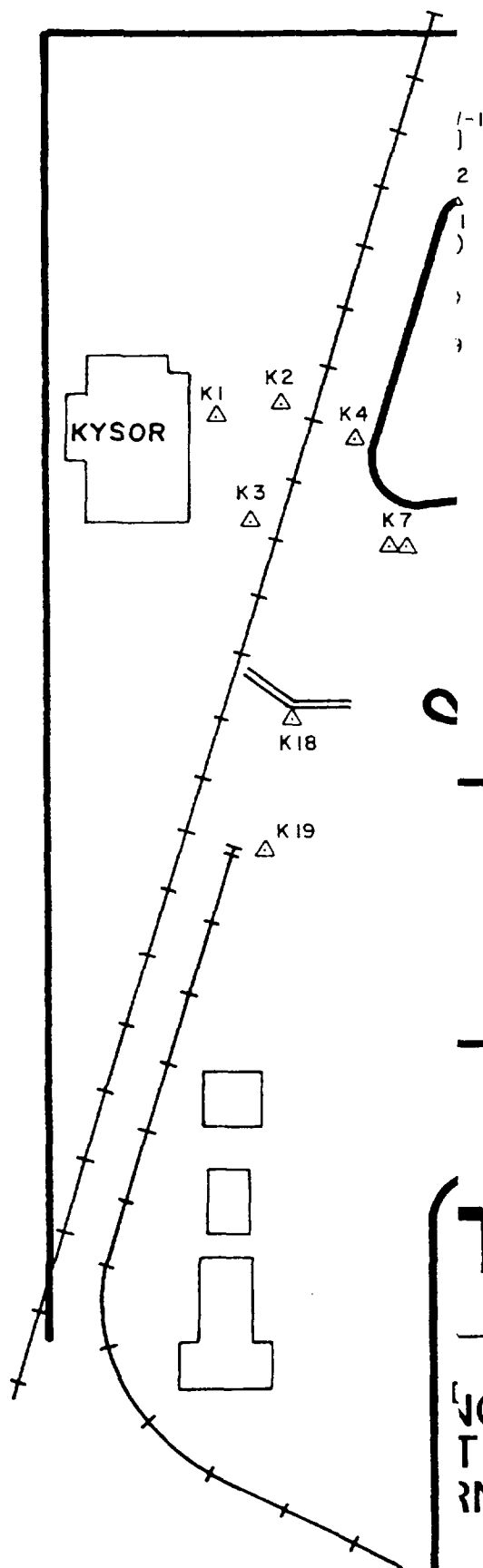
Initial studies of the groundwater during the RI by Jordan included the installation of six shallow monitoring wells and two deeper monitoring wells to provide data for a preliminary investigation of the hydrogeology, geology and extent of groundwater contamination. Data from analyses of samples from these and other selected monitoring wells did not indicate the presence of Northernair-related contamination in the shallow, water-table aquifer. However, hex-chrome was detected in the groundwater at concentrations exceeding the drinking water standard (0.05 mg/l) at a depth of approximately 150 feet in the confined to semi-confined, intermediate aquifer in two wells (MW-7 and K-15) sampled during the investigation. A second confining clay layer of

largely unknown properties separates this intermediate aquifer from the deeper aquifer in which the public water supply wells are screened.

The previous and supplemental studies by Jordan, and studies of contamination problems in the area by others for Kysor International (Kysor) and Four Star, Inc. (Four Star), have found that volatile organic contamination is widespread in both the water-table and intermediate aquifers of the Industrial Park area. Volatile contamination data obtained through the Supplemental Northernnaire RI are discussed briefly and are presented as tables and figures in this report. The concern for the presence of volatiles coexistent with the hex-chrome lies in the determination of appropriate remedial measures which must consider both organic and inorganic contaminants, if present. The volatile organic contaminant presence will be addressed in more detail as part of the Cadillac Area Remedial Investigation scheduled for implementation in 1986 by MDNR.

The geologic and hydrogeologic data obtained in the previous studies for Kysor and Four Star, however, have been incorporated with data collected during the Northernnaire RI in order to expand the understanding of the hydrogeology beyond the immediate Northernnaire study area. For reference, Figure 2-1 locates and identifies all known existing monitoring wells in the Industrial Park of Cadillac. Several of these wells have been used to provide groundwater quality and piezometric data for the Northernnaire RI.

A
K17



LEGEND

- (-1) NORTHERNAIRE WELLS
- 2 KYSOR WELLS
- (1) FOUR STAR WELLS
-) CITY WATER SUPPLY WELLS
-) DNR WELLS

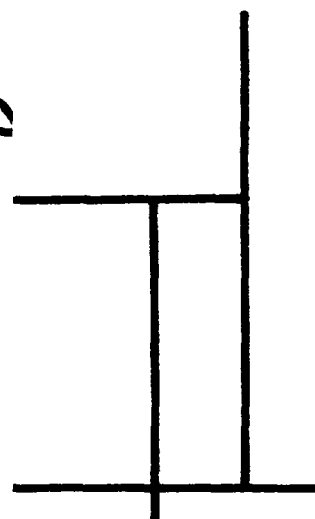


FIGURE 2-1
WELL LOCATION PLAN
AND CURRENT STUDIES
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
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In the 1984-1985 remedial investigation of Northernair by Jordan, groundwater flow in the intermediate aquifer was found to be generally in a northwesterly direction (as other studies had also suggested). Thus, the public supply well field is situated upgradient of the known hex-chrome contamination. However, data gaps and factors which concerned the plume contaminant size, distribution, and fate, as well as uncertainties concerning the upper confining clay layer, and the lower confining clay layer and underlying aquifer, necessitated this continued groundwater investigation. These factors and uncertainties included:

- o hex-chrome is present in the intermediate aquifer at concentrations above drinking water standards and the concentration distribution was not defined;
- o the plume is in the intermediate aquifer and may be migrating toward the deeper aquifer;
- o the direction and rate of movement of the plume was not well-defined;
- o the extent of the plume was not known; and
- o the mechanism by which the hex-chrome may have migrated through the upper clay aquitard had not been established.

2.3 SUPPLEMENTAL REMEDIAL INVESTIGATION

The initial remedial investigation of the Northernnaire site by Jordan identified several data gaps concerning the contamination of groundwater by hex-chrome. It is the purpose of the present investigation supplement, which has included the completion of several additional monitoring wells, to supply needed information to allow for a more accurate assessment of the hex-chrome contamination. To protect the lowest aquifer from contaminants (known and unknown) in the intermediate aquifer, MDNR determined that the lower aquitard should not be penetrated at this time. It was thought that the additional information regarding the extent and concentration of the hexavalent chromium plume together with additional hydrogeologic data would make possible relative assessments of the probable impacts of the Northernnaire derived contaminants on the lowest aquifer. The scope of work for the continuation of the RI is outlined in the site supplemental Work Plan, dated October, 1985 (revised December 1985), approved by the MDNR and USEPA.

The primary objective of the supplemental study was the definition of both the vertical and horizontal extent of the hex-chrome plume in the intermediate aquifer. This was achieved by installing 10 new monitoring wells into this aquifer. Vertical distribution of contamination was determined from testing of groundwater samples for hex-chrome at 10-foot intervals as each boring was advanced. This information was supplemented with data from the analysis of groundwater samples collected from both the newly installed wells and selected existing wells. In addition to water quality information, the installation of these wells provided important geologic and hydrogeologic data. Two

piezometers were also installed to investigate vertical hydraulic gradients within the intermediate aquifer and to provide groundwater elevation data on a somewhat more regional basis to evaluate the potential impact of the plume as a result of groundwater movement.

To summarize, the specific objectives of the supplemental work were the following:

1. to explore the geology and stratigraphy of the site in order to better understand the aquifer/aquitard systems;
2. to obtain vertical and horizontal groundwater quality information to define the extent of the hex-chrome plume;
3. to define the groundwater flow regimes of the shallow and intermediate aquifers;
4. to investigate the hydraulic communication between the shallow and intermediate aquifers, and between intermediate and deeper aquifers insofar as possible with limited data;
5. to collect data to aid in evaluating the environmental impact and ultimate fate of the hex-chrome contamination; and
6. to develop sufficient data to conduct a feasibility study evaluating and recommending remedial alternatives.

This report documents the methods and findings of the field investigations conducted during the supplemental Northernnaire RI and provides an assessment of the hex-chrome contamination, specifically focusing on the objectives outlined above.

3.0 SITE INVESTIGATION

3.1 INTRODUCTION

Data were collected during the field investigations in order to assess the hex-chrome contamination, its potential impact on groundwater quality and to provide an understanding of the site sufficient to evaluate remedial action alternatives. The primary mechanisms of data acquisition were through observations of borings installed with hollow-stem augers and by obtaining groundwater samples for analysis by means of screened-augers. These methods have provided geologic, groundwater quality, and, to some degree, hydrogeologic information needed to evaluate the hexavalent chromium contamination in area groundwater. In addition to data obtained during the drilling program, subsequent groundwater sampling and the collection of water levels have contributed to the assessment of the Northernaire site for this RI.

The following sections describe the methods and procedures employed throughout the course of the field program.

3.2 DRILLING PROGRAM

Drilling of boreholes for monitoring well and piezometer installation was accomplished using a Gus Pech Brat drill rig owned and operated by Keck Consulting Services, Inc. Installations were monitored by a Jordan geologist.

The hollow stem augers used were approximately 6.5-inch outside diameter and 4.5-inch inside diameter. A five-foot screened auger was used as a lead auger in which a removable teflon plug at the bottom of the auger was placed to keep out sediment during drilling.

The location of each monitoring well was discussed by Jordan and MDNR project coordinators prior to drilling. Some initial well locations were pre-determined based on the information gathered in the earlier RI work of 1984-1985. Several new well locations, however, were determined by incorporating recent groundwater flow data from both existing and newly installed wells. Once the ideal location of a well was selected, it was subject to practical considerations such as the presence of underground utilities, granting of access by property owner, and/or physical surface barriers. As a result, not all wells could be positioned in what were considered optimal locations.

Drilling began at location MW-9 on November 22, 1985. The well installation phase, consisting of the 10 new monitoring wells, MW-9 through MW-18, and two piezometers, P-1 and P-2, were completed on January 21, 1986. The approximate location of each of the wells is shown on Figure 3-1.

Prior to drilling near road sides or buildings, Miss Dig (an agency that clears locations of utility lines for contractors and others) was contacted to verify that the well location was a safe distance from any buried utility lines. Also, due to winter conditions during much of the field program, it was often necessary to have a local contractor clear the site of snow to allow access by the drilling rig.

During the drilling of each borehole, the geology of the hole was logged by observing spoils as they came up on the outside flights of the augers. The soils, which will be discussed in more detail in a following section, are primarily sands and clays. When clay was encountered, there was a noticeable change in the rate of drilling, thus allowing for a relatively accurate determination of the depth at which strata changes occurred. Drilling of the borehole stopped when the deep clay layer separating the intermediate and deeper aquifers was encountered. This lower clay layer was present generally at depths of between 155 and 183 feet.

The boring logs describing the soils encountered with depth are provided in Appendix A.

3.3 SCREENED AUGER SAMPLING

Groundwater sampling and field screening for hex-chrome and other parameters were done during the drilling of each borehole for monitoring well installation. Beginning at from five to ten feet below the water table, samples were taken through the lead five-foot section of screened auger at 10-foot intervals until completion of the boring. At each interval, the well was developed until the groundwater was clear (usually 40 to 60 gallons), and then a submersible pump was used to purge the well. Initially, purging was done on a well volume basis; one well volume was purged before the groundwater sample was collected. However, with increasing depths, and with a pump rate of under one gallon per minute, the time involved in purging one well volume was steadily increasing.

For the purpose of determining minimum acceptable purge volumes, groundwater was collected and tested for hex-chrome every 10 to 20 gallons during purging while drilling wells MW-9 and MW-10. At the 162- to 167-foot interval in well MW-9, hex-chrome was detected after only 10 gallons had been purged and continued to be present in the groundwater after one well volume (95 gallons) had been purged (see boring log, Appendix A). This indicated that the well was recharging fast enough so that the pump was drawing groundwater mainly through the screen and not evacuating the water column in the hollow-stem augers. To further verify this, the water level in the augers was monitored by tape measure during purging. This revealed that no significant drawdown was occurring. In light of these experiments and the attendant time savings, it was agreed by MDNR and Jordan that an adequate purge volume would be 20 gallons. Purge waters contained little contamination and were, upon testing, poured onto the ground to allow infiltration through the soils at the drilling location. An alternative considered was to release purge water to the municipal sanitary system, but generally the distance to the nearest sewer was prohibitive.

Field testing of groundwater samples was conducted to determine the elevations of the well screens and to provide data on the vertical distribution of contaminants. All samples were screened for temperature, pH, and specific conductivity using a TriPar three-function meter, and for hex-chrome using a HACH colorimetric test kit. When hex-chrome was positive in a field test, a sample was collected for submittal to the USEPA Contract Laboratory Program (CLP). Other samples for CLP were taken at 10-foot intervals starting at a predetermined depth, generally when approaching the deep, confining clay layer. CLP lab space was limited during the time of the investigation, but most wells

provided four CLP samples each for volatile organics, total dissolved metals, and hex-chrome. Between each 10-foot sampling interval, the pump was decontaminated with a trisodium phosphate (TSP) solution and then rinsed with potable water.

The boring logs in Appendix A summarize the pertinent data obtained from the screened auger groundwater sampling and analysis.

3.4 WELL INSTALLATION

Once the deep clay layer was encountered, drilling was stopped and preparations were made for installing the well. The well screens were placed in the zone of highest hex-chrome contamination for a particular boring. If hex-chrome was not detected, the screen was set at an appropriate level based on a comparison with adjacent upgradient and downgradient wells. Once the well screen level was determined, the augers were withdrawn to that level, allowing the hole below to collapse.

Ten-foot sections of AW-rod were screwed together and placed inside the augers in order to remove the teflon plug at the bottom of the screened auger. The well was then emplaced, using flush-coupled, ten-foot sections of Schedule 80, 2-inch I.D. PVC casing and a five-foot section of No. 10 slotted PVC well screen. Once the casing and screen were set, 15 feet of augers were removed which allowed the sand to collapse around the well screen and casing to form a natural sand filter pack. A bentonite grout was then pumped into the annulus

between the PVC casing and augers to provide a seal inhibiting migration of groundwater and sediment through the disturbed backfill material. The grouting was also necessary in some wells to seal the borehole through the upper clay layer to prevent unnatural groundwater mixing between the intermediate and water-table aquifers. The grout was completed to within approximately one to two feet below ground level. Steel well protective casings with locking caps were then placed over the wells. Figure 3-2 illustrates a typical well installation. Appendix A provides the completed specifications for each monitoring well.

Monitoring wells were developed by hydraulic surging with an air compressor. Development continued until the groundwater was free of fine sediment.

After drilling and installation of the well, all down-hole drilling equipment plus the rig were decontaminated with a high-pressure steam generator at a designated decontamination area adjacent to the site support trailer on Northernnaire property.

3.5 PIEZOMETER INSTALLATION

Two piezometers, P-1 and P-2, were installed into the upper levels of the water table aquifer in areas where the upper clay aquitard is absent. Piezometer P-1 is about 10 feet west of MW-11, and P-2 is located north of St. Ann's church on 13th Street. The piezometers were not intended for groundwater quality

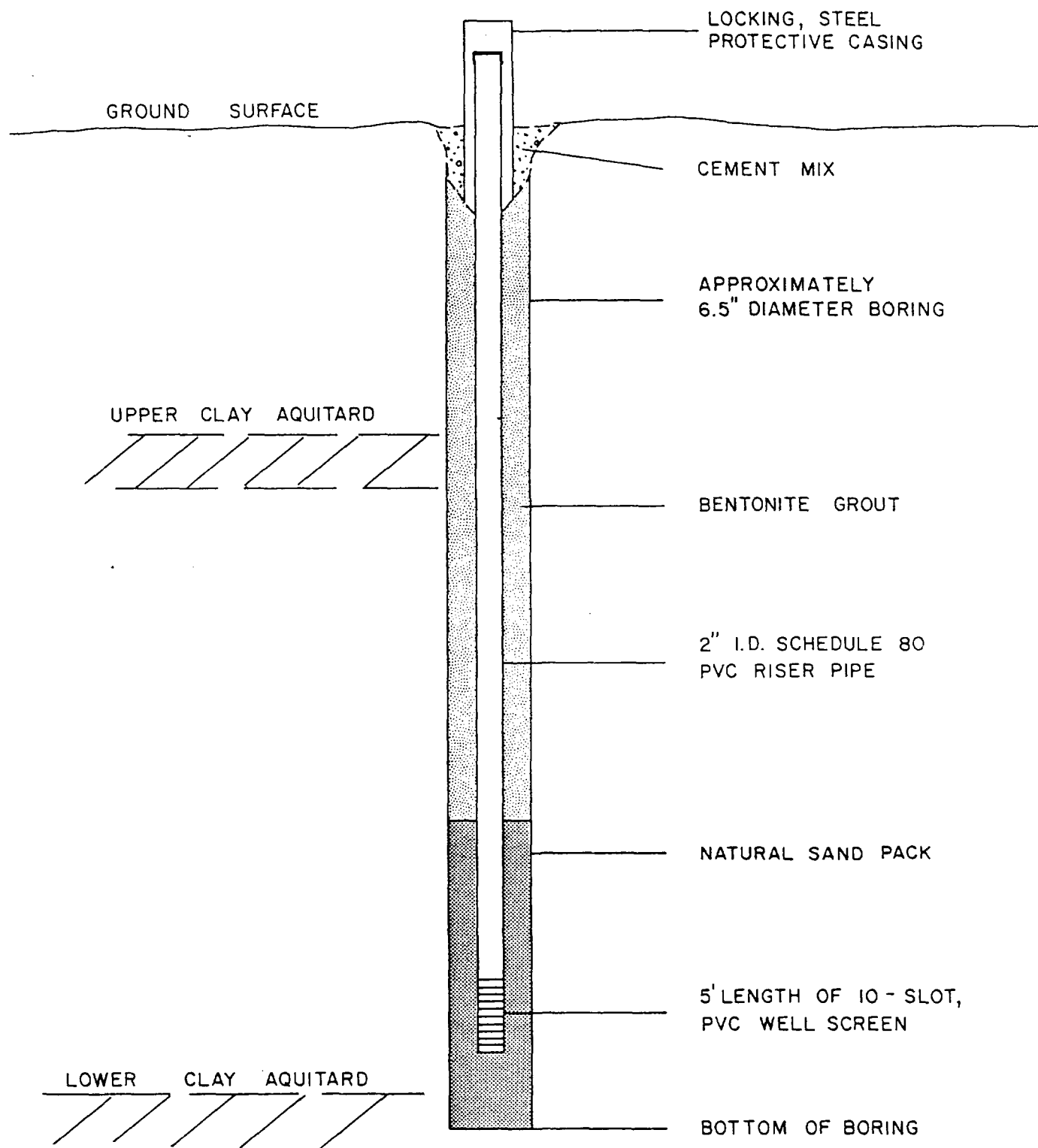


FIGURE 3-2
TYPICAL MONITORING
WELL CONSTRUCTION
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN

sampling, but provided water level data to help identify vertical flow gradients (P-1 and MW-11) and regional flow direction (P-2) at the northern end of the study area. Each PVC piezometer has a five-foot screen section set from approximately 45 to 50 feet below ground surface. The installations were backfilled with the drill cuttings rather than a bentonite grout since the piezometer would not be used as a monitoring well to collect samples for determinations of groundwater quality. The boring logs and well design specifications are included in Appendix A.

3.6 SURVEYING

Throughout the project, the elevation of each new monitoring well pipe was surveyed by Keck and Jordan personnel using standard surveying equipment. This was done so that groundwater elevations would be immediately available and could be used to determine flow directions to aid in the placement of subsequent wells. Approximate horizontal locations of the new and selected existing wells were also determined using a compass and tape measure. After the drilling program was completed, the MDNR provided a licensed survey crew to determine more precise vertical elevations and horizontal positions of the wells. These data are listed on Table 3.1 along with other boring and monitoring well information. These more accurate determinations were necessary inasmuch as the hydraulic gradients in the intermediate aquifer are small and even slight vertical measurement errors could significantly affect the interpretation of groundwater flow direction.

TABLE 3-1

BORING AND MONITORING WELL INFORMATION
SUPPLEMENTAL NORTHERNAIRE RI

Monitoring Well	Elevation		Aquitard Depth		Screen Interval
	Ground	TOC	Upper	Lower	
MW-1D	ND	1300.46*	63	NE	59-64
MW-1S	ND	Bent	NE	NE	35-40
MW-2	ND	1297.67	67	NE	48-53
MW-3	ND	1297.27	50	NE	38-48
MW-4	ND	1294.91	NE	NE	26-31
MW-5	ND	1297.28	50	NE	38-48
MW-6	ND	1296.91	54	NE	44-54
MW-7	1299.3	1301.32	NP	169	157.5-162.5
MW-8	ND	1295.49	(51-61)***	188	183-188
MW-9	1299.1	1301.86	(58-61)	173	162-167
MW-10	1298.4	1301.40	NP	156	150-155
MW-11	1299.4	1301.62	NP	169	161-166
MW-12	1297.1	1299.60	(36-67)	175	149-154
MW-13	1299.2	1301.01	NP	159	154-159
MW-14	1298.7	1301.06	NP	183	153-158
MW-15	1297.4	1299.60	(42-73)	180	160-165
MW-16	1298.5	1301.17	(54-63)	175	164-169
MW-17	ND	**	(40-49)	177	144-149
MW-18	1298.3	1300.97	NP	155	149-154
P-1	1299.3	1301.83	NP	NE	45-50
P-2	ND	**	NP	NE	44-49

Elevations in feet above Mean Sea Level (as surveyed by MDNR)

ND = No Data

NE = Not Encountered (but probably present)

NP = Not Present

* Suspected Error

** Not Surveyed

*** (51-61) Indicates Depth Interval of Upper Aquitard

4.0 SITE CHARACTERIZATION

4.1 GEOLOGY

The geology was determined by logging the spoils that came from the borehole during drilling and noting depths at which significant changes in torque required to turn the augers occurred. The recent data confirm previous descriptions of the geologic sequence and provide additional detail in the area of the contaminant plume.

Essentially, only four major distinct stratigraphic units were encountered at the site. These were two relatively thick units of sand, and two layers of clay. All of these materials are glacio-lacustrine in origin. Silt and gravel lenses were present on occasion. The sand deposits are generally brown, well-sorted, well-rounded, and medium-grained with traces of coarser sand. The clay is either red or gray-brown in color, firm, and often silty. Gravel was present in some clay strata, either as distinct seams, as in MW-12 and MW-15, or mixed in a clay matrix, as in MW-17.

These deposits of sand and clay alternately form two aquifers and two aquitards. A third sand unit underlying the lower aquitard, forms the deep aquifer which supplies the municipal water wells. Figures 4-1 and 4-2 are geologic cross-sections, representing the vertical configuration of these units. These interpretive cross-sections were derived from data acquired from the Northernair RI investigation as well as previous studies conducted for

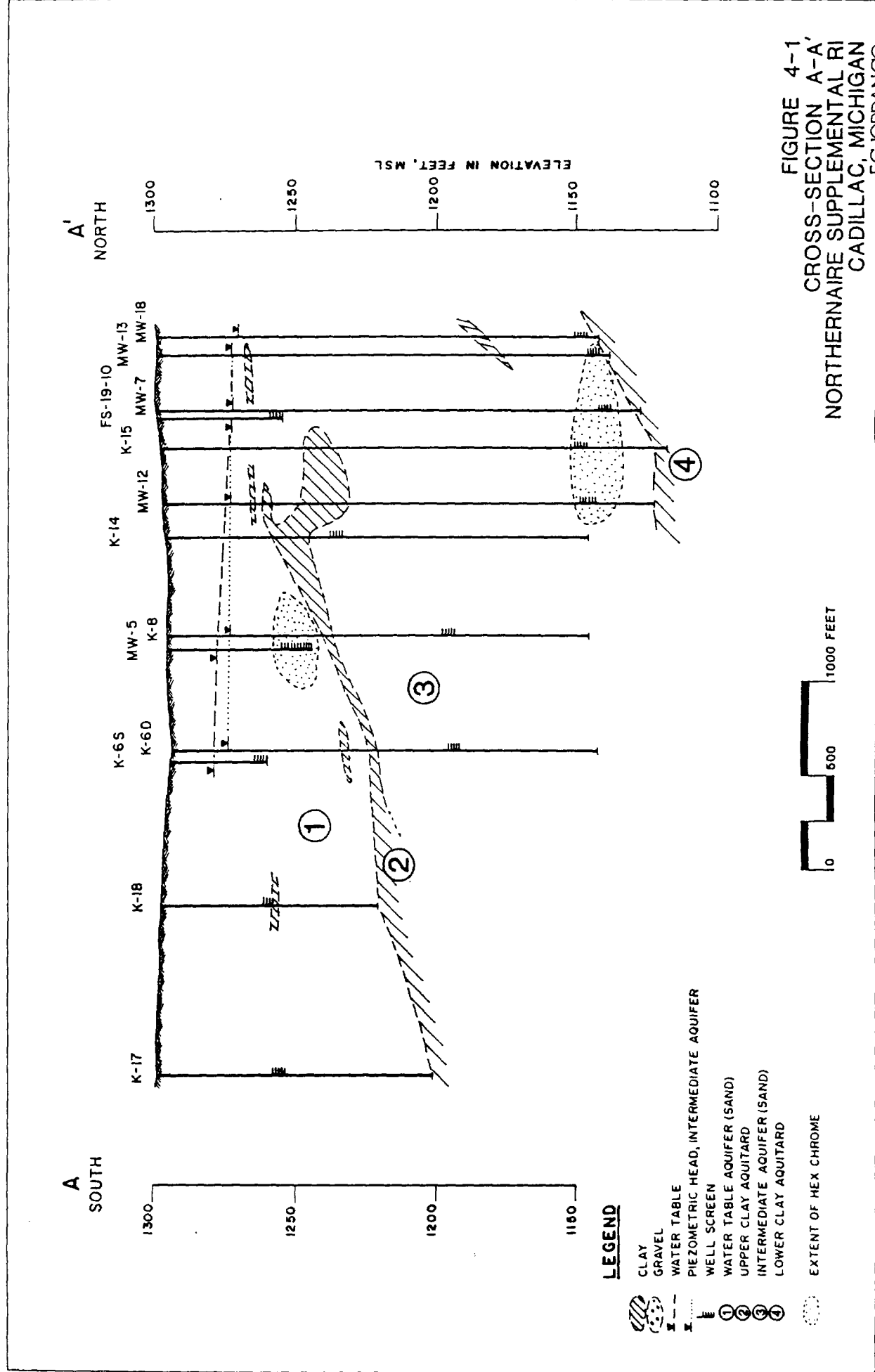


FIGURE 4-1
CROSS-SECTION A-A'
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

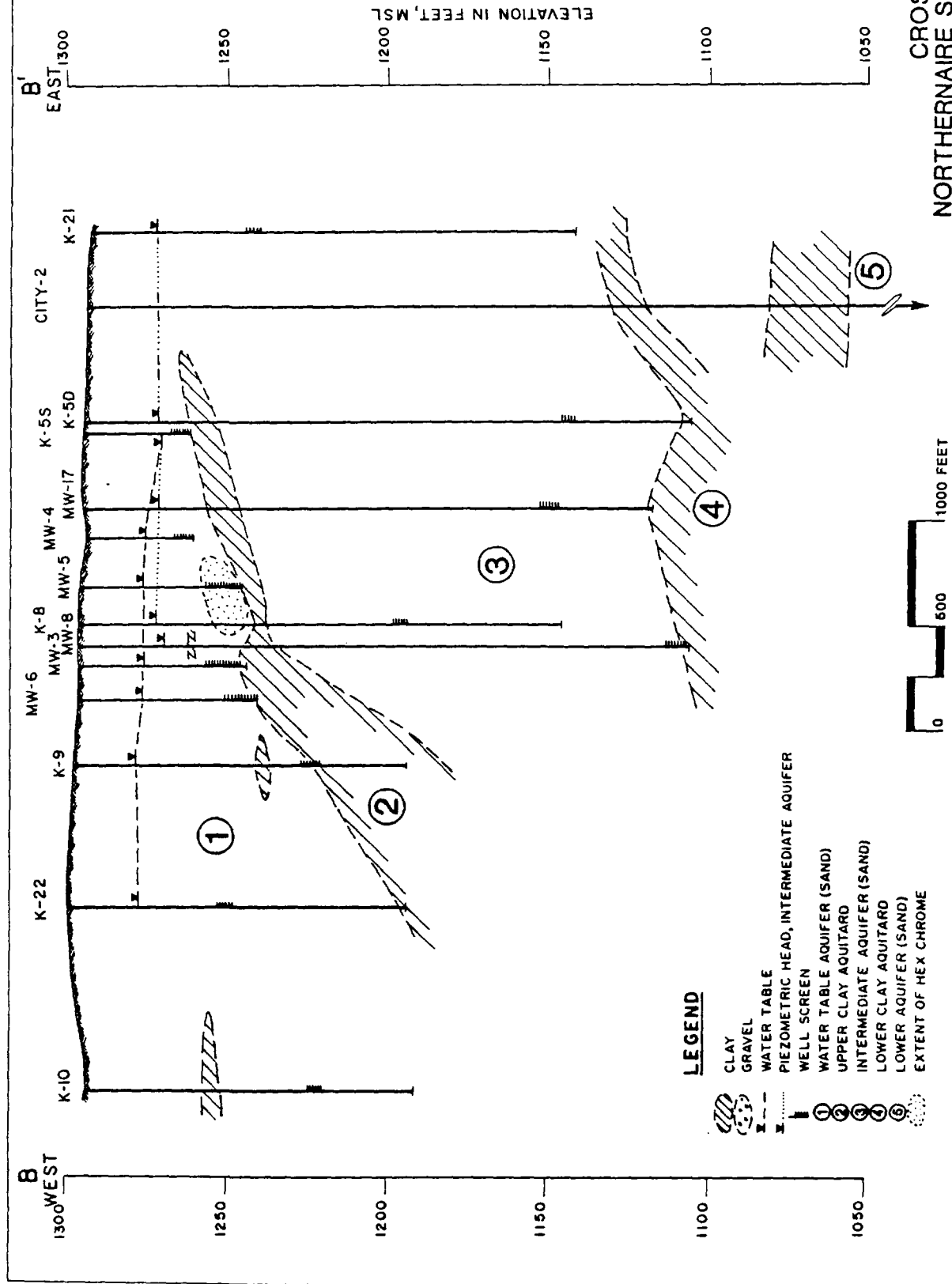


FIGURE 4-2
CROSS-SECTION B-B'
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
EC JORDANCO

Kysor and Four Star. Figure 2-1 shows the locations of the cross sections relative to the study area.

The uppermost sandy aquifer exists under water-table conditions and ranges in thickness across the study area from approximately 30 to 60 feet. The water table was generally encountered at a depth of from 30 to 35 feet. The first substantial clay layer encountered ranged in thickness from 2 to 18 feet across the southern and central parts of the study area but was absent in the northerly explorations. Figures 4-3 and 4-4 show plan views of the depth and thickness, respectively, of this upper clay unit. Because this unit is compact and composed predominantly of clay and some silt, it acts as an aquitard between the upper two water-bearing sand units. Where this clay layer ceases to be present to the northeast of the Northernnaire site, the upper sand aquifer merges with the intermediate aquifer to become a single unit. This is illustrated on the cross-sections of Figures 4-1 and 4-2.

In eight of the ten new borings drilled for the monitoring wells, and in well MW-8 (drilled previously), a thin clay layer was encountered at a depth of approximately 32 feet. This layer is shown on the cross-sections. The continuity of this layer over the site is uncertain in that it was not noted in several previous borings. This, however, may be a result of the fact that, when present, the clay is only 0.5 to 1.5 feet thick. During drilling, groundwater was encountered immediately above this clay layer. The hydraulic influence of this clayey zone is probably slight, due to its thinness and likely discontinuous presence.

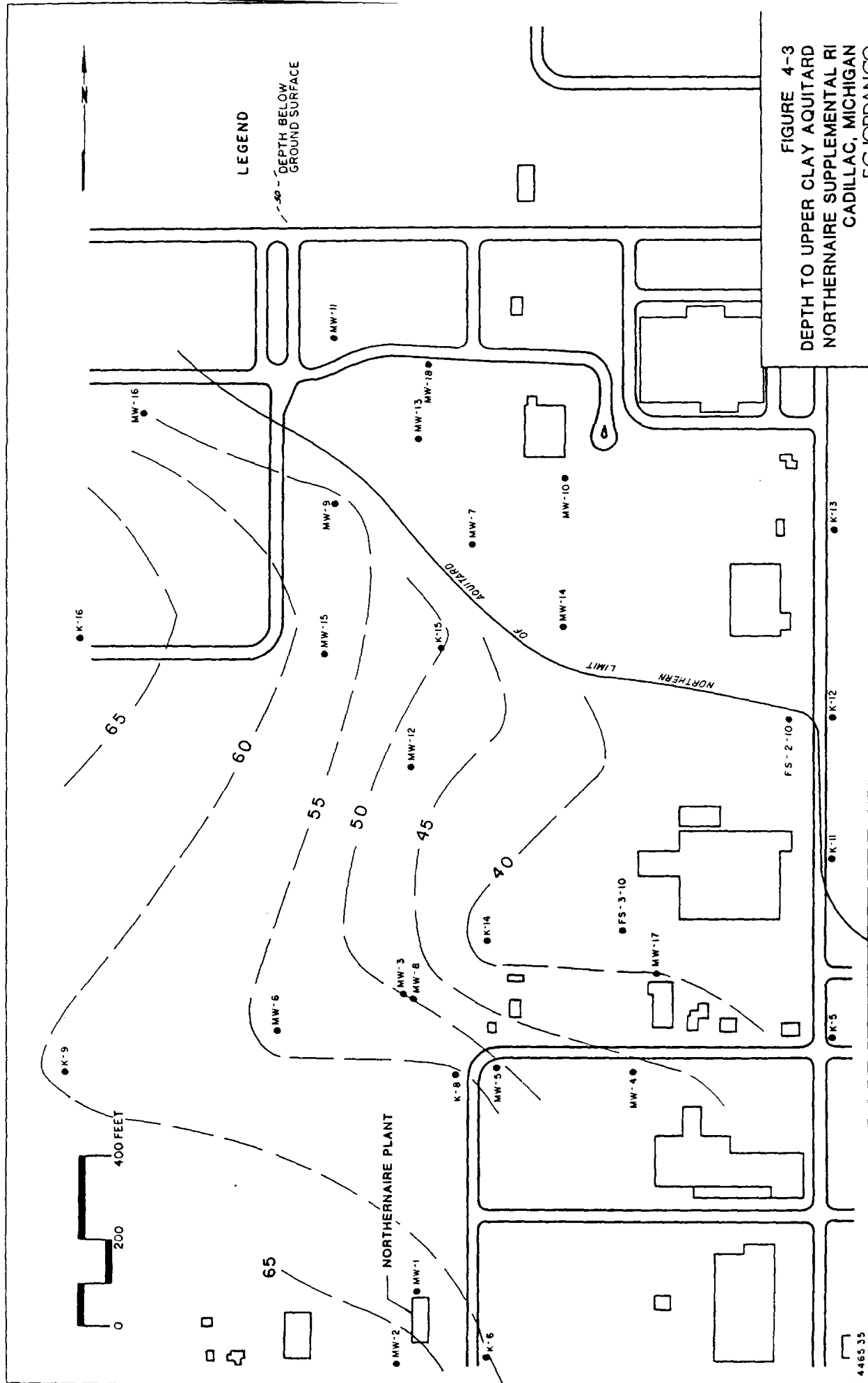


FIGURE 4-3
DEPTH TO UPPER CLAY AQUITARD
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

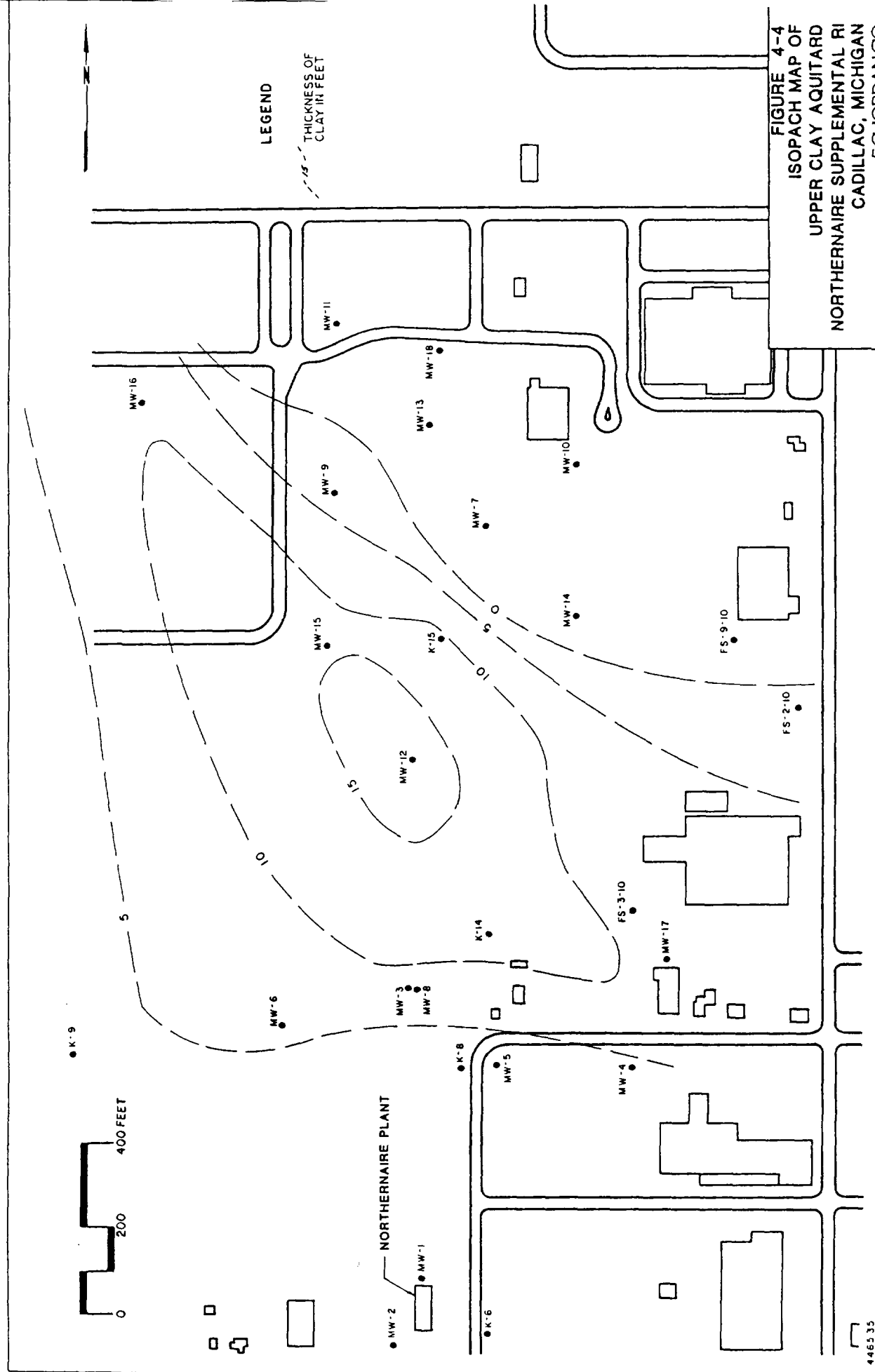
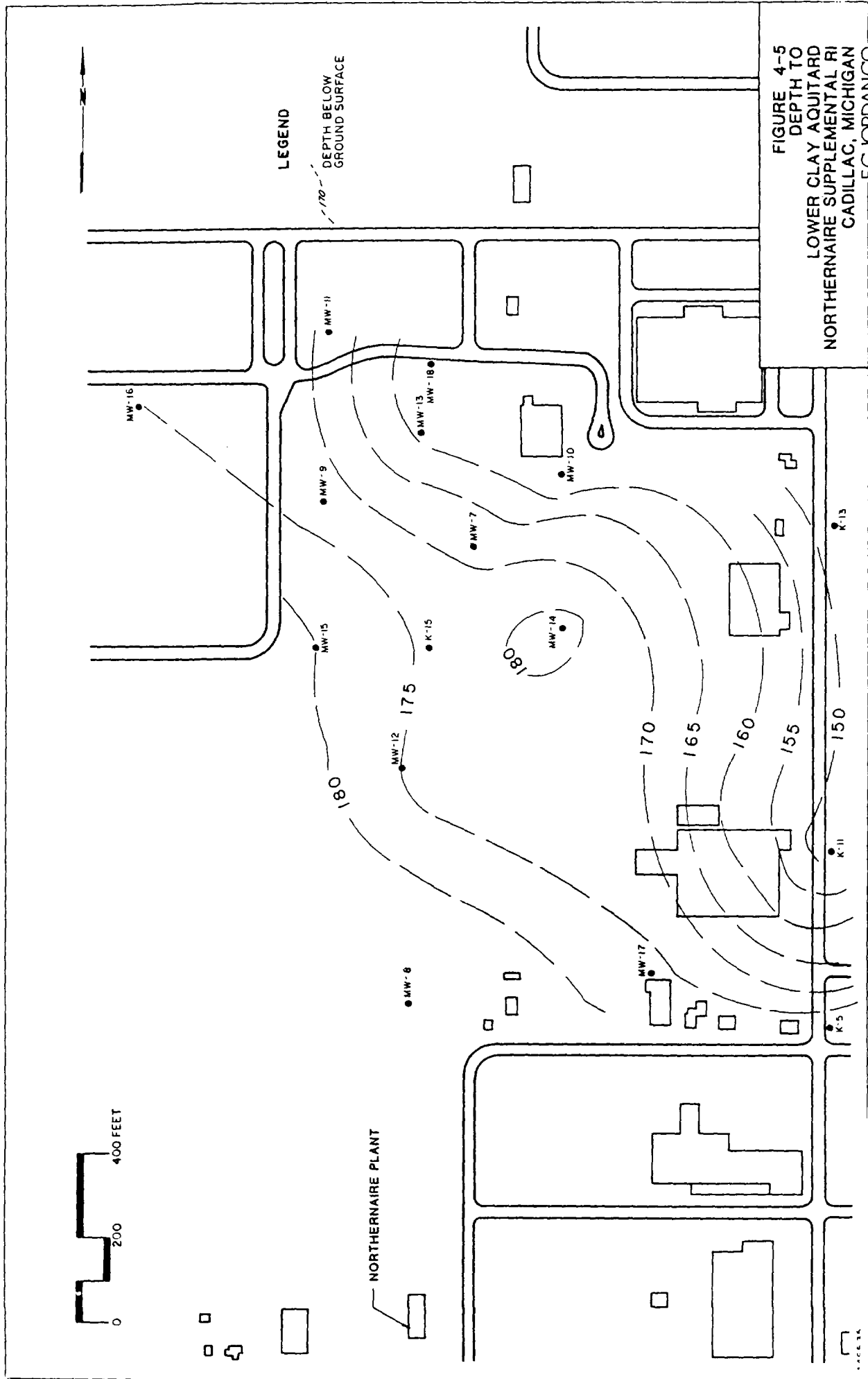


FIGURE 4-4
ISOPACH MAP OF
UPPER CLAY AQUITARD
NORTHERNNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
EC JORDANCO

Below the first clay aquitard lies the intermediate aquifer. This aquifer consists of a brown, well-sorted sand similar to that of the water-table aquifer. The thickness of the intermediate aquifer ranges from 100 to 130 feet. Where the upper clay aquitard is not present in the north and northeast of the Northernnaire site, the aquifer becomes unconfined and may be as much as 170 feet thick. The bottom of the intermediate aquifer is defined by another silty clay layer probably with a relatively low hydraulic conductivity which separates this aquifer from the still deeper aquifer which supplies the municipal wells. The lower clay aquitard was encountered during the supplemental work at a depth of 155 to 183 feet (Figure 4-5). A zone high in silt content was typically present just above the deeper clayey aquitard. This was most apparent when developing the boring during screened auger sampling. The thickness of this lower clay unit is not generally known. This lower clay layer was augered into by as much as four feet (boring MW-10). Boring logs from the city wells penetrating this unit suggest that this clay aquitard may be a series of several clay and sand units of various thicknesses and, in areas, may be as much as 30 feet thick. In order to eliminate any possibility of creating a conduit between the two aquifers, MDNR decided that this layer should not be breeched during the investigation. Figure 4-2 incorporates information from one of the boring logs of the city wells. The logs, however, do not allow for a detailed interpretation of the deeper aquitard structure.

The composition and character of the water-table and intermediate aquifer materials are remarkably uniform across the site. This simplifies interpretations of hydraulic characteristics since they should reflect similar uniform-



ity. The clayey aquitards are somewhat less predictable, particularly in regard to areal extent. The first (upper) major aquitard pinches out along a northwest-southeast line north of Kysor monitoring well K-15 and, as noted previously, exhibits a wide range in thickness where encountered (Figure 4-4). This aquitard also varies compositionally in that gravel may be present which may greatly increase the permeability of the unit.

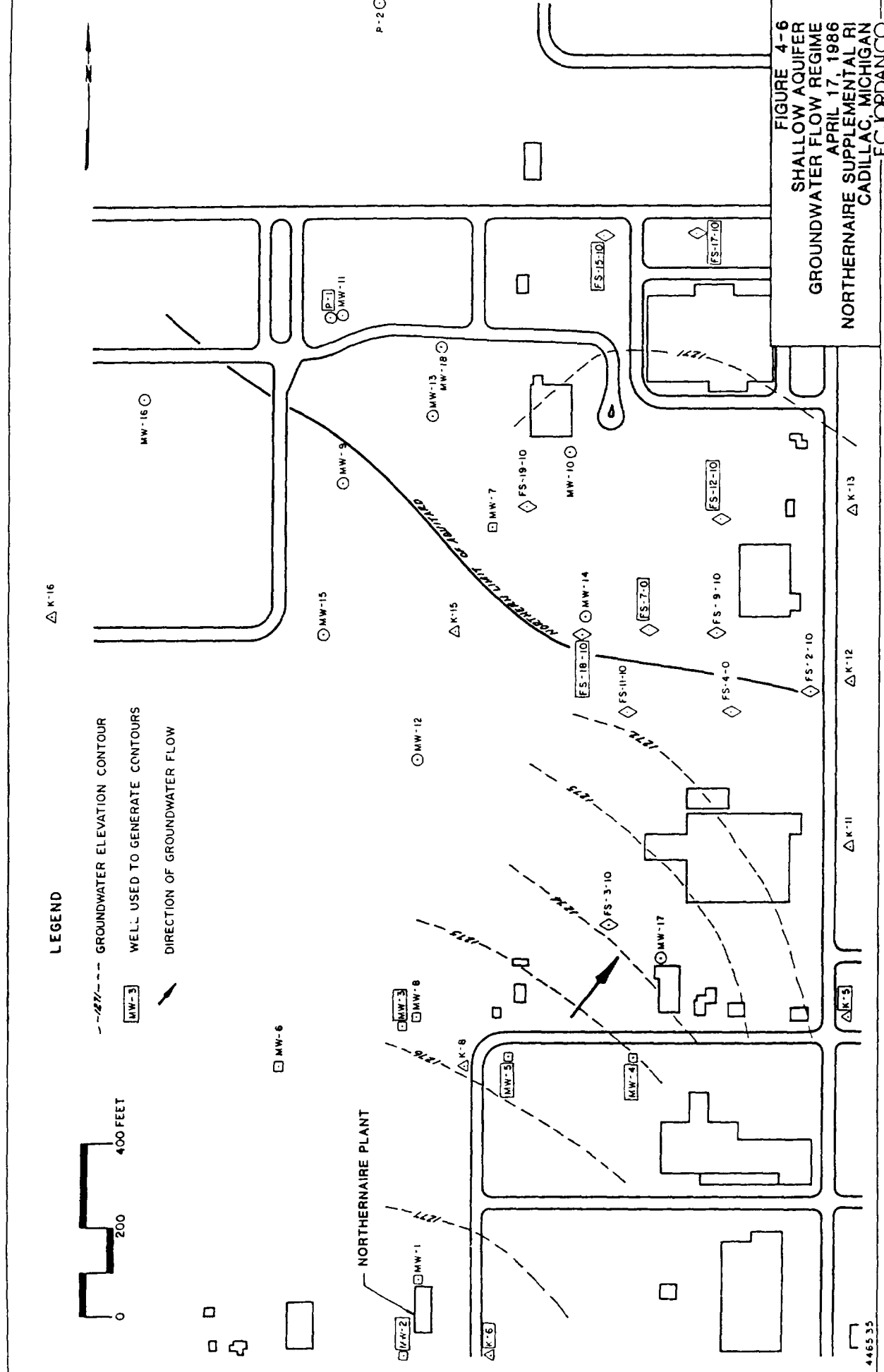
The lower clay aquitard was encountered at all locations across the site and has an apparent dip to the southwest. Figure 4-5 illustrates the depth at which this clay has been encountered. One deviation to the general southwest dip of this unit is near the center of the site at monitoring well MW-14. Here the clay appeared to be present slightly deeper than in adjacent wells.

Small clay lenses and more extensive clay layers were also encountered in several borings. The thin clay layer at a depth of 32 feet has already been discussed. A small clay lens at a depth of 111 to 121 feet was encountered in wells MW-18 and MW-13, respectively (Figure 4-1). Small clay lenses were encountered at shallow depths in monitoring wells MW-12 and MW-15.

4.2 HYDROGEOLOGY

Water Table Aquifer:

Figure 4-6 presents an interpretation of groundwater flow in the shallow



LEGEND

- - - 1271 - - - GROUNDWATER ELEVATION CONTOUR
- [MW-3] WELL USED TO GENERATE CONTOURS
- DIRECTION OF GROUNDWATER FLOW

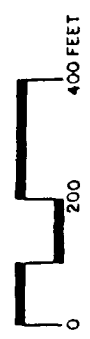


FIGURE 4-6
SHALLOW AQUIFER
GROUNDWATER FLOW REGIME
APRIL 17, 1986
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

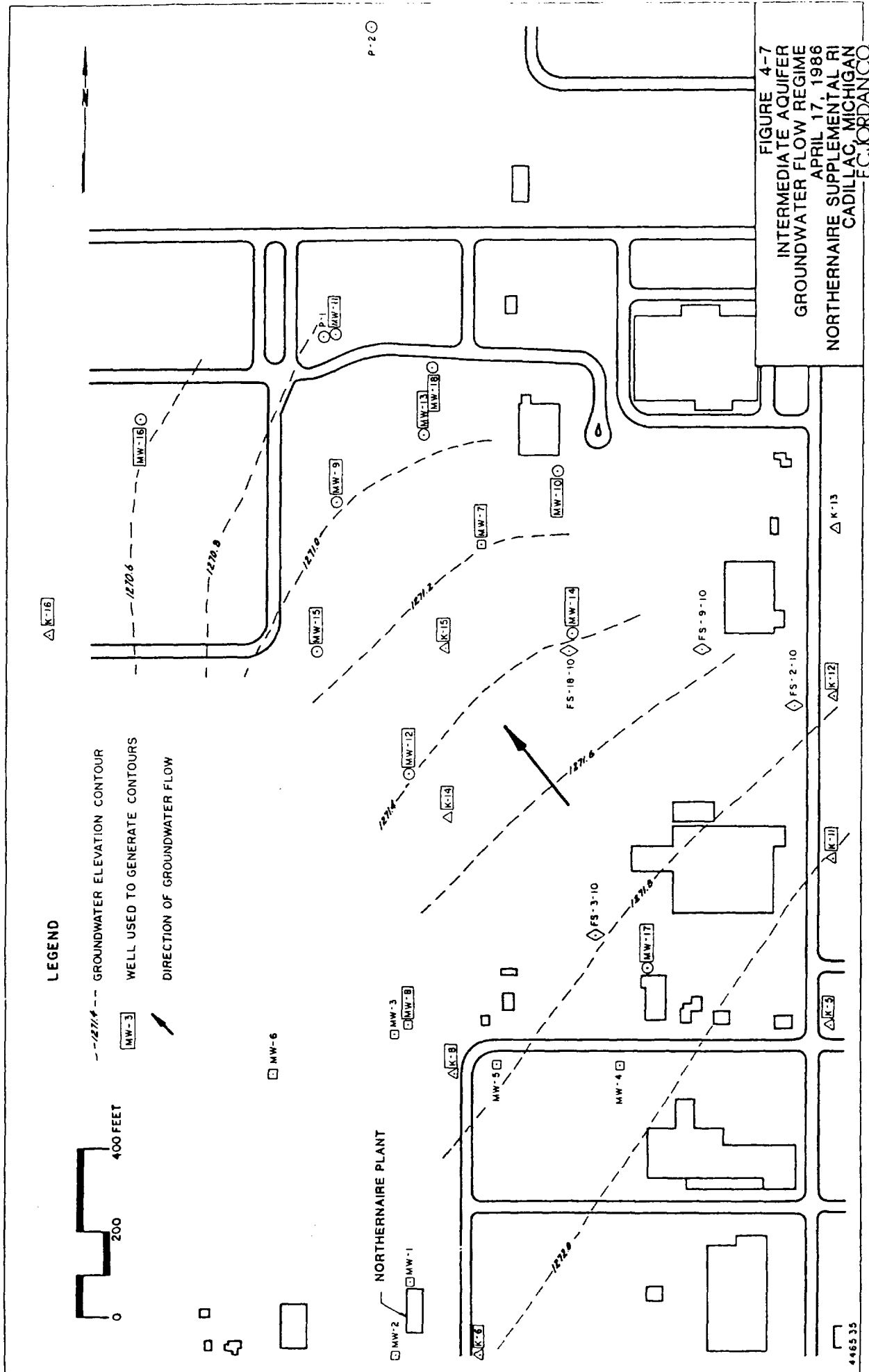
aquifer based on water level data obtained on April 17, 1986. Groundwater flow is to the north-northeast which is consistent with previous interpretations. North of where the aquitard pinches-out, the flow direction changes to a more northerly direction due to the merging of the upper aquifer with the intermediate aquifer which flows to the northwest. The horizontal gradient in the shallow aquifer averages approximately 0.004 feet per foot above where the upper clay aquitard is present. This gradient appears to gradually increase to the north east as the clay aquitard rises (Figs. 4-1 and 4-2). The gradient quickly becomes less steep (approximately 0.001 feet per foot) when the aquifers merge. The increasingly steeper gradient above the aquitard is due to the clay sloping upward from southwest to northeast in the direction of groundwater flow thus decreasing the saturated thickness (cross-sectional area) of the aquifer. In order to maintain a constant quantity of water per unit time with a constant permeability assumed for the aquifer and a decreasing cross-sectional area, the hydraulic gradient must increase. However, the increase in gradient does not offset the decreased cross sectional area in the direction of flow. This indicates probable significant leakage through the aquitard. This will be discussed in more detail later.

Intermediate Aquifer:

The direction of groundwater flow in the intermediate aquifer was monitored periodically throughout the course of the supplemental drilling program primarily to aid in the placement of monitoring wells in relation to known contamination. (The hex-chrome is highly soluble and, with little opportunity for

attenuation by the sandy soils, its movement should closely match groundwater flow rates.) Once all the wells were emplaced, they were re-surveyed by MDNR to obtain more precise vertical elevations. On April 17, 1986, water levels were obtained again. Figure 4-7 illustrates the groundwater flow of the intermediate aquifer on that date. Groundwater flow in this aquifer is generally to the northwest and measurements taken on previous dates reflect similar flow patterns. There appear to be some anomalies to the general northwesterly flow in the vicinity of MW-7, MW-9, MW-13 and MW-15. Flow in this area appears to have two distinct components with flow from MW-9 northerly to MW-11 and flow near MW-15 westerly toward K-16. With the very flat gradient, it is unclear what influences may be causing these apparent or real flow pattern variations. Additional data to be collected for the area-wide Cadillac groundwater study may clarify the flow pattern and resolve the questions regarding other possible hydrogeologic influences. Table 4-1 summarizes the groundwater elevations obtained throughout the course of the field work.

The horizontal hydraulic gradient in the intermediate aquifer is quite small across the Northernnaire site. Over an 1800-foot distance separating wells MW-17 and MW-16, the hydraulic head in the intermediate aquifer drops only 1.12 feet, which is a gradient of approximately 0.00062 foot per foot. This corresponds well to the 0.00068 horizontal hydraulic gradient cited in the May 1985 RI report for the intermediate aquifer.



Vertical Gradients:

With the addition of the new deep monitoring wells, and by using several existing wells from previous studies, vertical hydraulic gradients between the shallow and intermediate aquifers were determined at the following well pairs: K-5S and K-5D; K-6S and K-6D; K-16S and K-16D (Kysor study); MW-3 and MW-8; MW-5 and K-8; and MW-17 and FS-3-10 (Four Star study). Where the upper aquitard is absent, several shallow and deep well pairs exist where vertical gradients across the expanded water-table aquifer can be determined. These pairs include: MW-7 and FS-19-20; MW-11 and P-1; MW-14 and FS-18-10; and K-12 and FS-2-10.

Cross-sections in Figures 4-1 and 4-2 illustrate the hydraulic head differences between the water table and intermediate aquifers. It is apparent that where the upper aquitard is present, a significant downward gradient exists. Near the south-central portions of the site by monitoring well pair K-6S and K-6D, and by monitoring wells MW-5 and K-8, downward vertical potentials of over 5.5 and 3 feet exist, respectively. These head differences gradually diminish as the edge of the upper aquitard is approached. Much smaller downward vertical potentials on the order of 0.01 to 0.20 feet exist in the shallow and deep well pairs penetrating the water table aquifer north of the limit of the upper clay aquitard.

TABLE 4-1

GROUNDWATER ELEVATIONS
SUPPLEMENTAL NORTHERNAIRE RI

Monitoring Well	TOC	Elevation		
		1-21-86	2-25-86	4-17-86
MW-1i	1300.46*	ND	1279.83	1281.07
MW-2	1297.67	ND	1276.59	1277.89
MW-3	1297.27	ND	1274.56	1275.82
MW-4	1294.91	ND	1274.07	1275.07
MW-5	1297.28	ND	1274.53	1275.65
MW-6	1296.91	ND	1274.44	1275.63
MW-7	1301.32	1270.91	1270.50	1271.20
MW-8	1298.00	1271.22	1270.89	1271.64
MW-9	1301.86	1270.72	1270.38	1271.00
MW-10	1301.40	1270.77	1270.40	1271.06
MW-11	1301.62	1270.54	1270.17	1270.83
MW-12	1299.60	1271.05	1270.72	1271.39
MW-13	1301.01	1270.68	1270.44	1270.98
MW-14	1301.06	1271.07	1270.33	1271.38
MW-15	1299.60	1270.79	1270.42	1271.13
MW-16	1301.17	1270.26	1270.05	1270.57
MW-17	**	ND	ND	ND
MW-18	1300.97	1270.59	1270.29	1270.87
P-1	1301.83	1270.63	1270.16	1270.90
P-2	**	ND	ND	ND
K-5S	1297.58	ND	1270.72	1271.52
K-5D	1295.16	ND	1271.13	1272.06
K-6S	1297.07	ND	1276.66	1277.74
K-6D	1294.90	ND	1271.28	1271.99
K-8	1296.64	1271.33	1270.89	1271.75
K-9	1297.94	ND	ND	1277.30
K-11	1295.85	ND	1271.13	1272.00
K-12	1299.89	ND	1270.89	1271.75
K-13	1299.16	ND	ND	1271.38
K-14	1297.82	ND	ND	1271.55
K-15	1299.78	1270.96	ND	1271.33
K-16S	1299.46	ND	ND	1273.54
K-16D	1299.60	1270.07	ND	1270.39
K-20	1295.05	ND	ND	1272.95
K-21	1293.53	ND	ND	1272.60
K-22	1301.68	ND	ND	1276.02
K-23	1296.30	ND	ND	1272.19
DNR-1	1297.97	ND	1271.99	1272.99
DNR-2	1300.36	ND	ND	1271.41

TABLE 4-1
(continued)

GROUNDWATER ELEVATIONS
SUPPLEMENTAL NORTHERNAIRE RI

Monitoring Well	TOC	Elevation		
		1-21-86	2-25-86	4-17-86
FS-4-0	1296.86	ND	ND	1271.71
FS-7-10	1301.39	ND	ND	1271.48
FS-11-10	1301.73	ND	ND	1271.85
FS-12-10	1299.96	ND	ND	1271.32
FS-15-10	1301.54	ND	1269.90	1270.69
FS-17-10	1300.07	ND	ND	1270.69
FS-18-10	1300.98	ND	1271.17	1271.52
FS-18-20	1300.42	ND	1270.20	1271.45
FS-19-10	1300.54	ND	ND	1271.33
FS-19-20	1300.47	ND	ND	1271.21

Elevations in feet above Mean Sea Level

* Suspected Error

** Not Surveyed

ND = No Data

The presence of substantial downward vertical gradients between the upper and intermediate aquifers is significant in that the gradients provide a mechanism by which contaminants can be transported from the upper to the lower aquifer through the clay aquitard. This will be discussed in more detail in a following section. Also, although vertical gradients are small in the unconfined intermediate aquifer, they are generally as great or larger than the horizontal gradients in much of the aquifer. Thus, in areas, vertical components of the groundwater flow may be greater than horizontal components and tend to carry contaminants deep into the intermediate aquifer.

There is evidence from the hydraulic gradients measured and their patterns for the presence of vertical seepage through both aquitards. With regard to the shallow aquifer, the uniform horizontal gradient in the presence of a rising lower boundary (the upper aquitard) suggests a seepage rate at least equal to the rate of rise the groundwater would encounter as it flows northeasterly toward the rising clay aquitard. In a water balance sense, the constant velocity of the flow through a narrowing cross-section must be balanced by a loss of water, probably through the lower aquitard. Further, this loss must also accommodate the recharge rate the upper aquifer experiences through infiltration. A similar argument holds for the intermediate aquifer. Here the gradient and cross section for flow are relatively constant (as is the water table elevation over the years). Hence, seepage rates through the lower aquitard must approximate the recharge rate through infiltration of precipitation. This probably is in the range of 10-30 inches per year given the sandy nature of the soils.

5.0 CONTAMINATION ASSESSMENT

5.1 GROUNDWATER QUALITY

Screened Auger Sampling:

The procedures for obtaining groundwater samples by way of screened augers have been discussed in Section 3.3. In field testing the screened auger samples, hex-chrome was detected in only three of the ten new monitoring wells installed. Table 5-1 summarizes, by well, the presence of hex-chrome and total chromium from the screened auger sampling and CLP data. The highest concentration of hex-chrome was detected in monitoring well MW-12. Two successive intervals in this well contained hex-chrome, one from a depth of 148 to 153 feet at 0.15 parts per million (ppm), and the other from 158 to 163 feet at 0.08 ppm. As shown on Figure 4-6, MW-12 is upgradient of well K-15 where hex-chrome contamination was detected earlier (0.93 ppm, January 1985). Monitoring well MW-8 (installed in January, 1985), which is closer to Northernnaire than MW-12, showed no hex-chrome contamination. Therefore, the edge of the plume in the intermediate aquifer probably exists between monitoring wells MW-12 and MW-8.

Hex-chrome was also detected by screened auger sampling in two wells, MW-9 and MW-13, which are downgradient from the previously detected contamination at wells K-15 and MW-7. The concentration of hex-chrome in MW-9 was 0.10 ppm at a

TABLE 5-1

CHROMIUM DETECTION LEVELS
SUPPLEMENTAL NORTHERNAIRE RI

Monitoring Well	HACH* Winter 1985/86	CLP: 2/86 Chromium		HACH* 4/86	CLP: 5/86 Chromium	
		Total	Hex		Total	Hex
MW-5	ND	1.55	1.60	1.10	0.43	0.54
MW-7	ND	0.49	0.50	ND	0.29	0.34
MW-9	0.10	0.10	0.05	ND	0.08	0.09
MW-12	0.15	**	**	**	**	**
MW-13	<0.05	0.04	<0.01	ND	0.03	0.04
K-15	ND	1.97	2.20	ND	ND	ND

Results in Parts Per Million (PPM)

Monitoring wells not listed either did not contain chromium or did not contain chromium at levels above associated field blanks.

* HACH test kit detects hex-chrome only.

** Could not obtain a sample due to a bent well casing.

ND = No Data

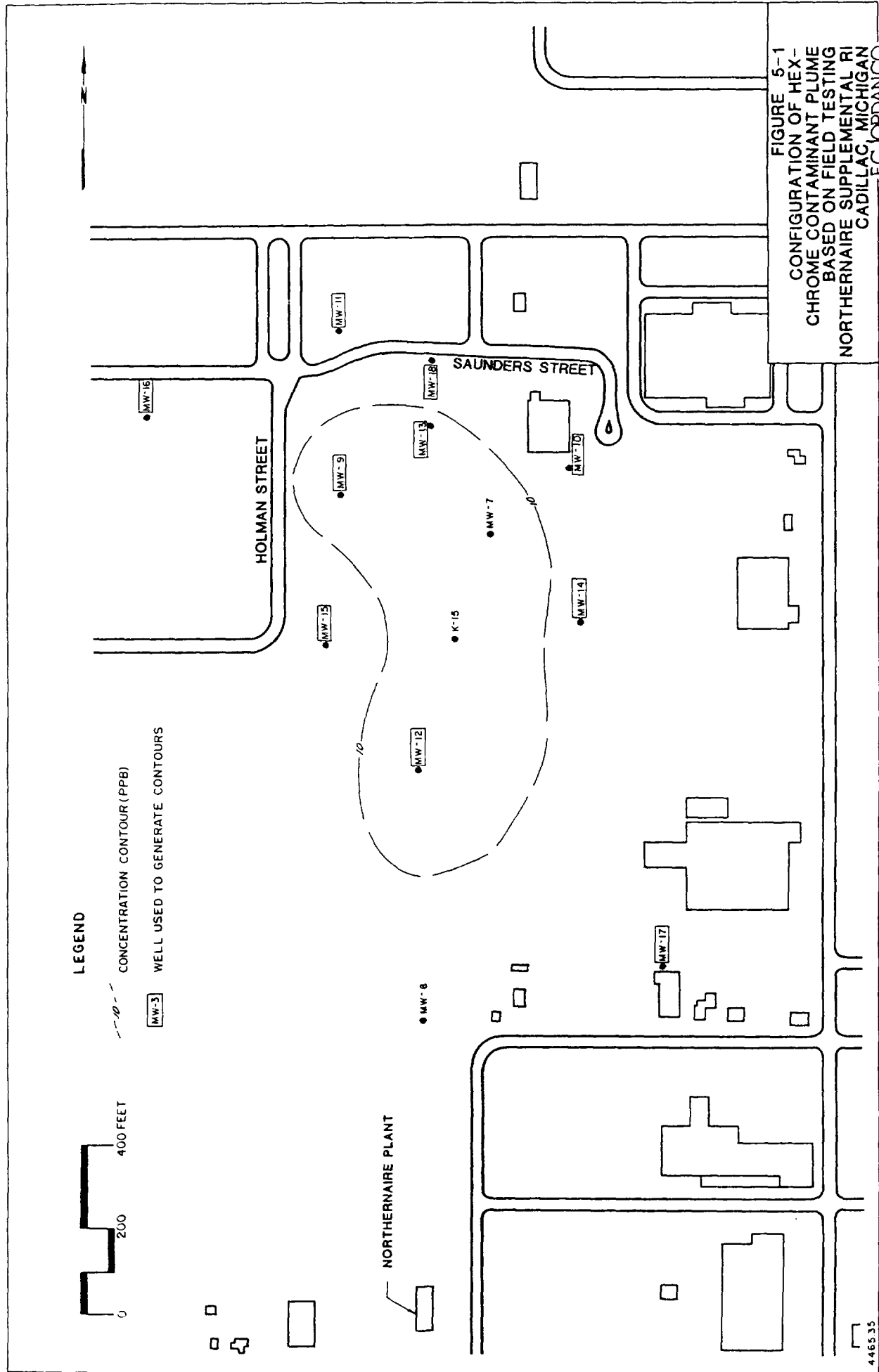
depth of from 162 to 167 feet. Hex-chrome was detected in MW-13, but at a concentration too low to quantify by the colorimetric technique used with the HACH test kit. It is estimated that the concentration encountered was less than 0.05 ppm at a depth of 153 to 158 feet. With the two data points at MW-9 and MW-13, plus data at wells MW-16, MW-11, and MW-18 located farther down-gradient and showing no hex-chrome contamination, the northern, leading edge of the plume has been defined.

Monitoring wells MW-10, MW-14, and MW-17 were placed upgradient of the known contamination and no hex-chrome was detected. Monitoring well MW-15, located west of K-15, also showed no hex-chrome contamination.

Figure 5-1 shows an interpretation of the contaminant plume boundaries as defined by the screened auger sampling of the new monitoring wells. It appears that the plume lies beneath the undeveloped portions of the industrial park. The leading edge likely has not crossed Holman Street to the west or Saunders Street to the north. In the south, the trailing edge is north of Eighth Street and is probably west of the Four Star building. (Due to legal issues, a deep well for assessing hex-chrome contamination could not be placed on Four Star property.)

CLP Sampling:

After allowing the aquifers to stabilize after the drilling program, groundwater sampling events were conducted in February and May, 1986



to obtain samples from the newly installed wells plus several existing wells installed for the previous Northernair RI, Kysor and Four Star studies. Selected samples were submitted to the CLP for analysis of metals and volatile organic compounds. Table 5-1 also summarizes the wells in which hex-chrome was detected and the concentrations reported for the February and May, 1986 sampling. Tables 5-2 and 5-3 summarize the concentrations detected for metals in February and May, respectively. Tables 5-4 and 5-5 summarize the concentrations of volatile organic compounds for February (CLP) and May (Jordan Lab), respectively.

Horizontal Extent of Chromium: Figures 5-2 and 5-3 present, in plan view, interpretations of the distribution of total chromium for February, 1986 in the upper and intermediate aquifers, respectively. The February data were used because more wells were involved in this sampling event than for the May event. These two figures were constructed on the basis that any detection of chromium in monitoring wells at concentrations in the same range as associated field blanks are considered background detections and are thus not considered part of the contaminant plume. Also, the groundwater samples collected in February, 1986 were analyzed for both total chromium and hex-chrome. By comparing the results for total and hex-chrome in Table 5-2, it is apparent that the concentrations do not vary significantly and thus it can be deduced that the total chromium value is almost entirely attributable to the presence of hex-chrome. One exception to this is the analysis for MW-13 where total chromium was detected at a concentration of 39 parts per billion (ppb) and hex-chrome was not detected above the minimum detection limit of 10 ppb. However, because hex-chrome was detected using the Hach test kit during installation of moni-

TABLE 5-2

METALS
FEBRUARY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

Metal	MW-1D	MW-2	MW-3	MW-4	MW-5	MW-6	MW-7	MW-8	MW-9	MW-10	MW-11	MW-13
Aluminum	(19) (29)E	(91) (28)E	-- (34)E	(37) (29)E	(27) (42)E	-- (27)E	-- (27)E	-- (26)E	(55) (35)E	(64) (20)E	(27) (24)E	(119) (29)E
Barium	(0.3)	--	--	(0.3)	(0.3)	(0.2)	(0.2)	(0.2)	(0.4)	(0.3)	(0.3)	(0.3)
Beryllium	--	--	--	--	52.9	--	--	--	--	--	--	--
Cadmium	72,470 (10)	38,610 (8)	88,990 (10)	93,340 11	75,330 1,545	82,500 13	62,830 489	48,050 (8)	60,740 97	59,860 13	49,360 12	63,170 39
Chromium	--	--	--	--	41	--	--	--	--	--	--	--
Copper	494	(31)	(14)	(12)	(31)	(13)	(38)	(18)	(35)	(54)	(22)	(81)
Iron	--	--	--	--	--	--	9.7s	--	--	--	--	--
Lead	14,680	8,602	17,620	20,900	16,160	17,760	13,570	11,590	13,910	13,580	11,450	14,550
Magnesium	159	(1)	359	(3)	(1)	(2)	(3)	24	(7)	(12)	15	(7)
Manganese	--	--	--	--	(21)	--	--	--	--	--	--	--
Nickel	5,516 (3.3)	(1,937)	(2,732)	(1,132)	(3,885)	(875)	(685)	--	(2,039)	(2,173)	(1,726)	(3,090)
Potassium	--	--	(3.9)	(3.7)	(2.9)	(3.2)	(2.1)	--	(6.8)	(7.9)	(5.9)	(6.1)
Silver	72,300 (6.5)	(2,871)E (4.3)	44,350 (6.9)	47,810E (7.4)	151,500 (6.9)	27,760 (6.1)	45,090 (4.9)	7,395 (4.4)	59,100E (9.3)	21,920E (10)	13,240E (8.2)	20,360E (8.8)
Sodium	24	21	(5)	(18)	25	(6)	--	(4)	(6)	(5)	(7)	(12)
Vanadium	<10	ND	<10	ND	1,600	<10	500	<10	50	<10	<10	<10
Zinc	--	--	--	--	--	--	--	--	--	--	--	--
Hex-cr	--	--	--	--	--	--	--	--	--	--	--	--

All measurements in PPB

() = Greater or equal to instrument detection limit but less than contract required detection limit.

s = determined by method of standard addition.

E = estimated or not reported due to the presence of interference.

ND = No Data

-- = Below minimum detection limit.

2.86.88T
0005.0.0

TABLE 5-2
(continued)

METALS
FEBRUARY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

Metal	MW-14	MW-15	MW-16	MW-17	MW-18	K-5S	K-5D	K-6S	K-8	K-15	DNR-1	MW-100*
Aluminum	(82)	--	--	(128)	--	(18)	--	--	(50)	--	(18)	--
Barium	(24)E	(7)E	(18)E	(26)E	(26)E	(10)	(26)	(131)E	(31)	(23)	(30)	--
Beryllium	(0.3)	--	--	(0.3)	--	--	--	(0.4)	--	--	--	(0.2)
Cadmium	--	--	--	--	--	--	--	--	--	--	--	--
Calcium	62,360	45,160	51,780	58,020	70,540	34,790	68,840	246,600	60,780	48,680	69,620	(36)
Chromium	12	(5)	(5)	10	(7)	(5)	--	14	(6)	1,974	(6)	(7)
Copper	--	--	--	--	--	--	--	--	--	--	(4)	--
Iron	(21)	(4)	--	(53)	--	--	--	(42)	(70)	(4)	--	(21)
Lead	--	--	--	--	--	--	--	--	--	--	--	--
Magnesium	13,710	9,766	13,040	12,260	16,690	11,680	14,330	44,660	14,380	11,760	15,720	--
Manganese	(11)	(6)	(8)	(12)	(6)	56	(4)	(3)	18	(2)	(1)	(1)
Nickel	--	--	--	--	--	--	--	--	--	--	--	--
Potassium	(2,834)	--	--	(1,162)	(2,485)	(4,173)	(3,397)	12,500	1,535	3,856	(3,211)	--
Silver	(5.5)	--	--	(3.0)	--	--	--	(6.6)	--	--	--	--
Sodium	51,370E	9,401E	(3,831)E	34,460E	7,846E	14,170	20,600	306,300E	17,170	11,300	45,800	(86)
Vanadium	(8.3)	--	--	(6.4)	(3.1)	--	--	(11.4)	(1.8)	(1.9)	(1.8)	(3.3)
Zinc	26	(4)	(5)	(8)	(15)	25	399	542	(15)	62	381	(7)
Hex-cr	<10	<10	<10	<10	<10	<10	<10	<10	<10	2,200	ND	<10

All measurements in PPB

() = Greater or equal to instrument detection limit but less than contract required detection limit.

s = determined by method of standard addition.

E = estimated or not reported due to the presence of interference.

ND = No Data

* Field blank for MW-3, MW-5, MW-6, and MW-8.

-- = Below minimum detection limit.

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0006.0.0

TABLE 5-2
(continued)

METALS									
FEBRUARY 1986 CLP DATA									
SUPPLEMENTAL NORTHERNAIRE RI									
Metal	MW-101*	MW-103*	MW-104*	MW-105*	MW-106*	MW-107*	FS-15-10	FS-18-10	FS-18-20
Aluminum	(37)	(91)	--	--	--	--	(91)	--	--
Barium	(26)E	--	(29)E	--	--	--	(7)E	(6)E	(9)
Beryllium	(0.2)	(.02)	--	--	(0.4)	--	(0.4)	--	--
Cadmium	--	--	--	--	--	--	--	--	--
Calcium	48,250	(87)	60,580	--	(28)	(21)	41,490	19,590	25,420
Chromium	(8)	(6)	31	(3)	11	(4)	13	(4)	(6)
Copper	--	97	--	--	--	--	(5)	--	--
Iron	(16)	(34)	(5)	--	(15)	(20)	(21)	(42)	--
Lead	--	--	--	--	--	--	--	--	--
Magnesium	11,630	(60)	14,460	--	--	--	8,908	(4,872)	7,353
Manganese	24	(1)	(5)	(0)	(1)	(0)	(2)	(1)	(1)
Nickel	--	--	--	--	--	--	--	--	--
Potassium	--	--	(1,589)	--	(1,285)	(896)	(1,988)	(1,672)	5,253
Silver	--	--	--	--	(6.4)	--	(7.1)	--	--
Sodium	7,421	(137)	20,400E	(171)E	(85)E	(125)	5,620E	(783)E	(749)
Vanadium	(5.1)	(3.1)	--	--	(7.8)	--	(9.3)	--	(1.7)
Zinc	(5)	34	(11)	(10)	(8)	(18)	532	302	177
Hex-cr	<10	<10	<10	<10	<10	<10	ND	ND	ND

All measurements in PPB

() = Greater or equal to instrument detection limit but less than contract required detection limit.

s = determined by method of standard addition.

E = estimated or not reported due to the presence of interference.

ND = No Data

-- = Below minimum detection limit.

* MW-101: Duplicate of MW-8

MW-103: Field blank for MW-11, MW-7, MW-9, MW-10, MW-11, and MW-15.

MW-104: Duplicate for MW-13

MW-105: Field blank for MW-2, MW-4, MW-13, MW-14, MW-16, MW-17, MW-18, K-65.

MW-106: Field blank for FS-15-10, FS-18-10

MW-107: Field blank for K-5S, K-5D, K-8, K-15, DNR-1, FS-18-20

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TABLE 5-3

METALS
MAY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

Metal	MW-1D	MW-2	MW-3	MW-4	MW-5	MW-6	MW-7	MW-8	MW-9	MW-10	MW-10	MW-11
Aluminum	205	(103)	(86)	4,130	359	(116)	(155)	(143)	(166)	(133)	(128)	(108)
Barium	--	--	--	(61)	--	--	--	--	--	--	--	--
Beryllium	--	--	--	--	--	--	--	--	--	--	--	--
Cadmium	--	--	--	--	18	--	--	--	--	--	--	--
Calcium	20,200	33,700	84,000	102,500	77,700	77,700	57,800	44,800	59,700	69,400	62,100	51,900
Chromium	--	--	--	--	431	--	286	--	80	--	--	--
Copper	--	25	--	--	27	--	(13)	--	--	--	--	--
Iron	211	(83)	(75)	4,500	441	193	(53)	(90)	(45)	406	(11)	(58)
Lead	--	--	--	74	--	--	--	(4)	--	--	--	5
Magnesium	9,100	7,160	16,700	23,500	16,700	15,800	11,800	10,300	13,000	13,500	13,400	11,300
Manganese	(10)	--	379	87	16	--	--	(15)	--	147	--	--
Mercury	--	--	--	--	--	--	--	--	--	--	--	0.2
Nickel	--	--	--	--	--	--	--	--	--	--	--	--
Potassium	(2,110)	(1,840)	(2,500)	(2,620)	(3,840)	(1,450)	(1,110)	(747)	(1,170)	5,080	(837)	(627)
Silver	--	--	--	--	--	--	--	--	--	--	--	--
Sodium	47,200	(3,160)	37,500	57,500	79,600	23,100	41,700	8,380	56,100	38,900	19,100	13,500
Vanadium	--	--	--	--	--	--	--	--	--	--	--	--
Zinc	60	(16)	30	32	140	23	--	40	24	39	(18)	(13)
Cyanide	--	--	--	--	11	--	--	--	--	--	--	--
Hex-cr	--	--	--	--	539	--	345	--	91	--	--	--

All measurements in PPB

() = result is greater than or equal to the instrument detection limit but less than the contract-required detection limit.

-- = Below minimum detection limit.

2.86.88T
0011.0.0

TABLE 5-3
(continued)

METALS
MAY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

Metal	MW-13	MW-14	MW-15	MW-16	MW-17	MW-18	MW-DP*	MW-BK*
Aluminum	(154)	(178)	(186)	(129)	(138)	(156)	(129)	(118)
Barium	--	--	--	--	--	--	--	--
Beryllium	--	--	--	--	--	--	--	--
Cadmium	--	--	--	--	--	--	--	--
Calcium	62,400	62,100	52,500	53,900	53,600	64,700	58,000	19,600
Chromium	32	--	--	--	--	--	282	--
Copper	(11)	--	--	--	--	--	--	(11)
Iron	(53)	109	(58)	(37)	133	(90)	(40)	(59)
Lead	12	--	--	--	--	--	--	--
Magnesium	13,700	12,900	10,400	12,200	10,900	13,800	11,800	8,830
Manganese	147	--	--	--	--	--	--	(10)
Mercury	0.2	--	--	--	--	--	--	--
Nickel	--	--	--	--	--	--	--	--
Potassium	(1,600)	(1,610)	(566)	(508)	(1,430)	(1,690)	(1,170)	(2,710)
Silver	--	--	(6)	--	--	--	--	--
Sodium	19,300	40,800	6,470	5,040	33,00	8,760	41,900	48,400
Vanadium	--	--	--	--	--	--	--	--
Zinc	--	(14)	55	565	21	28	(16)	34
Cyanide	--	--	--	--	--	--	--	--
Hex-cr	40	--	--	--	--	--	35	--

All measurements in PPB

() = result is greater than or equal to the instrument detection limit but less than the contract-required detection limit.
-- = Below minimum detection limit.

* MW-DP = Duplicate of MW-7
MW-BK = Field Blank

2.86.88T
0012.0.0

TABLE 5-4

TOTAL VOLATILE ORGANICS
FEBRUARY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

COMPOUND	MDL (1)	MW-10	MW-2	MW-3*	MW-4	MW-5	MW-6*	MW-7	MW-8	MW-9	MW-10
Chloromethane	10	--	--	--	--	--	--	--	--	--	--
Bromomethane	10	--	--	--	--	--	--	--	--	--	--
Vinyl Chloride	10	--	--	--	--	--	--	--	--	--	--
Chloroethane	10	--	--	--	--	--	--	--	--	--	--
Methylene Chloride			2JB	1,200B	1JB	1J	130JB	1J	3JB	1JB	2JB
1,1-Dichloroethane	5	--	--	890	--	--	--	--	10	--	17
1,1-Dichloroethane	5	--	--	230J	--	--	1,300	--	2J	--	--
Trans-1,2-Dichloroethane	5	380	--	--	--	--	--	--	--	--	4J
Chloroform	5	--	--	--	--	--	--	--	--	--	3J
1,2-Dichloroethane	5	--	--	--	--	--	2,000	--	--	--	--
2-Butanone		11B	10B	1,200B	13B	12B	1,200B	12B	12B	12B	13B
1,1,1-Trichloroethane	5	22	2J	13,000	--	--	15,000	2J	140	--	190
Trichloroethane	5	450	5	5,600	1J	27	4,200	13	22	2J	3J
Benzene	5	1J	--	--	--	--	--	--	--	--	--
4-Methyl-2-Pentanone	10	--	--	260J	--	--	--	--	--	--	--
Tetrachloroethane	5	8	--	--	--	4J	--	--	--	--	--
Toluene	5	2J	--	400	--	--	--	--	--	--	--
Ethylbenzene	5	56	--	--	--	--	--	--	--	--	--
Total Xylenes	5	45	--	230J	--	--	--	--	--	--	--

(1) MDL = Minimum detection limit

* = MDL is (listed MDL x 100)

B = analyte found in blank

J = estimated

-- = Below minimum detection limit.

All measurements in PPB

2.86.88T
0008.0.0

TABLE 5-4
(continued)

TOTAL VOLATILE ORGANICS
FEBRUARY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

COMPOUND	MDL (1)	MW-11	MW-13	MW-14	MW-15	MW-16	MW-17	MW-18	FS-15-10	FS-18-10	FS-18-20
Chloromethane	10	--	--	--	--	--	--	--	--	--	--
Bromomethane	10	--	--	--	--	--	--	--	--	--	--
Vinyl Chloride	10	--	--	--	--	--	--	--	--	--	--
Chloroethane	10	--	--	--	--	--	--	--	--	--	--
Methylene Chloride		1JB	1JB	1JB	2JB	2JB	1JB	2JB	1JB	1JB	1JB
1,1-Dichloroethene	5	--	--	--	1J	--	--	--	--	--	--
1,1-Dichloroethane	5	--	--	--	--	--	--	--	--	--	--
Trans-1,2-Dichloroethene	5	--	--	--	--	--	--	--	--	--	--
Chloroform	5	--	--	--	--	--	--	--	--	--	--
1,2-Dichloroethane		--	--	--	--	--	--	--	--	--	--
2-Butanone		13B	14B	12B	10B	11B	14B	11B	12B	13B	11B
1,1,1-Trichloroethane	5	1J	--	--	5	--	1J	--	--	--	--
Trichloroethene	5	420	90	3J	340	--	--	99	--	--	3J
Benzene	5	--	--	--	--	--	--	--	--	--	--
4-Methyl-2-Pentanone	10	--	--	--	--	--	--	--	--	--	--
Tetrachloroethene	5	--	4J	--	--	--	--	2J	--	--	--
Toluene	5	--	--	--	--	--	--	--	--	--	--
Ethylbenzene	5	--	--	--	--	--	--	--	--	--	--
Total Xylenes	5	--	--	--	--	--	--	--	--	--	--

(1) MDL = Minimum detection limit

* = MDL is (listed MDL x 100)

B = analyte found in blank

J = estimated

-- = Below minimum detection limit.

All measurements in PPB

2.86.88T
0009.0.0

TABLE 5-4
(continued)

TOTAL VOLATILE ORGANICS
FEBRUARY 1986 CLP DATA
SUPPLEMENTAL NORTHERNAIRE RI

COMPOUND	MDL (1)	MW-100**	MW-101**	MW-104**	MW-105**	MW-106**	DNR-1
Chloromethane	10	--	--	--	--	--	--
Bromomethane	10	--	--	--	--	--	--
Vinyl Chloride	10	--	--	--	--	--	--
Chloroethane	10	--	--	--	--	--	--
Methylene Chloride	5	1JB	3JB	2JB	2JB	1JB	1JB
1,1-Dichloroethene	5	--	10	--	--	--	--
1,1-Dichloroethane	5	--	1J	--	--	--	--
Trans-1,2-Dichloroethene	5	--	--	--	--	--	--
Chloroform	5	--	--	--	--	--	--
1,2-Dichloroethane	5	--	--	--	--	--	--
2-Butanone	5	12B	11B	13B	10B	13B	--
1,1,1-Trichloroethane	5	--	130	--	--	--	11B
Trichloroethene	5	--	20	86	--	--	1J
Benzene	5	--	--	--	--	--	--
4-Methyl-2-Pentanone	10	--	--	--	--	--	--
Tetrachloroethene	5	--	--	4J	--	--	--
Toluene	5	--	--	--	--	--	--
Ethylbenzene	5	--	--	--	--	--	--
Total Xylenes	5	--	--	--	--	--	--

(1) MDL = Minimum detection limit

* = MDL is (listed MDL x 100)

B = analyte found in blank

J = estimated

-- = Below minimum detection limit.

All measurements in PPB

** MW-100: Field blank for MW-3, MW-5, MW-6, and MW-8

MW-101: Duplicate of MW-8

MW-104: Duplicate of MW-13

MW-105: Field blank for MW-2, MW-4, MW-13, MW-16, MW-17, and MW-18

MW-106: Field blank for FS-15-10 and FS-18-10

2.86.88T

0010.0.0

TABLE 5-5

SELECTED VOLATILE ORGANICS
MAY 1986, JORDAN LAB DATA
SUPPLEMENTAL NORTHERNAIRE RI

COMPOUND	MW/D	MW-2	MW-3	MW-4	MW-5	MW-6	MW-7	MW-8	MW-10	MW-11	MW-17	MW-18
Carbon Tetrachloride	--	--	410	--	--	330	--	19	36	--	--	--
Methylene Chloride	--	--	1,110	--	--	--	--	--	--	--	--	--
Tetrachloroethylene	--	--	11	--	--	30	--	--	--	--	--	--
1,1-Dichloroethane	--	--	150	--	--	16	--	--	--	--	--	--
1,1-Dichloroethene	--	--	620	--	--	550	--	--	--	--	--	--
1,1,1-Trichloroethane	--	19	3,800	12	19	3,000	--	120	220	--	--	--
1,1,2-Trichloroethane	--	--	71	--	--	26	--	10	--	--	--	--
1,2-Dichloroethane	--	--	260	--	--	700	--	--	--	--	--	--
Trans-1,2-Dichloroethene	170	--	--	--	--	--	--	--	--	--	--	--
Trichloroethene	430	42	2,600	16	112	1,600	24	37	--	610	--	110

All Measurements in PPB

-- = Below minimum detection limit

NR = Not Reported

2.86.88T
0013.0.0

TABLE 5-5
(continued)

SELECTED VOLATILE ORGANICS
MAY 1986, JORDAN LAB DATA
SUPPLEMENTAL NORTHERNAIRE RI

COMPOUND	FS-12-10	FS-13-10	FS-13-20	FS-14-10	FS-14-20	FS-15-10	FS-19-10	FS-19-20	FS-20
Carbon Tetrachloride	--	--	--	--	--	--	--	--	--
Methylene Chloride	NR	NR	NR	NR	NR	NR	NR	NR	NR
Tetrachloroethylene	NR	NR	NR	NR	NR	NR	NR	NR	NR
1,1-Dichloroethane	NR	NR	NR	NR	NR	NR	NR	NR	NR
1,1-Dichloroethene	--	--	--	--	--	--	--	--	--
1,1,1-Trichloroethane	--	--	10	--	--	--	--	--	--
1,1,2-Trichloroethane	NR	NR	NR	NR	NR	NR	NR	NR	NR
1,2-Dichloroethane	NR	NR	NR	NR	NR	NR	NR	NR	NR
Trans-1,2-Dichloroethene	NR	NR	NR	NR	NR	NR	NR	NR	NR
Trichloroethene	--	--	7,100	77	--	--	47	77	--

All Results in PPB

-- = Below minimum detection limit

NR = Not Reported

2.86.88T
0014.0.0

TABLE 5-5
(continued)

SELECTED VOLATILE ORGANICS
MAY 1986, JORDAN LAB DATA
SUPPLEMENTAL NORTHERNAIRE RI

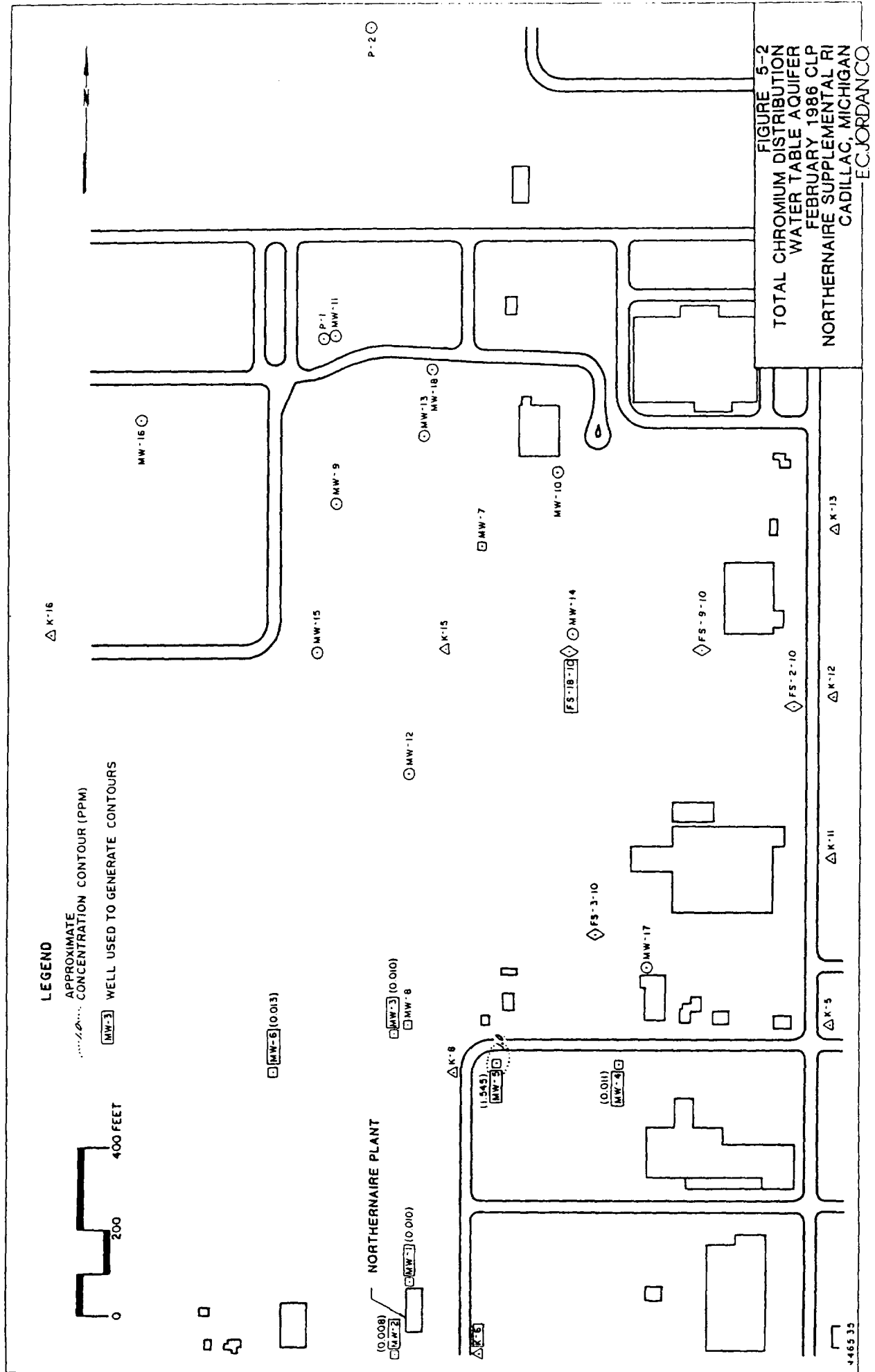
COMPOUND	DNR-1	DNR-5	DNR-7	DNR-9	DNR-10S	DNR-10D	DNR-11	DNR-13	DNR-14S	DNR-14D	DNR-15
Carbon Tetrachloride	--	58	--	--	--	--	--	--	--	--	--
Methylene Chloride	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Tetrachloroethylene	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
1,1-Dichloroethane	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
1,1-Dichloroethene	--	22	--	--	--	--	--	--	--	--	--
1,1,1-Trichloroethane	--	400	--	--	--	--	--	--	--	--	--
1,1,2-Trichloroethane	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
1,2-Dichloroethane	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Trans-1,2-Dichloroethene	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Trichloroethene	--	1,300	--	--	17	25	66	65	62	--	--

All Results in PPB

-- = Below minimum detection limit

NR = Not Reported

2.86.88T
0015.0.0



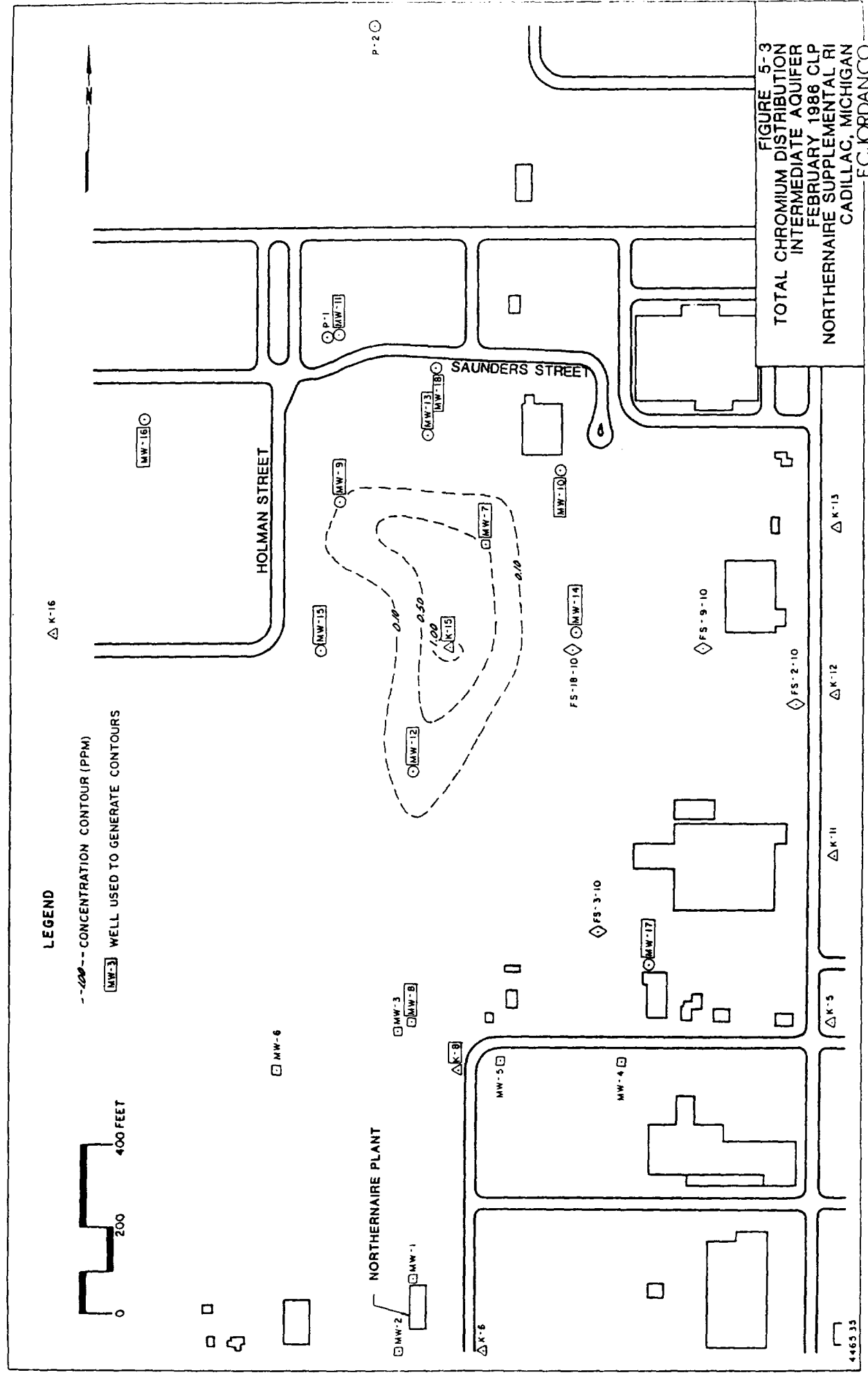


FIGURE 5-3
TOTAL CHROMIUM DISTRIBUTION
INTERMEDIATE AQUIFER
FEBRUARY 1986 CLP
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

toring well MW-13, it is believed that the total chromium value reported by CLP for the February sampling of MW-13 may be more accurate than the hex-chrome value. Thus, for consistency and because, where found, the total chromium concentrations did not vary significantly from the hex-chrome concentrations, the total chromium values of the February sampling event will be used for all the wells when discussing chromium contamination.

In the shallow aquifer, 1.5 parts per million (ppm) of total chromium was detected in monitoring well MW-5 in February, whereas all other shallow wells sampled during this event showed chromium in the same range of that detected in the field blanks. The detection of chromium in MW-5 was later verified using the HACH test kit for hex-chrome during the collection of groundwater levels in April. Cadmium was also detected in this well sample at 0.053 ppm. The May sampling event also detected total chromium and cadmium in MW-5 but in significantly reduced concentrations of 0.43 ppm and 0.018 ppm, respectively. Cadmium is the other major contaminant of concern at Northernair, and MW-5 was the only well to show cadmium contamination. All previous samplings of groundwater in MW-5 did not detect significant chromium or cadmium. However, inasmuch as MW-5 is directly downgradient of the Northernair facility (Figure 4-7) it is expected that, if hex-chrome or cadmium had been released from the Northernair facility, MW-5 would become contaminated. The absence of chromium and cadmium in MW-11, adjacent to the northern side of the building, suggests that the principal source of the contamination has been eliminated both by the closure of the Northernair plant in 1981 and the removal of wastes by EPA and MDNR in 1983. Because MW-5 had the only detectable chromium and cadmium, accurate plume contouring of the areal extent of the contamination is not possible

beyond noting that the chromium and cadmium were not detected in K-8 to the west or MW-4 to the east. The decrease in concentrations of chromium and cadmium from February to May is probably a result of the plume being very limited in areal extent and moving quickly through the upper aquifer. This small plume may eventually be detected in a well downgradient of MW-5 but in much lesser concentrations due to additional groundwater mixing.

The February, 1986 CLP results for chromium in the intermediate aquifer were used to construct Figure 5-3. These results confirm the screened auger sampling data used to determine the extent of hex-chrome contamination in the field (Figure 5-1). It appears that the leading edge of the chromium contamination has not extended beyond the undeveloped area north of the Northernair building and that the trailing edge is situated between MW-8 and MW-12. Also, in both Figures 5-1 and 5-3, the plume is skewed toward the northwest (MW-9) indicating the influence of groundwater flow on plume migration. The May, 1986 analytical data indicate similar concentrations and extent of the chromium plume.

Vertical Distribution of Chromium:

The screened auger sampling conducted during drilling and the subsequent CLP sampling provides data by which the vertical distribution of hex-chrome in the upper and intermediate aquifers can be assessed. The geologic cross-sections of Figures 4-1 and 4-2 show the vertical extent of hex-chrome in both aquifers. In the upper aquifer, hex-chrome was detected only in well MW-5 which has a 10 foot screened interval from 38 to 48 feet below ground surface directly

above the upper clay aquitard. It appears that the hex-chrome has descended to the bottom of the water table aquifer and further descent to the intermediate aquifer may be moderated by the clay aquitard.

The north-south cross-section (Figure 4-1) shows that the hex-chrome contamination has migrated essentially to the bottom of the intermediate aquifer, near the top of the lower aquitard. It appears then, that the clay aquitards influence the vertical migration of the contaminants but, based on the configuration of the plumes shown on Figures 5-1, 5-2, 5-3, the groundwater flow, not the aquitards, dictates migration direction. The descent of the hex-chrome plumes within their respective aquifers is probably a result of downward hydraulic gradients within each unit.

Total Metals: Figures 5-4 and 5-5 show the distribution of total metals for February, 1986 in the upper and intermediate aquifers, respectively. Kysor well K-6 in the upper aquifer had the highest concentration of total metals (611 ppm). The remaining shallow wells in this area ranged anywhere from 52 to 250 ppm. The primary metals detected in all the monitoring wells sampled were calcium, sodium, magnesium and potassium. From Figure 5-4, it appears that the highest concentration of total metals occurs in wells adjacent to roads. Concentrations appear to decrease with increasing distance from the road. This may reflect the salting of the roads which would introduce sodium in winter and possibly calcium in summer into the subsurface.

Unlike the distribution of total metals in the upper aquifer, the distribution of metals in the intermediate aquifer is more uniform (Figure 5-5). The range in concentration is from 65 ppm in MW-15 to 136 ppm in MW-9.

Volatile Organic Compounds: Figures 5-6 and 5-7 represent the distribution of total volatile organic compounds for February, 1986 in the upper and intermediate aquifers, respectively. The highest concentrations of total volatiles in the upper aquifer were detected in monitoring wells MW-3 (23 ppm) and MW-6 (24 ppm). The primary contaminants in these two wells are 1,1,1-trichloroethane and trichloroethene (TCE). All other shallow wells sampled had less than 0.05 ppm total volatiles with the exception of MW-11 which had almost 1.0 ppm. Again, TCE is the primary contaminant in the monitoring wells. The analytical results for May, 1986 showed similar relative concentrations between monitoring wells, but volatile organic concentrations in MW-3 and MW-6 had significantly dropped to 9 and 6 ppm, respectively. Additional monitoring is required to determine if this decrease in concentration is a real trend or a seasonal fluctuation.

The plume encountered within the study area appears to be centered around wells MW-3 and MW-6 located northwest and thus upgradient of the Northernnaire site. From the distribution of volatile contaminants shown in Figure 5-6, it is evident that this contamination is more widespread than the immediate study area of this RI. From historical chemical use records and known chemical operations, it is believed that Northernnaire was not a contributor to the

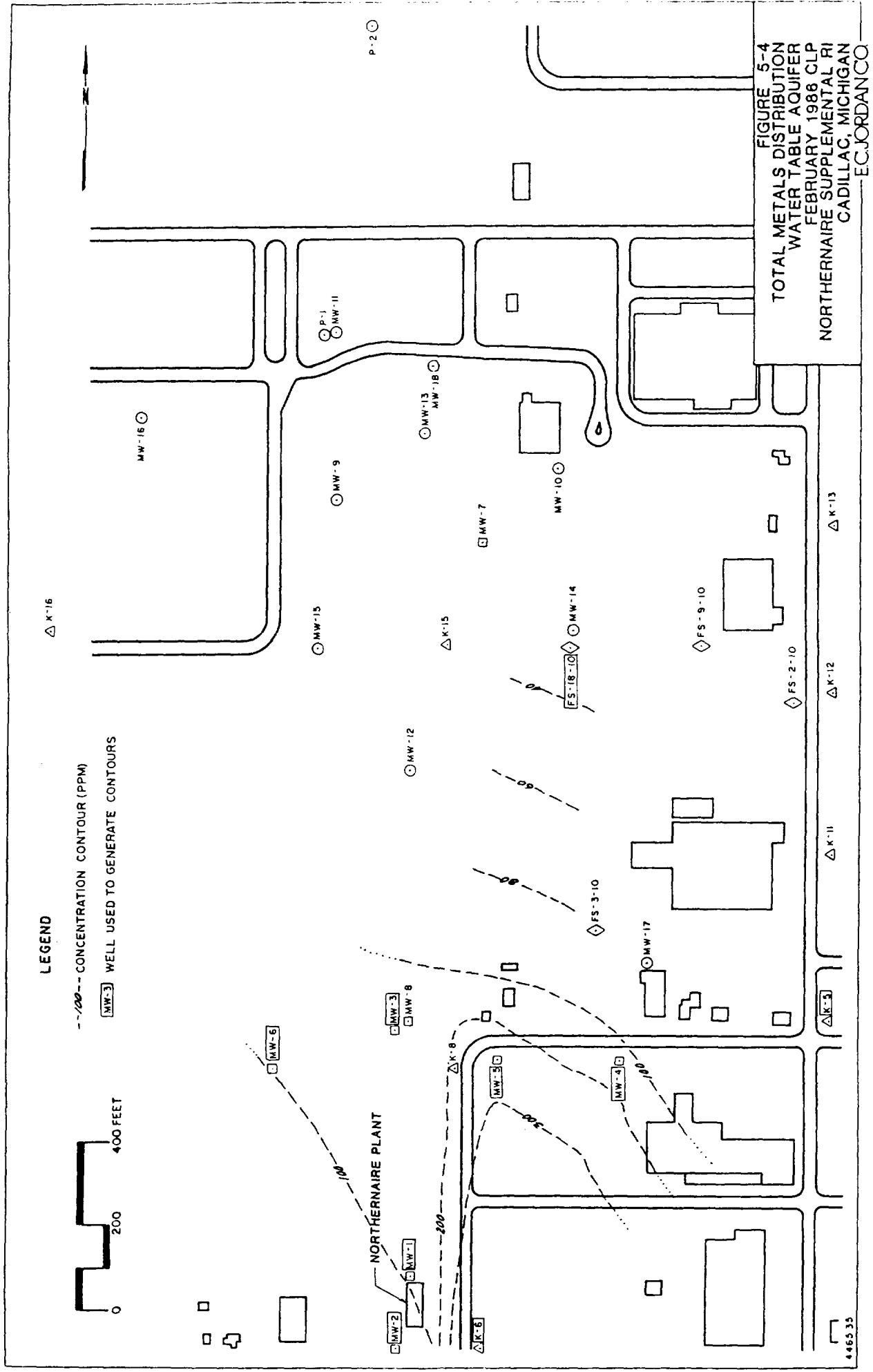


FIGURE 5-4
TOTAL METALS DISTRIBUTION
WATER TABLE AQUIFER
FEBRUARY 1988 CLP
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

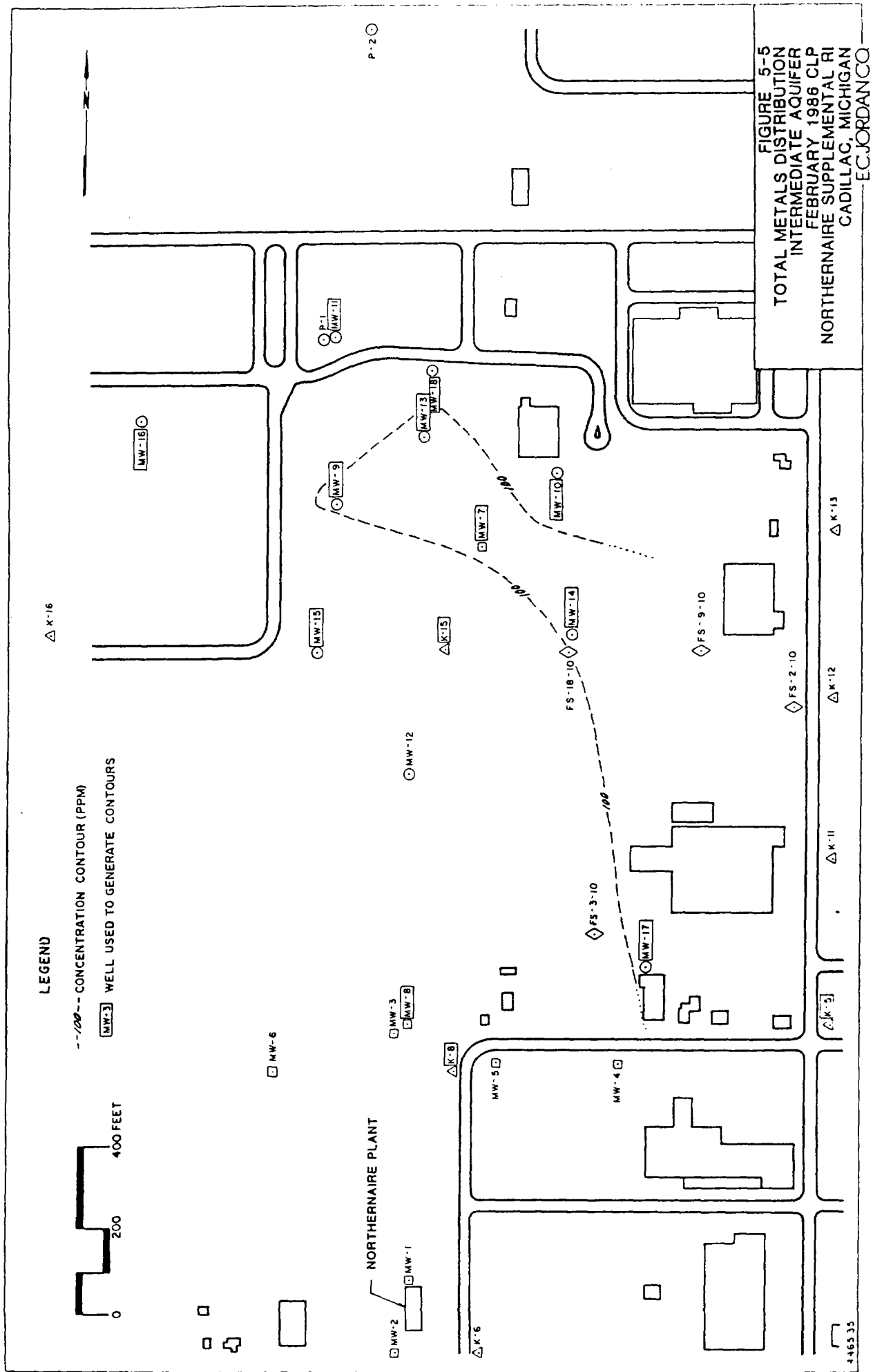


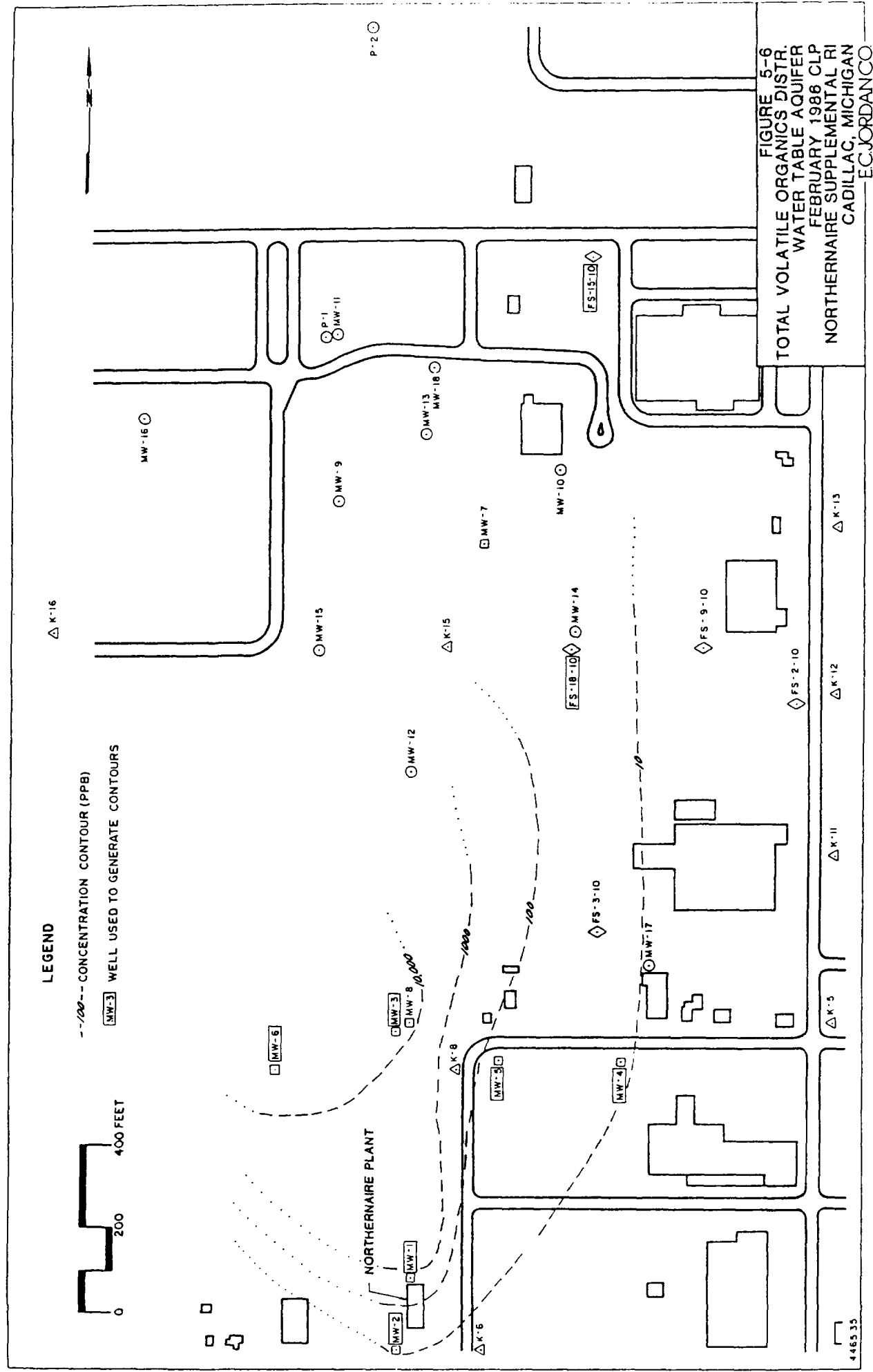
FIGURE 5-5
TOTAL METALS DISTRIBUTION
INTERMEDIATE AQUIFER
FEBRUARY 1986 CLP
NORTERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

volatile organic contamination found in the aquifer. These volatile organics data indicate a larger contamination problem and will be used to supplement the data to be obtained in the Cadillac Area 307 Study to be conducted later in 1986.

No distinct trends in the distribution of total volatiles in the intermediate aquifer are apparent (Figure 5-7). Concentrations of approximately 0.4 ppm were detected in wells MW-11 and MW-15. Wells near the center of the undeveloped area had concentrations generally less than .03 ppm. As in the upper aquifer, trichloroethene is the most prevalent volatile contaminant. The lack of a "classic" plume configuration in the data may reflect multiple sources and/or sporadic release rates. These data for volatile organic contaminants in the intermediate aquifer will also be used to supplement data obtained in the future Cadillac Area Study.

5.2 CONTAMINANT PLUME MIGRATION

Water Table Aquifer: Hex-chrome and cadmium has only recently been detected in MW-5 in the upper aquifer and the migration is apparently being controlled by groundwater flow which is to the northeast. The plume in this aquifer is situated near the top of the clay aquitard which separates the upper and intermediate aquifers. The edges of this shallow plume are not precisely defined but the plume does not seem to be horizontally extensive inasmuch as surrounding wells MW-11, MW-3, MW-4 and MW-17 (shallow screened auger samples) did not indicate the presence of hex-chrome or cadmium. Since hex-chrome or cadmium had not been detected in earlier sampling of MW-5, the present



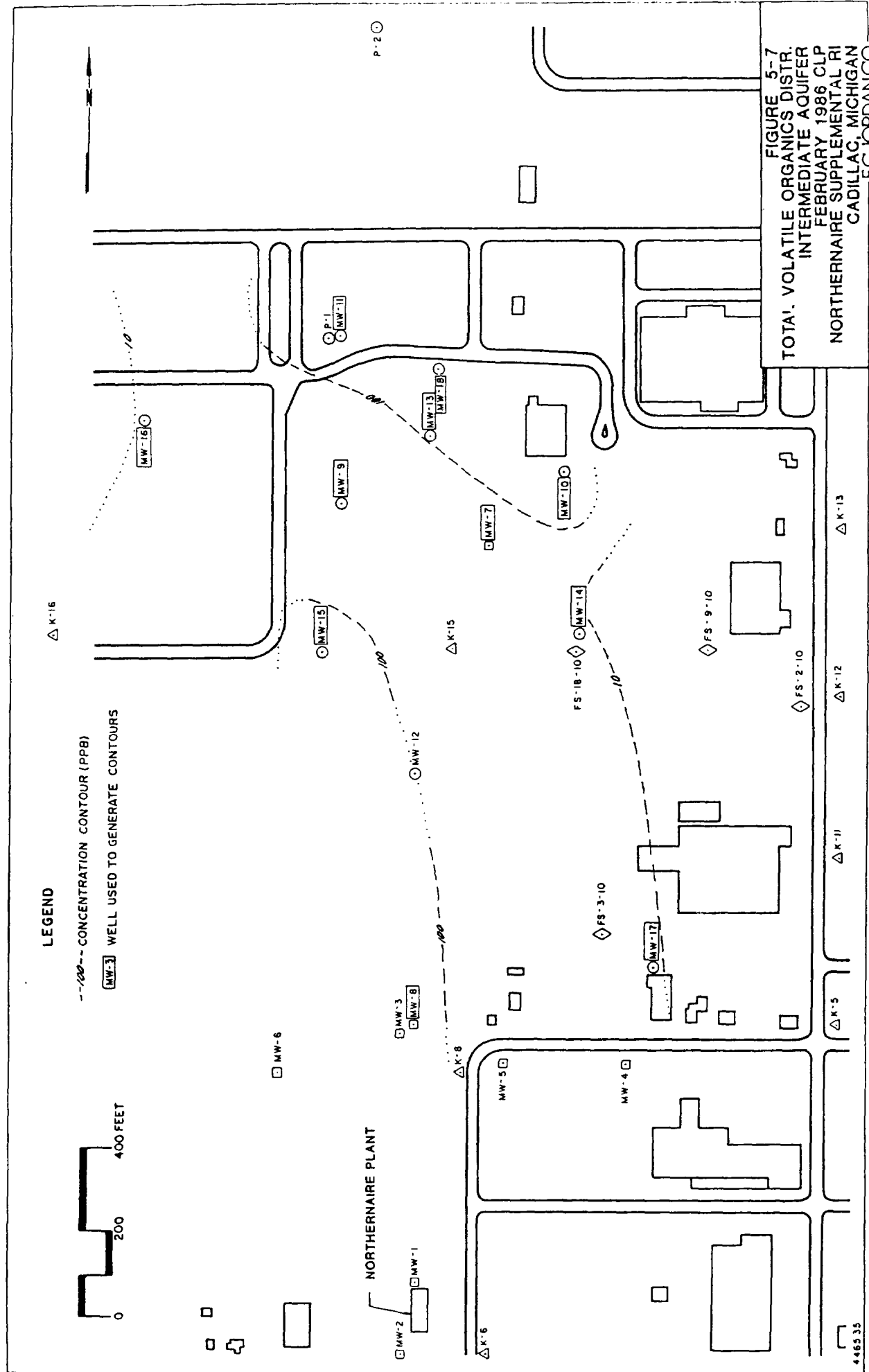


FIGURE 5-7
TOTAL VOLATILE ORGANICS DISTR.
INTERMEDIATE AQUIFER
FEBRUARY 1986 CLP
NORTHERNAIRE SUPPLEMENTAL RI
CADILLAC, MICHIGAN
ECJORDANCO

detection may be the result of a small isolated "slug" of chromium migrating through this area. Whether the release was from late operating or other activities near the Northernnaire building or from leaky sewers used by Northernnaire is not certain. However, with the small apparent extent of the slug and the rapid rate of groundwater movement through the area, it is anticipated that this release will be dissipated to low concentrations as it is moved through and out of the shallow aquifer.

Intermediate Aquifer: In the intermediate aquifer, migration of the contaminant plume is being controlled by the generally northwesterly flow of groundwater. This is evidenced by the shape of the plume which is skewed to the northwest. With the only known source of hexavalent chromium being the Northernnaire plant, the plume has migrated to its present-day position, conceivably beginning as early as 1971, the year Northernnaire began operations. With the closure of the plant in 1981 and the subsequent clean-up of waste materials by the USEPA and MDNR, the source of the hex-chrome in the groundwater has been curtailed for several years. This would explain why the plume in the water table and intermediate aquifers does not extend all the way back to the Northernnaire building. It is expected that the hex-chrome plume in the intermediate aquifer will continue to migrate in a generally northwesterly direction.

The avenue by which the hex-chrome has been introduced into the intermediate aquifer from the shallow, water-table aquifer cannot be positively identified, however, several valid mechanisms exist, any of which may be a factor to varying degrees. These include:

- o the relatively strong downward vertical gradients known to exist between the upper and intermediate aquifers would allow the hex-chrome to migrate through the clay to the intermediate aquifer; or
- o the upper clay unit is very thin in some areas (only three feet at K-8), and may not be continuous across the site and thus would allow the hex-chrome to migrate to the lower aquifer with only small downward vertical gradients present; or
- o since the actual permeability of the clay layer has not been determined and depending on the soil matrix of the unit (which on occasion contained gravel), the aquitard may be significantly more permeable than expected.

With the available data, it is most likely that the hex-chrome contamination is being transported through the upper clay aquitard where it is thinnest in the area of K-8 and K-6 by the substantial vertical gradients between the aquifers. The rate of migration through the clay will be enhanced where the clay becomes more silty or gravelly. The hex-chrome plumes in both aquifers have descended to the bottom of the aquifers just above their respective clay aquitards probably as a result of small downward vertical gradients which exist within the aquifers themselves as well as, in the case of the upper aquifer, the upward sloping (to the NE) surface of the upper aquitard.

5.3 ENVIRONMENTAL FATE

Water Table Aquifer:

The positive identification of hex-chrome in MW-5 in the shallow, water table aquifer was a reversal of conditions found in the earlier RI work. It further substantiates that the hex-chrome contamination was being released from the Northernnaire facility. This detection of hex-chrome in the shallow aquifer may also indicate that hex-chrome was released to the environment from the Northernnaire facility in sporadic, concentrated, releases in addition to more frequent discharges at lower concentrations (e.g., overflowing rinses).

The fate of hex-chrome contamination in the shallow aquifer will be to migrate in a northeastward direction with the groundwater flow, and, as hydrogeologic conditions become favorable, to migrate into the intermediate aquifer. In part, this migration into the intermediate aquifer is believed to be caused by the driving forces of a downward vertical gradient between aquifers across a thin, silty clay aquitard.

Part of the plume in the shallow aquifer may remain in that aquifer and continue travelling in a northeastward direction until the upper aquitard ends. No imminent threat should be posed by the hex-chrome inasmuch as all area residences are supplied by city water. Also, the plume should continue to be diluted with time and the distance travelled. Because the detection of hex-chrome in MW-5 was the first detection of chromium contamination in the shallow aquifer, MW-5 and adjacent wells should be closely monitored to better define

plume movement, particularly the groundwater flow rate in the water table aquifer.

Intermediate Aquifer:

Once the hex-chrome has reached the intermediate aquifer, it travels with the groundwater in a north-northwestward direction. The plume also moves deeper within the aquifer, probably due to small but significant downward vertical gradients. However, downward movement of the plume below the lower clay aquitard can not be assessed because no data are available at these depths. The plume in the intermediate aquifer should continue moving in a generally northwestward direction. This movement of the plume can be monitored by the existing downgradient monitoring wells (MW-11, MW-15, MW-16, and MW-18). The apparent environmental threat this plume could pose is if it were able to migrate through the lower clay aquitard into the lower aquifer where the city water supply wells are screened. The potential for contamination reaching the municipal wells would depend on the radius of influence developed by the municipal well, the seepage rate through the aquitard, the concentration in the plume and the relative fraction of flow to the well that the contaminant plume would present. Note that since flow to a well occurs radially and the plume is thin relative to the screen length, the well would draw in most of its water from clean areas which would dilute the hexavalent chromium present. Whether or not the resultant concentration would exceed drinking water standards (0.05 mg/l) is not known for certain at the present time. However, based on probable leakage rates, continued dilution and dispersion and the probable limited radius of influence of any one municipal well, it is likely that the

contaminant levels detected for the hexavalent chromium would not significantly impact the municipal well field water quality.

6.0 CONCLUSIONS AND RECOMMENDATIONS

6.1 CONCLUSIONS

The consolidation of geological and piezometric data from the current Northernnaire RI and past area studies has permitted an understanding of the hydraulic relationships that exist between the water table (shallow) aquifer and intermediate aquifer and the intervening clay aquitard. Also, the acquisition of groundwater quality data through both screened auger sampling and a separate groundwater sampling event for CLP analysis has provided data to assess the extent of chromium contamination. The following conclusions have been reached:

1. Recent groundwater level data confirm previous interpretations that groundwater flow in the shallow aquifer is primarily in a northeasterly direction.
2. Piezometric data from monitoring wells screened in the intermediate aquifer indicate a generally northwesterly direction of flow in that aquifer.
3. Groundwater sampling in February, 1986, provided for the first time during the RI, positive identification of hex-chrome and cadmium in the shallow aquifer (in MW-5). Since MW-5 is directly downgradient from Northernnaire with no intervening sources, the contribution by the Northernnaire facility

to the hexavalent chromium contamination in the aquifer is now more directly confirmed.

4. The hex-chrome and cadmium contamination in MW-5 in February, 1986 is apparently part of a small plume migrating in a northeastward direction. MW-5 was tested previously with no chromium or cadmium detected, and the concentrations of these two metals were significantly reduced in May, 1986. Indications are that sporadic releases may have occurred in addition to more constant releases over time such as discharges of contaminated rinse waters into leaking sewers.
5. Geologic data indicate that the upper clay aquitard is thin immediately north of the Northernnaire building. In some areas of the upper aquitard, the clay also becomes silty and gravelly which would significantly increase the permeability of the geologic unit.
6. Significant downward vertical gradients exist between the shallow and intermediate aquifers where the aquitard is thin north of the Northernnaire building (near wells K-6 and K-8). The combination of these gradients and a thin aquitard diminish the effectiveness of the aquitard to restrict groundwater flow and provide a mechanism by which hex-chrome could migrate from the shallow to the intermediate aquifer.
7. Smaller, yet significant downward vertical gradients exist within the upper and intermediate aquifers. This may be the reason the hex-chrome contamination is found deep in the intermediate aquifer just above the

confining layer.

8. The fate of the hex-chrome plume in the shallow aquifer will be to continue in a northeasterly direction, eventually to become diluted and to merge with the flow in the intermediate aquifer. Moreover, due to the relatively large vertical gradients, much of the contamination will reach the intermediate aquifer and begin to migrate in a generally northwesterly direction as it moves toward the lower clay aquitard.
9. The leading edge of the plume of hexavalent chromium contamination in the intermediate aquifer presently does not extend beyond Holman Street to the west and Saunders Street to the north in the undeveloped area of the Industrial Park north of the Northernaire facility. The trailing edge lies somewhere between MW-8 and MW-12.
10. The fate of the hex-chrome plume in the intermediate aquifer is to continue to migrate in a generally northwesterly direction at a depth of from approximately 150 to 180 feet. This plume should also continue to be diluted as it moves through the aquifer.
11. An accurate assessment of the potential impact of the hex-chrome plume on the municipal wells, which may occur through seepage of the hex-chrome plume through the lower aquitard into the deeper aquifer, is not possible with available data. However, enough information is known to be able to deduce that the potential for such migration does exist. Interpretations of the hydrogeologic data from the study combined with applications of

principals of fluid dynamics and mass balance leads to this conclusion. These factors and principals include: 1) a relatively uniform horizontal gradient in the intermediate aquifer; 2) a relatively even thickness of the aquifer across the site; and 3) the necessity that a mass balance be maintained through flow, recharge and leakage on some unit volume of the aquifer. For this balance to be maintained with a uniform horizontal flow implies that the volume of leakage through the bottom of the aquifer must approximate the gain through recharge and infiltration of precipitation.

12. The lower aquitard is believed to provide some protection to the lower aquifer by retarding the rate of seepage of the hex-chrome plume through it. There will be an undetermined time before the chromium can pass completely through this protective layer. Also, the clay will determine the seepage rate at which the contamination reaches the deep aquifer. Here, if the radius of influence of the municipal well field is great enough or if groundwater flow in the lowest aquifer were to be easterly, the plume may be drawn toward the municipal wells. However, even if this occurs, the dilution available in the aquifer may prevent exceeding water quality standards for hexavalent chromium (0.05 mg/l). However, the potential for impact exists, and the quality and thickness of the aquitard in the vicinity of the plume is not known.

13. Volatile organic contamination is widespread in both the shallow and intermediate aquifers. The investigation and assessment of the contamination by volatile organics (solvents) will be addressed in the Cadillac area groundwater study scheduled to begin in the summer of 1986.

6.2 RECOMMENDATIONS

1. There should be continued, periodic groundwater monitoring particularly of MW-5 and adjacent wells to monitor the latest indications of contamination in the shallow aquifer and to monitor its rate of movement and fate.
2. Continue to monitor the plume in the intermediate aquifer utilizing selected wells to indicate the possible dilution of the plume, as well as its possible migration into areas which are presently clean. These wells currently downgradient of the plume will serve as an effective monitoring network for continued migration.
3. The remedial action selected for the hexavalent chromium plume should be determined in conjunction with the assessment and remediation of the wider problem of volatile organic contamination.

APPENDIX A
BORING LOGS
AND
MONITORING WELL DIAGRAMS

2.86.88
0083.0.0

KEY TO ABBREVIATIONS:

PVC	=	Polyvinyl Chloride
CUT	=	Cuttings from Auger Flights
SAS	=	Screened Auger Samples
Med	=	Medium
Tr.	=	Trace
SU	=	Standard Units
Gals.	=	Gallons
NA	=	Not Applicable/Not Available
CLP	=	Contract Laboratory Program

See Figure 3-1 for monitoring well locations.

On logs MW-11 through MW-18, specific conductivity is reported as 1/10 uhos/cm (i.e., reported value = 266; actual value = 2,660).

NORTHERNAIRE GROUNDWATER STUDY		JOB NO. 4465-35 DRILL RIG/METHOD Gus Pech/HSA OPERATOR P. Buell/Keck INSPECTOR T. Taylor/ECJ		BORING I.D. MW-9 LOCATION See Figure DATE STARTED 11-22-85 DATE COMPLETED 11-25-85	
FIELD BORING LOG COMPANY E.C. JORDAN					

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ ID.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS					
FROM	TO					HEAD SPACE	HEX CR.	pH	SPEC. COND.	TEMP.	OTHER
0	20	cut			Med. to dark brown, med. to fine grained, med. dense, well sorted, quartz sand.						
20	30	cut			Med. to light brown, fine grained, med. dense, sand, tr. fine to coarse gravel.						
30	31	cut			Small seam of gravel.						
35	36	cut		///	Small clay lens.						
37	42	SAS	1		Developed 50 gals. until clear. Purged 15 gals.	0	0	7.3393	8.1		
47	52	SAS	2 *		Developed 55 gals. until clear. Purged 15 gals.	0	0	7.4650	7.0		
58	61	cut		///	Gray, firm, clay, little sand and silt.						
62	67	SAS	3 *		Developed 28 gals. until clear. Purged 23 gals.	0.1	0	7.2581	7.5		
72	77	SAS	4		Developed 55 gals. until clear. Purged 30 gals.	0	0	7.3577	8.3		
	87	cut			Med. brown, med. grained sand, tr. to little gravel.						
82	87	SAS	5		Developed 55 gals. until clear. Purged 37 gals.	0	0	7.4503	9.1		
92	97	SAS	6		Developed 55 gals. until clear Purged 44 gals.	0	0	7.7428	8.4		
102	107	SAS	7 *		Developed 55 gals. until clear. Purged 54 gals.		0	7.3356	7.6		

GROUNDWATER ENCOUNTERED AT 35.8 FT.

ppm	ppmSU	**	°C
UNITS			

NOTES:

* Sample sent to CLP.

Density of sands and consistency of clays are estimates only.

** Specific Conductivity = uhos/cm.

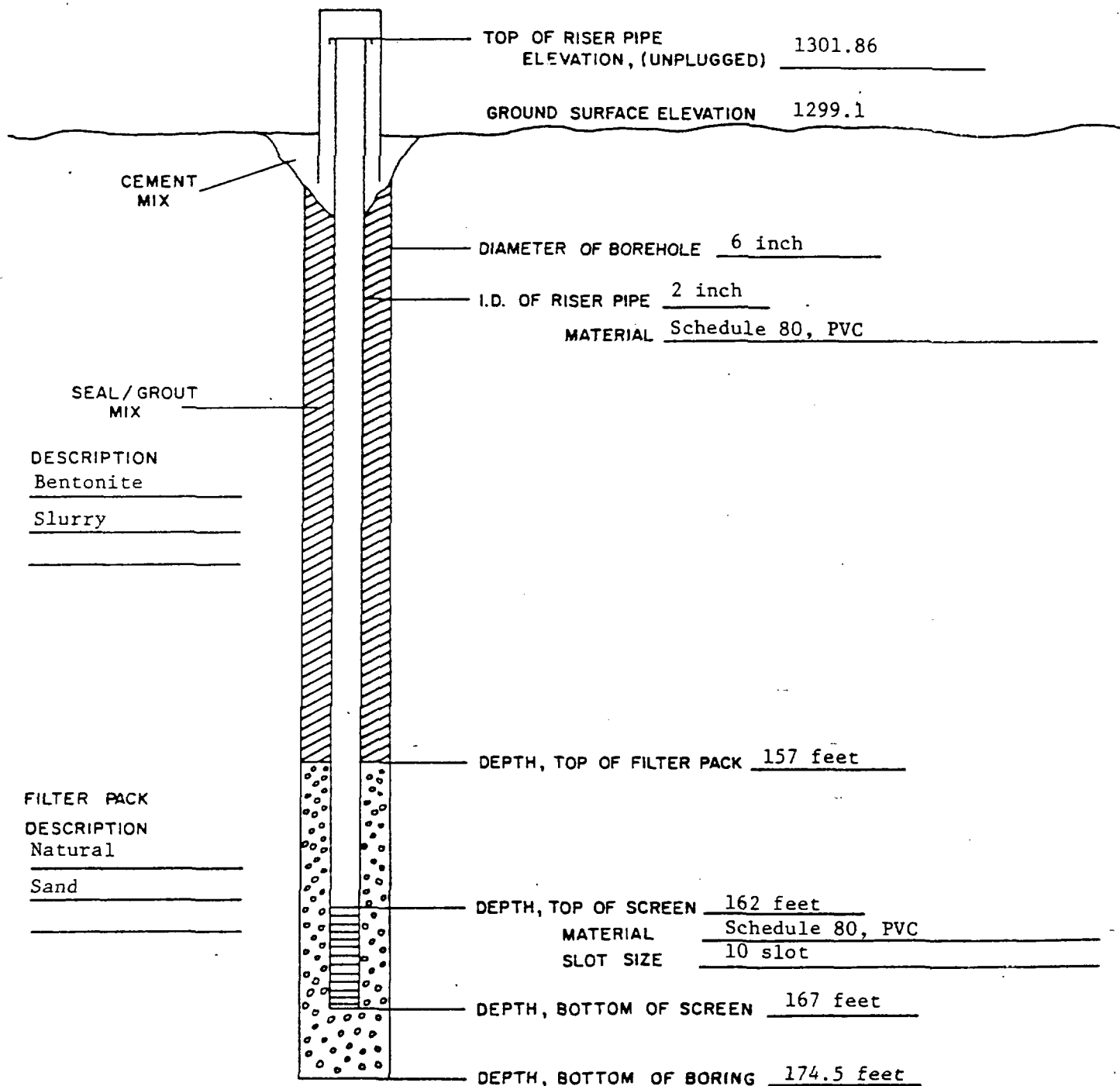
TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

TOTAL DEPTH 174.5

PAGE 1 OF 2

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-9



AQUITARD INFORMATION

DEPTH TO UPPER 58 feet THICKNESS 3 feet

DEPTH TO LOWER 173 feet

NORTHERNAIRE GROUNDWATER STUDY				JOB NO. <u>4465-35-</u>		BORING I.D. <u>MW-10-</u>			
FIELD BORING LOG				DRILL RIG/METHOD <u>Gus Peck/HSA</u>		LOCATION <u>See Figure</u>			
COMPANY <u>E.C. JORDAN</u>				OPERATOR <u>M. Allen/Keck</u>		DATE STARTED <u>11-26-85</u>			
				INSPECTOR <u>T. Taylor/ECJ</u>		DATE COMPLETED <u>12-4-85</u>			

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ ID.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS						
FROM	TO					HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER	
0	10	cut			Med to dark brown, med to fine grained, med dense, well sorted quartz sand.							
10	30	cut			Med to light brown, fine grained, med dense sand.							
30	32	cut		///	Small clay layer.							
37	42	SAS	15*		Developed 55 gals. until clear. Purged 7 gals.	0	0	7.2	308	9.1		
47	52	SAS	16*		Developed 55 gals. until clear. Purged 14 gals.	0	0	7.3	313	9.1		
57	62	SAS	17		Developed 40 gals. until clear Purged: 10 gals. 22 gals.	0	0	7.3	200	8.0		
						0	0	7.4	208	9.1		
67	72	SAS	17		Developed 40 gals. until clear. Purged: 10 gals. 20 gals. 29 gals.	0	0	7.5	122	8.8		
						0	0	7.6	130	8.6		
						0	0	7.6	136	8.9		
77	82	SAS	18		Developed 50 gals. until clear. Purged: 3 gals. 10 gals. 30 gals.			7.8	156	8.7		
								7.7	152	8.8		
								7.7	151	8.9		
87	92	SAS	19		Developed 40 gals. until clear. Purged: 3 gals. 20 gals.			7.6	165	8.5		
								7.6	158	8.5		
97	102	SAS	20		Developed until clear. Purged: 3 gals. 25 gals.			7.9	195	8.3		
								7.9	203	8.3		
107	112	SAS	21*		Developed until clear Purged: 3 gals. 20 gals. 30 gals.			7.5	261	8.3		
								7.5	265	8.5		
								7.5	202	8.6		

GROUNDWATER ENCOUNTERED AT <u>32</u> FT.						PPM	PPM	SU	**	°C
NOTES: * Sample sent to CLP Density of sands and consistency of clays are estimates only. ** Specific Conductivity = uhos/cm.						UNITS				

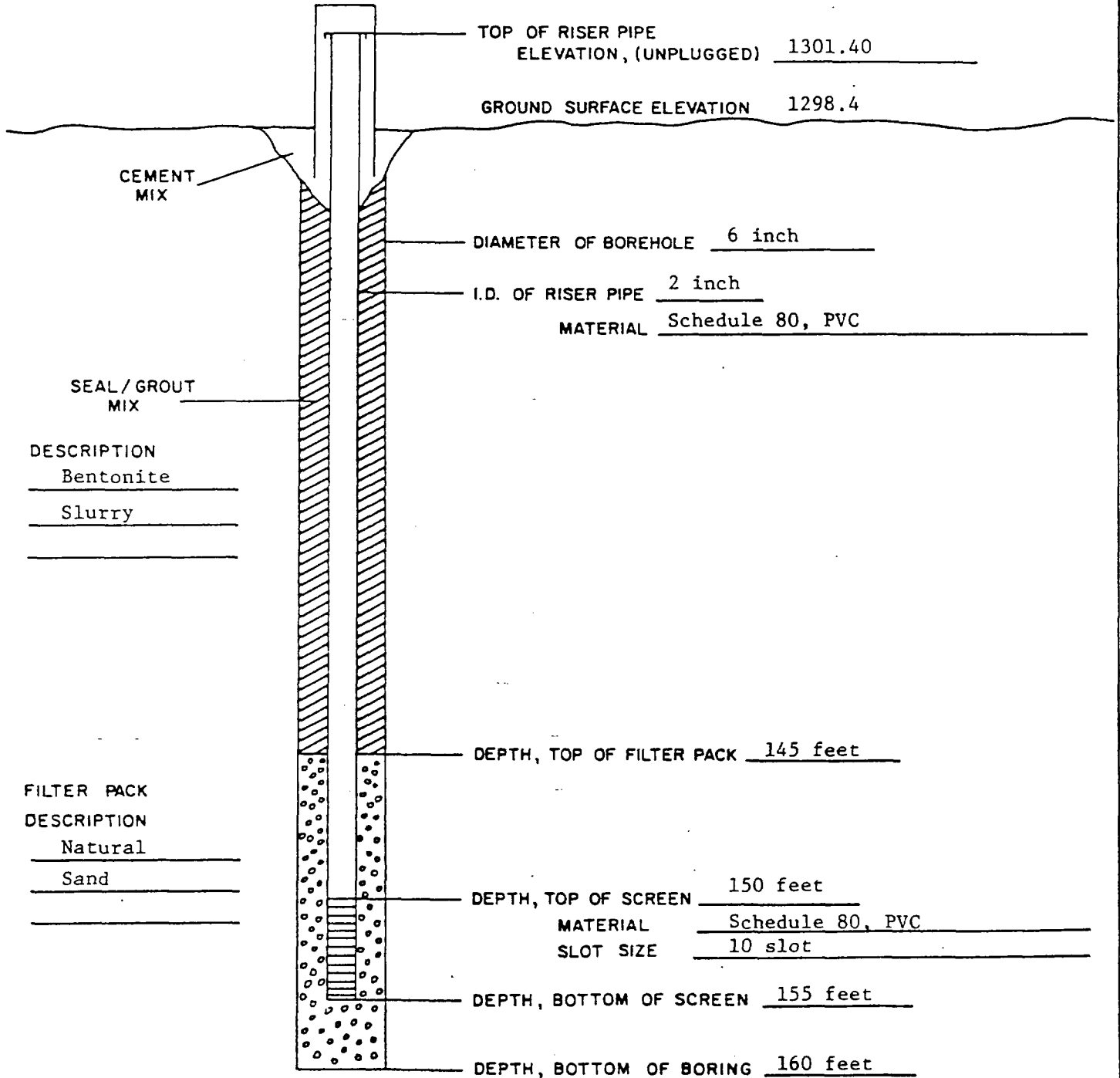
TOTAL DEPTH <u>160</u>					
TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%					
PAGE 1 OF 2					

BORING I.D. MW-10
LOCATION See Figure
DATE STARTED 11-26-85
DATE COMPLETED 12-4-85

PAGE 2 OF 2

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-10



AQUITARD INFORMATION

DEPTH TO UPPER	<u>NA</u>	THICKNESS	<u>0</u>
DEPTH TO LOWER	<u>156 feet</u>		

ECJORDANCO

NORTHERNAIRE GROUNDWATER STUDY						JOB NO.	4465-35	BORING I.D.	MW-11					
FIELD BORING LOG						DRILL RIG/METHOD	Gus Pech/HSA	LOCATION	See Figure					
COMPANY E.C. JORDAN						OPERATOR	M. Allen/Keck	DATE STARTED	12-5-85					
						INSPECTOR	T. Taylor/ECJ	DATE COMPLETED	12-9-85					
SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ ID.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS								
FROM	TO					HEAD SPACE	HEX CR.	pH	SPEC. COND.	TEMP.	OTHER			
0	25	CUT			Medium to dark brown, medium to fine grained, medium dense, well sorted quartz sand.									
25	30	CUT			Medium to light brown, fine grained, medium dense, sand.									
30	32	CUT		/ / /	Red brown, soft to firm clay layer.									
33	38	SAS	28*		Developed until clear. Purged: 30 gals.		0	7.3	266	9.0				
43	48	SAS	29		Developed until clear. Purged: 10 gals.		0	7.5	146	9.2				
53	58	SAS	30		Developed until clear. Purged: 20 gals.		0	7.4	183	8.7				
63	68	SAS	31		Developed until clear. Purged: 20 gals.		0	7.4	185	8.5				
73	78	SAS	32		Developed until clear. Purged: 20 gals.		0	7.4	166	8.6				
	88	SAS	33		Developed until clear. Purged: 20 gals.		0	7.5	182	8.9				
93	98	SAS	34		Developed until clear. Purged: 20 gals.		0	7.5	186	8.8				
103	108	SAS	35		Developed until clear. Purged: 20 gals.		0	7.6	197	9.0				
113	118	SAS	36		Developed until clear. Purged: 20 gals.		0	7.5	171	8.4				
123	128	SAS	37		Developed until clear. Purged: 20 gals.									
GROUNDWATER ENCOUNTERED AT 32 FT.						PPM	SU	**	°c					
NOTES:						UNITS								
* Sample sent to CLP.														
Density of sands and consistency of clays are estimates only.														
** Specific conductivity = 1/10 uhos/cm.														
TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%						TOTAL DEPTH 172								
						PAGE 1 OF 2								

[illegible]

GROUNDWATER ENCOUNTERED AT 32 FT.

NOTES:

PPM	SU	**	P
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UNITS

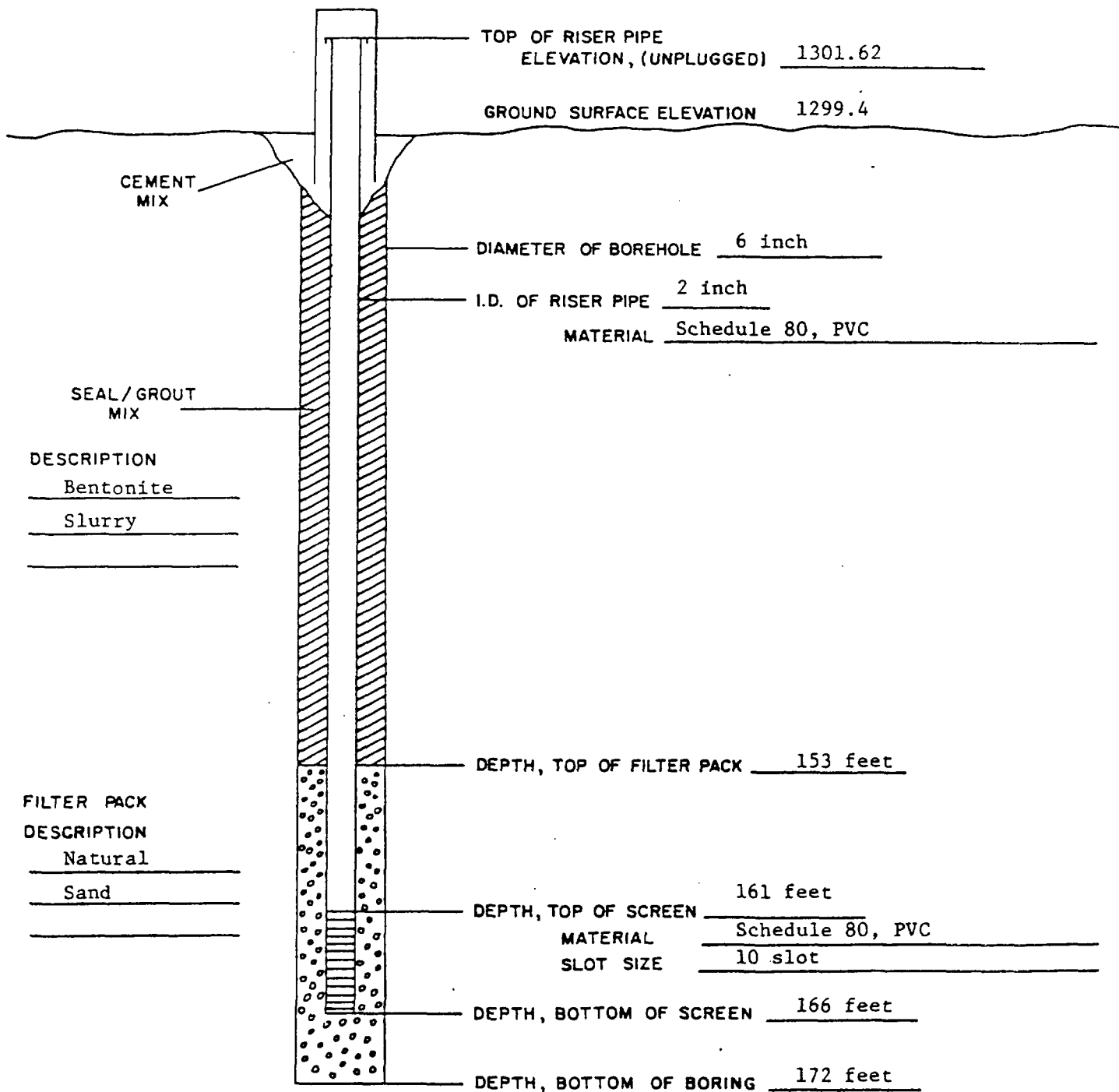
TOTAL DEPTH 172

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

PAGE 2 OF 2

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-11



AQUITARD INFORMATION

DEPTH TO UPPER	<u>NA</u>	THICKNESS	<u>0</u>
DEPTH TO LOWER	<u>169 feet</u>		

NORTHERNAIRE GROUNDWATER STUDY				JOB NO. <u>4465-35</u>		BORING I.D. <u>MW-12</u>			
FIELD BORING LOG				DRILL RIG/METHOD <u>Gus Pech/HSA</u>		LOCATION <u>See Figure</u>			
COMPANY <u>E.C. JORDAN</u>				OPERATOR <u>M. Allen/Keck</u>		DATE STARTED <u>12-10-85</u>			
				INSPECTOR <u>D. Argentati/ECJ</u>		DATE COMPLETED <u>12-12-85</u>			

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ LD.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS						
FROM	TO					HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER	
0	10	CUT			Medium to dark brown, medium to fine grained, medium dense, well sorted quartz sand.							
10	33	CUT			Medium to light brown, fine grained, medium dense, sand.							
33	34	CUT		///	Red brown, soft to firm, clay layer.							
34	36	CUT			Light brown, medium to fine, sand.							
36	40	CUT		///	Gray, soft to firm clay.							
40	49	CUT			Light brown, medium to fine, sand.							
49	58	SAS	42	///	Clay Developed until clear. Purged: 7 gals.							
58	60	CUT		•••	Fine, Gravel.							
60	67	CUT		///	Dark brown, firm, silty clay; grades to a red brown clay.							
68					Light brown, fine grained, medium dense, sand.							
68	73	SAS	43		Developed until clear. Purged: 30 gals.							
78	83	SAS	44		Developed until clear. Purged: 30 gals.							
88	93	SAS	45		Developed until clear. Purged: 20 gals.							
98	103	SAS	46		Developed until clear. Purged: 20 gals.							
108	113	SAS	47		Developed until clear. Purged: 20 gals.							

GROUNDWATER ENCOUNTERED AT <u>28</u> FT.						PPM	-SU	**	°c
NOTES: * Sample sent to CLP. Density of sands and consistency of clays are estimates only. ** Specific conductivity = 1/10 uhos/cm.						UNITS			

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%						TOTAL DEPTH <u>176</u>			
						PAGE <u>1</u> OF <u>2</u>			

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. 4465-35	BORING I.D. MW-12
	DRILL RIG/METHOD Gus Pech/HSA	LOCATION See Figure
FIELD BORING LOG	OPERATOR M. Allen/Keck	DATE STARTED 12-10-85
COMPANY E.C. JORDAN	INSPECTOR T. Taylor/ECJ	DATE COMPLETED 12-12-85

INSPECTOR T. Taylor/ECJ

DATE COMPLETED 12-12-85

COMPANY E.C. JORDAN

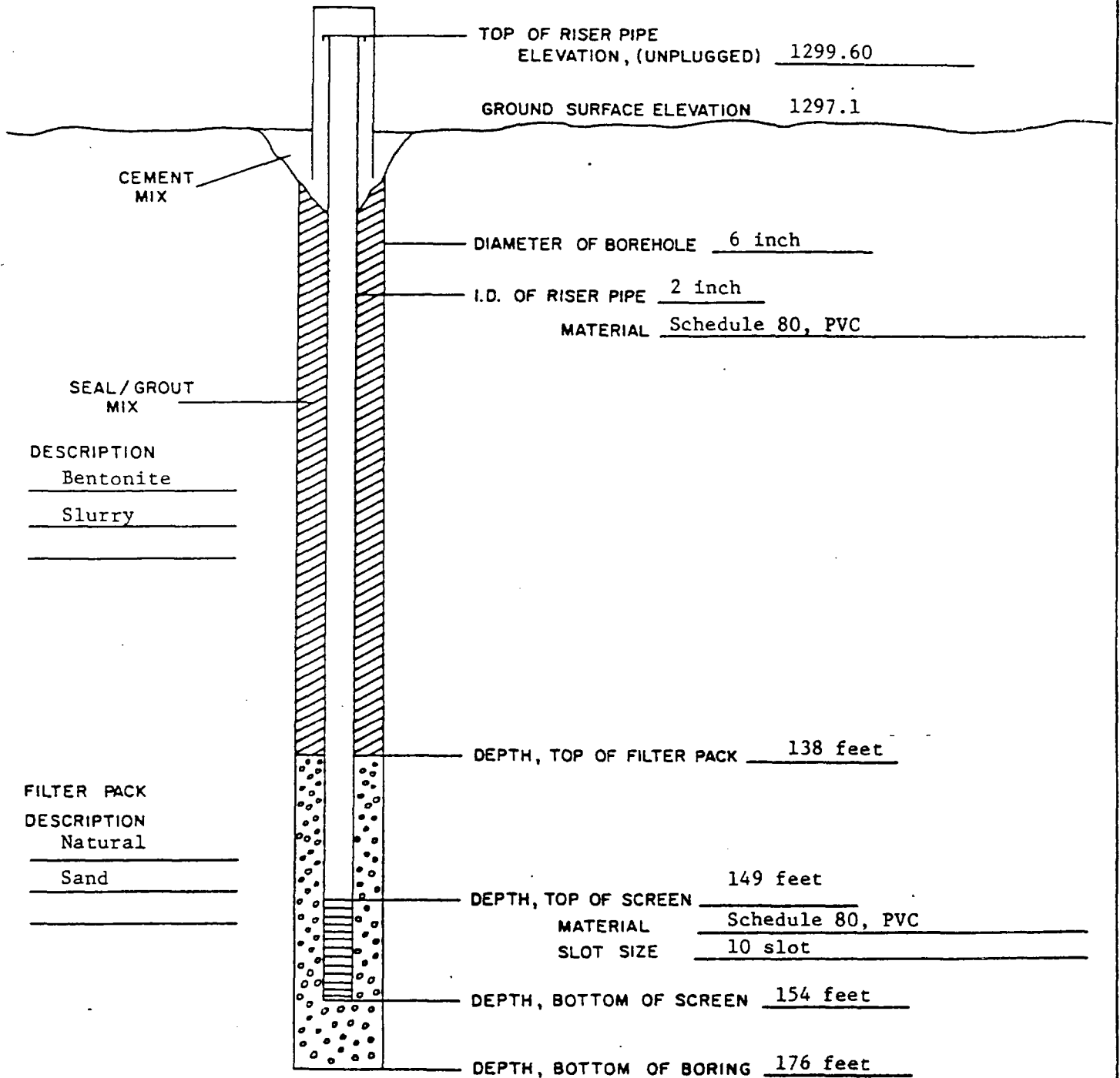
NOTES:

PAGE 2 OF 2

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-12



AQUITARD INFORMATION

DEPTH TO UPPER	<u>49 feet</u>	THICKNESS	<u>18 feet</u>
DEPTH TO LOWER	<u>175 feet</u>		

NORTHERNAIRE GROUNDWATER STUDY				JOB NO. 4465-35 DRILL RIG/METHOD Gus Pech/HSA OPERATOR M. Allen/Keck INSPECTOR T. Taylor/ECJ				BORING I.D. MW-13 LOCATION See Figure DATE STARTED 12-13-85 DATE COMPLETED 12-15-85							
FIELD BORING LOG COMPANY E.C. JORDAN															
SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ LD.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME			FIELD PARAMETERS							
FROM	TO							HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER		
0	10	CUT			Medium to dark brown, medium to fine grained, medium dense, well sorted, quartz sand.										
10	15	CUT			Medium to light brown, fine grained, medium dense, sand.										
15	20	CUT			Same sand, with little gravel.										
32	33	CUT		///	Brown, soft to firm, clay.										
33	38	SAS	54		Developed until clear. Purged: 10 gals.										
43	48	SAS	55		Developed until clear. Purged: 15 gals.										
53	58	SAS	56		Developed until clear. Purged: 18 gals.										
63	68	SAS	57		Developed until clear. Purged: 20 gals.										
73	78	SAS	58		Developed until clear. Purged: 20 gals.										
83	88	SAS	59		Developed until clear. Purged: 20 gals.										
93	98	SAS	60		Developed until clear. Purged: 20 gals.										
103	108	SAS	61		Developed until clear. Purged: 20 gals.										
113	118	SAS	62		Developed until clear. Purged: 20 gals.										
121	123			///	Clay.										
GROUNDWATER ENCOUNTERED AT 32.5 FT.								PPM	SU	**	OC				
NOTES:								UNITS							
* Sample sent to CLP. Density of sand and consistency of clays are estimates only. ** Specific conductivity = 1/10 uhos/cm.															
TOTAL DEPTH 161															
TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%												PAGE 1 OF 2			

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. 4465-35	BORING I.D. MW-13
	DRILL RIG/METHOD Gus Pech/Auger	LOCATION See Figure
FIELD BORING LOG COMPANY E.C. JORDAN	OPERATOR M. Allen/Keck INSPECTOR T. Taylor/ECJ	DATE STARTED 12-13-85 DATE COMPLETED 12-15-85

BORING I.D. MW-13

LOCATION See Figure

DATE STARTED 12-13-85

DATE COMPLETED 12-15-8

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. 4465-35	BORING I.D. MW-13
	DRILL RIG/METHOD Gus Pech/Auger	LOCATION See Figure
FIELD BORING LOG COMPANY E.C. JORDAN	OPERATOR M. Allen/Keck INSPECTOR T. Taylor/ECJ	DATE STARTED 12-13-85 DATE COMPLETED 12-15-85

[illegible]

PPM	SU	**	°C
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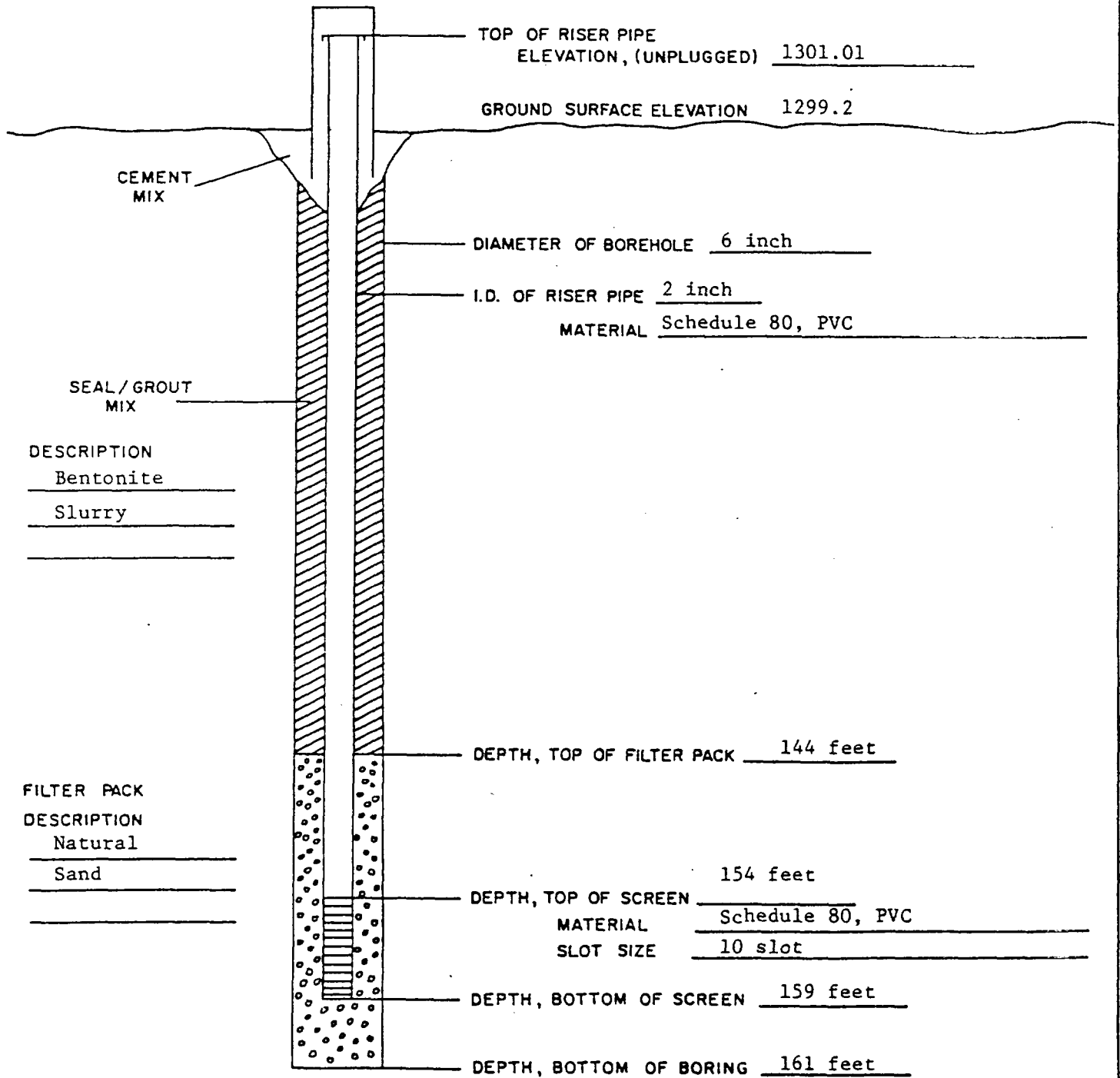
UNITS

PAGE 2 OF 2

[illegible]

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-13



AQUITARD INFORMATION

DEPTH TO UPPER

NA

THICKNESS

0

DEPTH TO LOWER

159 feet

ECJORDANCO

NORTHERNAIRE GROUNDWATER STUDY				JOB NO. 4465-35				BORING I.D. MW-14			
FIELD BORING LOG COMPANY E.C. JORDAN				DRILL RIG/METHOD Gus Pech/HSA				LOCATION See Figure			
				OPERATOR M. Allen/Keck				DATE STARTED 12-16-85			
				INSPECTOR D. Argentati/ECJ				DATE COMPLETED 12-19-85			

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ I.D.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS						
FROM	TO					HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER	
0	35	CUT			Medium to dark brown, medium to fine grained, medium dense, well sorted, quartz sand.							
35	35.5	CUT			Clay lens.							
38	43	SAS			Water too silty for field testing.							
43	48	SAS	67		Developed until clear. Purged: 20 gals.			0	8.0	66	7.8	
53	58	SAS	68		Developed until clear. Purged: 20 gals.							
63	68	SAS	69		Developed until clear. Purged: 20 gals.			0	8.0	33	8.6	
73	78	SAS	70		Developed until clear. Purged: 20 gals.			0	7.7	55	8.6	
83	88	SAS	71		Developed until clear. Purged: 20 gals.			0	8.0	48	8.1	
93	98	SAS	72		Developed until clear. Purged: 20 gals.			0	7.9	47	8.6	
103	108	SAS	73		Developed until clear. Purged: 20 gals.			0	7.7	56	8.8	
113	118	SAS	74		Developed until clear. Purged: 20 gals.			0	7.8	64	8.8	
123	128	SAS	75*		Developed until clear. Purged: 20 gals.			0	7.9	86	8.6	
133	138	SAS	76*		Developed until clear. Purged: 20 gals.			0	7.9	65	8.9	
143	148	SAS	77*		Developed until clear. Purged: 20 gals.			0	7.8	109	9.0	

GROUNDWATER ENCOUNTERED AT <u>35</u> FT.						PMM	SU	**	°C
						UNITS			

NOTES: * Sample sent to CLP.
 Density of sands and consistency of clays are estimates only.
 ** Specific conductivity = 1/10 uhos/cm.

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%		TOTAL DEPTH PAGE OF
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[illegible]

PPM	SU	**	°C
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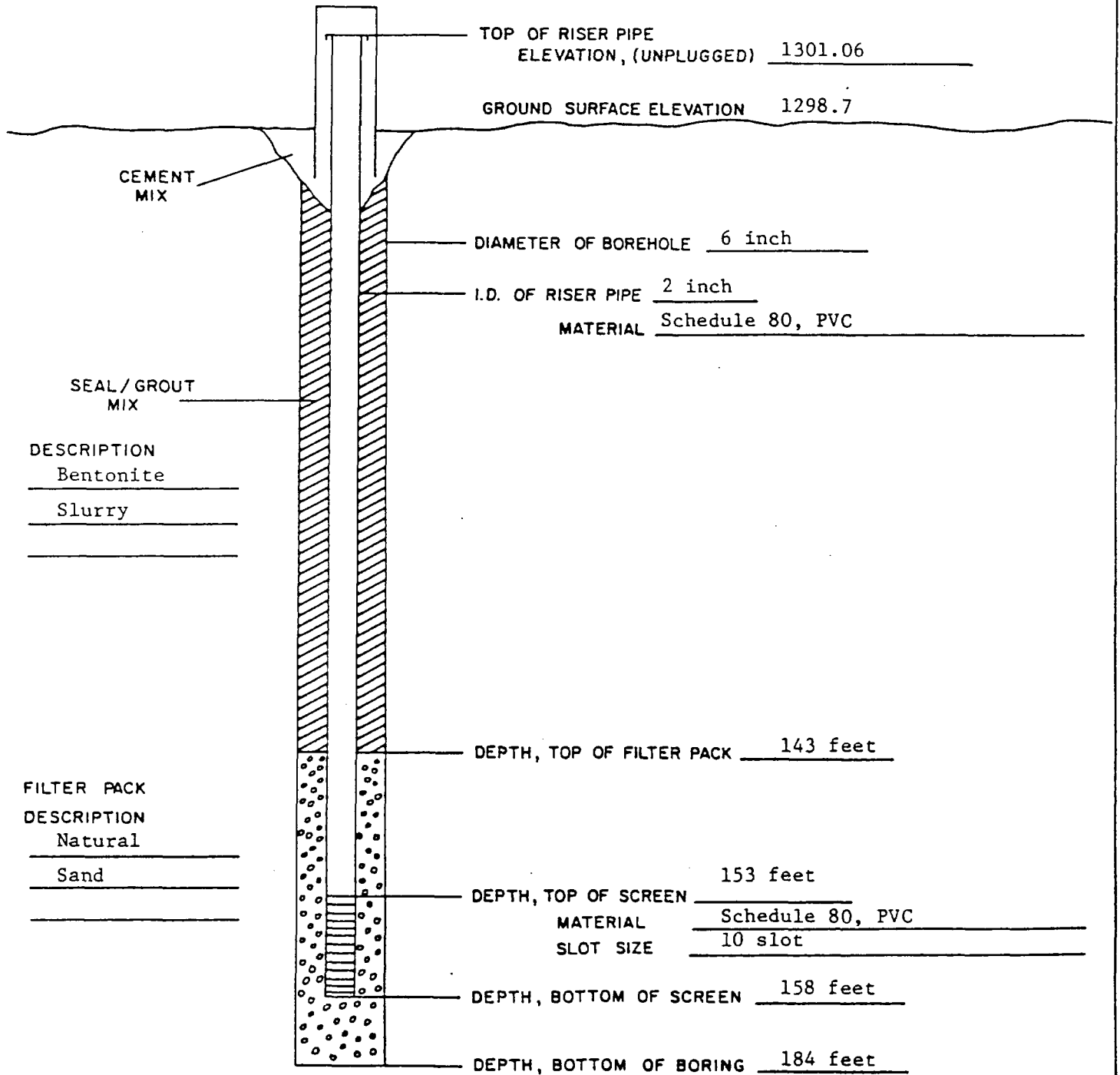
UNITS

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 36-50%

PAGE 2 OF 2

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-14



AQUITARD INFORMATION

DEPTH TO UPPER	<u>NA</u>	THICKNESS	<u>0</u>
DEPTH TO LOWER	<u>183 feet</u>		

EC.JORDAN CO

NORTHERNAIRE GROUNDWATER STUDY		JOB NO. <u>4465-35</u> DRILL RIG/METHOD <u>Gus Pech/Auger</u> OPERATOR <u>M. Allen/Keck</u> INSPECTOR <u>D. Argentati/ECJ</u>		BORING I.D. <u>MW-15--</u> LOCATION <u>See Figure</u> DATE STARTED <u>1-3-86</u> DATE COMPLETED <u>1-7-86</u>	
FIELD BORING LOG COMPANY <u>E.C. JORDAN</u>					

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ LD.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS					
FROM	TO					HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER
0	10	CUT			Dark brown, medium grained, medium dense, well sorted, quartz sand, tr. gravel.						
10	29	CUT			Medium to light brown, fine grained, well sorted, sand.						
29	30	CUT		///	Clay layer.						
	37	SAS	81		Developed until clear. Purged: 20 gals.			0	7.5	118	9.5
42	49	CUT		///	Gray, soft to firm clay.						
49	52	CUT			Medium brown, fine grained sand.						
52	57	SAS	82		Developed until clear. Purged: 20 gals.			0	7.7	174	9.3
57	58	CUT		///	Clay layer.						
58	59	CUT			Brown sand.						
59	66	CUT		///	Clay.						
66	67	CUT			Sand.						
	73			///	Clay.						
73	78	SAS	83		Developed; remained silty. Purged: 30 gals.			0	7.9	158	8.7
83	88	SAS	84		Developed; remained silty. Purged: 20 gals.			0	7.8	153	8.5
93	98	SAS	85		Developed until clear. Purged: 20 gals.			0	7.9	118	8.7
103	108	SAS	86		Developed until clear. Purged: 20 gals.			0	7.8	178	8.3

GROUNDWATER ENCOUNTERED AT 31 FT.

NOTES:

- * Sample sent to CLP.
- Density of sand and consistency of clay are estimates only.
- ** Specific conductivity = 1/10 uhos/cm.

PPM SU	** P _c
UNITS	

TOTAL DEPTH 183

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

PAGE 1 OF 2

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. <u>4465-35</u>	BORING I.D. <u>MW-15</u>
	DRILL RIG/METHOD <u>Gus Pech/HSA</u>	LOCATION <u>See Figure</u>
FIELD BORING LOG COMPANY <u>E.C. JORDAN</u>	OPERATOR <u>M. Allen/Keck</u> INSPECTOR <u>D. Argentati/ECJ</u>	DATE STARTED <u>1-3-86</u> DATE COMPLETED <u>1-7-86</u>

BORING I.D. MW-15

LOCATION See Figure

DATE STARTED 1-3-86

DATE COMPLETED 1-7-86

COMPANY E.C. JORDAN

GROUNDWATER ENCOUNTERED AT 31 FT.

NOTES:

PPM	SU	**	°C
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UNIT 8

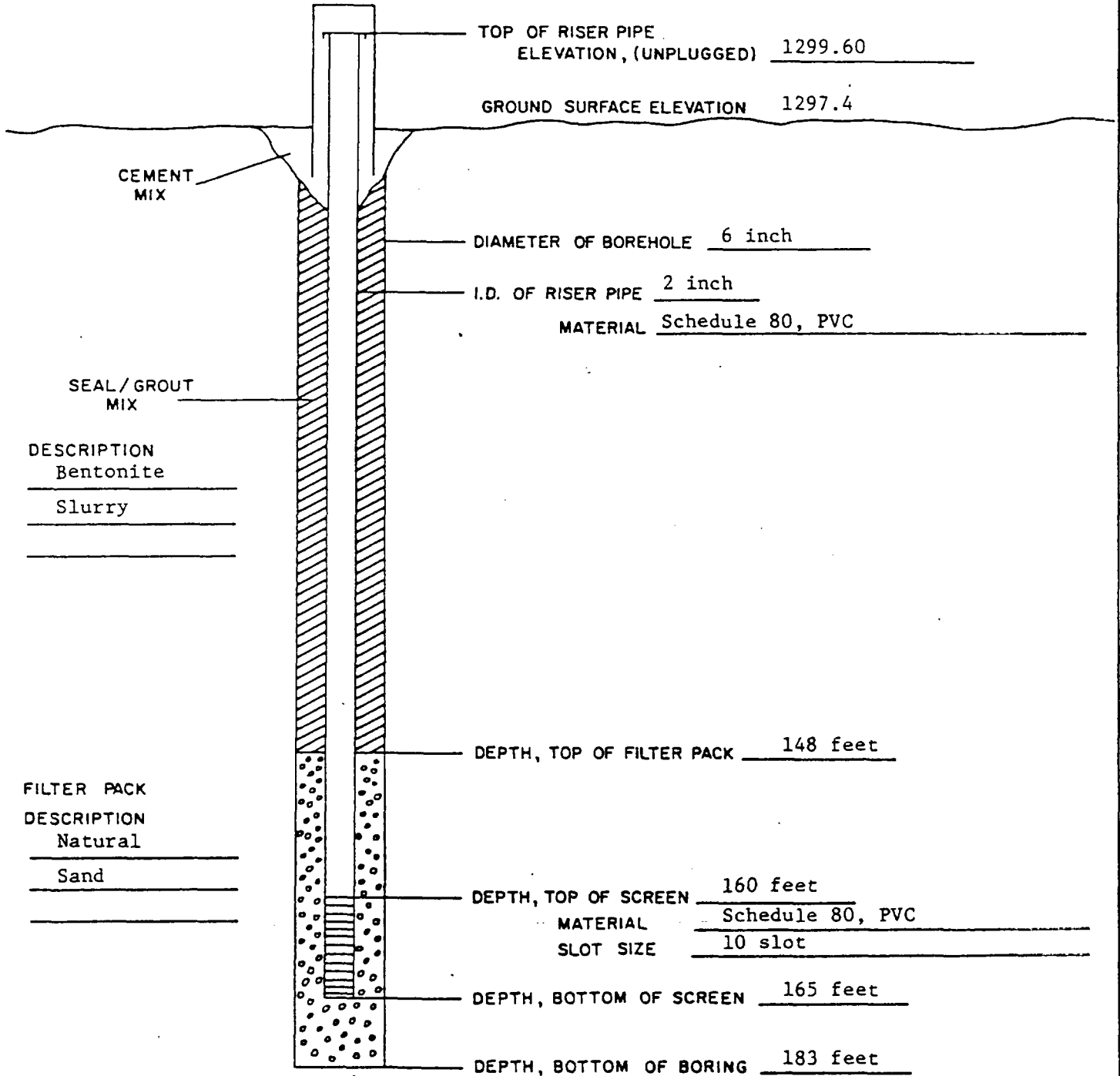
TOTAL DEPTH	183
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PAGE 2 OF 2

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-15



AQUITARD INFORMATION

DEPTH TO UPPER	<u>59 feet</u>	THICKNESS	<u>14 feet</u>
DEPTH TO LOWER	<u>180 feet</u>		

ECJORDANCO

NORTHERNAIRE GROUNDWATER STUDY				JOB NO. 4465-35 DRILL RIG/METHOD Gus Pech/HSA		BORING I.D. MW-16 LOCATION See Figure	
FIELD BORING LOG COMPANY E.C. JORDAN				OPERATOR M. Allen/Keck INSPECTOR D. Argentati/ECJ		DATE STARTED 1-10-86 DATE COMPLETED 1-16-86	

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ ID.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS						
FROM	TO					HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER	
0	25	CUT			Medium to dark brown, medium to fine grained, medium dense, well sorted, quartz sand.							
25	30	CUT			Medium to light brown, fine grained, well sorted, sand.							
32	32.5	CUT			Soft to firm clay.							
33	38	SAS	94		Developed until clear. Purged: 20 gals.			0	8.0	95	8.6	
43	48	SAS	95		Developed until clear. Purged: 20 gals.			0	8.0	150	7.8	
54	63			//	Clay.							
63	68	SAS	96		Developed until clear. Purged: 20 gals. Second phase product seen in development water.			0	8.0	131	7.5	
73	78	SAS	97		Developed until clear. Purged: 20 gals.			0	7.8	172	8.3	
83	88	SAS	98		Developed until clear. Purged: 20 gals.			0	7.8	136	9.0	
93	98	SAS	99		Developed until clear. Purged: 20 gals.			0	7.8	140	9.2	
103	108	SAS	100		Developed until clear. Purged: 20 gals.			0	7.8	156	9.6	
113	118	SAS	101		Developed until clear. Purged: 20 gals.			0	7.8	108	8.6	

GROUNDWATER ENCOUNTERED AT 30 FT.						PPM	SU	**	°C
						UNITS			

NOTES:

- * Sample sent to CLP.
- Density of sand and consistency of clay are estimates only.
- ** Specific conductivity = 1/10 uhos/cm.

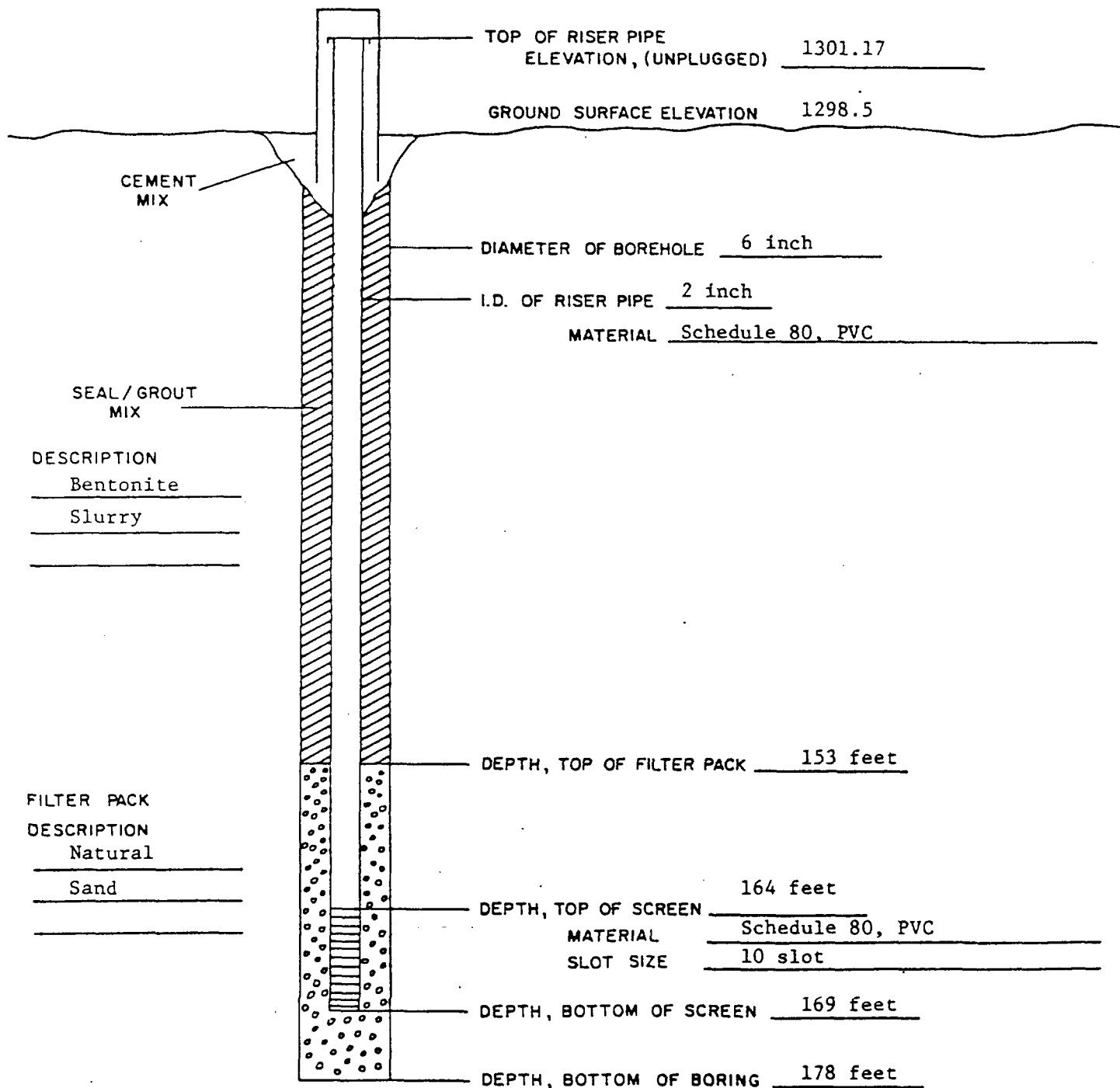
TOTAL DEPTH 178	
TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%	
PAGE 1	OF 2

BORING I.D. MW-16
LOCATION See Figure
DATE STARTED 1-10-86
DATE COMPLETED 1-16-86

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-16



DESCRIPTION
Bentonite
Slurry

FILTER PACK
 DESCRIPTION
Natural
Sand

AQUITARD INFORMATION

DEPTH TO UPPER 54 feet THICKNESS 9 feet
 DEPTH TO LOWER 175 feet

NORTHERNAIRE GROUNDWATER STUDY				JOB NO. 4465-35 DRILL RIG/METHOD Gus Pech/HSA OPERATOR M. Allen/Keck INSPECTOR T. Taylor/ECJ		BORING I.D. MW-17- LOCATION See Figure DATE STARTED 1-18-86 DATE COMPLETED 1-19-86	
FIELD BORING LOG COMPANY E.C. JORDAN							

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ LD.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS							
FROM	TO					HEAD SPACE	HEX CR	PH	SPEC COND.	TEMP.	OTHER		
0	30	CUT			Medium to dark brown, medium to fine grained, medium dense, well sorted, quartz, sand.								
33	38	SAS	108*		Developed until clear. Purged: 20 gals.				0	7.2	281	9.8	
49	49	CUT		//	Medium gray, firm, silty clay, some gravel and small interbedded sand seams.								
53	58	SAS	109*		Developed until clear. Purged: 20 gals.				0	7.5	161	9.7	
63	68	SAS	110		Developed until clear. Purged: 20 gals.				0	7.4	187	10.4	
64	88	CUT			Color of sand changes sharply to gray. Gradually changes back to brown at 88 feet.								
73	78	SAS	111		Developed until clear. Purged: 20 gals.				0	7.5	180	9.4	
83	88	SAS	112		Developed until clear. Purged: 20 gals.				0	7.6	111	9.5	
93	98	SAS	113		Developed until clear. Purged: 20 gals.				0	7.6	149	10.5	
103	108	SAS	114		Developed until clear. Purged: 20 gals.				0	7.6	177	9.7	
123	128	SAS	115		Developed until clear. Purged: 20 gals.				0	7.7	143	9.9	
123	128	SAS	116		Developed until clear.				0	7.7	182	9.4	
GROUNDWATER ENCOUNTERED AT 30 FT.						PPM	SU	**	Pc				
						UNITS							

NOTES: * Sample sent to CLP.
 Density of sand and consistency of clay are estimates only.
 ** Specific conductivity = 1/10 uhos/cm.

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%	TOTAL DEPTH 178 PAGE 1 OF 2
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FIELD BORING LOG
COMPANY E.C. JORDAN

JOB NO. 4465-35
DRILL RIG/METHOD Gus Pech/HSA
OPERATOR M. Allen/Keck
INSPECTOR T. Taylor/ECJ

BORING I.D. MW-17
LOCATION See Figure
DATE STARTED 1-18-86
DATE COMPLETED 1-19-86

[illegible]

GROUNDWATER ENCOUNTERED AT 30 FT.

PPM	SU	**	^o C
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NOTES:

UNITS

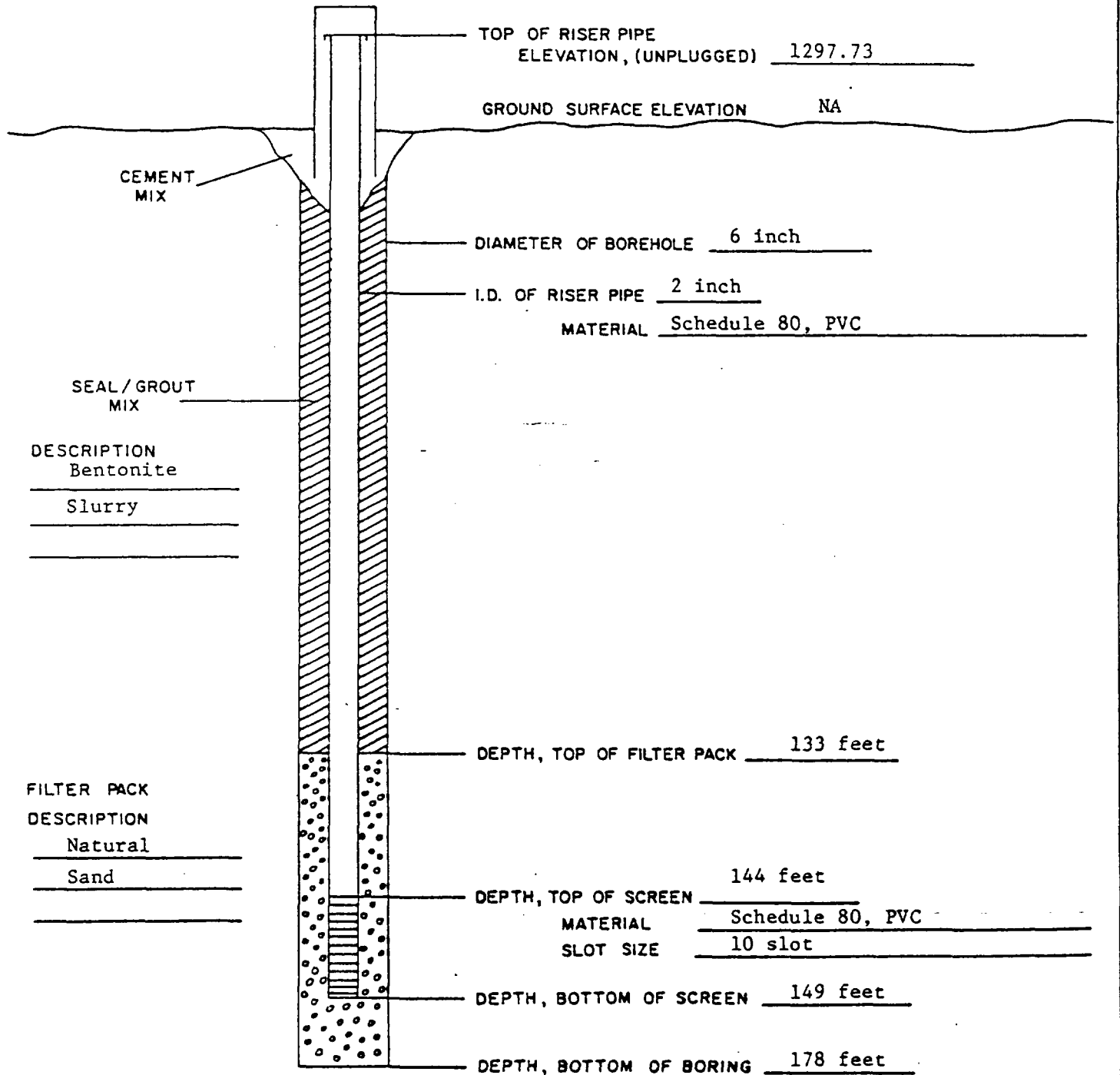
TOTAL DEPTH ±78

PAGE 2 OF 2

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-17



AQUITARD INFORMATION

DEPTH TO UPPER 40 feet THICKNESS 9 feet
DEPTH TO LOWER 177 feet

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. <u>4465-35</u> DRILL RIG/METHOD <u>Gus Pech/HSA</u> OPERATOR <u>M. Allen/Keck</u> INSPECTOR <u>T. Taylor/ECJ</u>	BORING I.D. <u>MW-18</u> LOCATION <u>See Figure</u> DATE STARTED <u>1-20-86</u> DATE COMPLETED <u>1-21-86</u>
FIELD BORING LOG COMPANY <u>E.C. JORDAN</u>		

SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ ID.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME	FIELD PARAMETERS						
FROM	TO					HEAD SPACE	HEX CR.	PH	SPEC. COND.	TEMP.	OTHER	
0	30	CUT			Medium brown, medium to fine, medium dense, well sorted, quartz sand.							
33	38	SAS	122		Developed; remained silty Purged: 10 gals.		0	7.9	34	10.2		
43	48	SAS	123		Developed until clear. Purged: 20 gals.		0	8.0	72	9.2		
53	58	SAS	124		Developed until clear. Purged: 20 gals.		0	7.7	118	9.5		
63	68	SAS	125		Developed until clear. Purged: 20 gals.		0	7.6	139	9.4		
73	78	SAS	126		Developed until clear. Purged: 20 gals.		0	7.6	149	8.7		
83	88	SAS	127		Developed until clear. Purged: 20 gals.		0	7.6	144	8.8		
93	98	SAS	128		Developed until clear. Purged: 20 gals.		0	7.7	122	8.7		
103	108	SAS	129		Developed until clear. Purged: 20 gals.		0	7.6	165	8.5		
111	115	CUT		///	Medium brown, firm, silty clay.							
113	118	SAS	130		Developed; remained silty. Purged: 20 gals.		0	7.6	164	9.0		
123	128	SAS	131		Developed until clear. Purged: 20 gals.		0	7.5	171	8.7		
133	138	SAS	132*		Developed until clear. Purged: 20 gals.		0	7.6	162	8.3		
GROUNDWATER ENCOUNTERED AT 32 FT.							PPM	SU	**	°C		
NOTES:						UNITS						

NOTES:

- * Sample sent to CLP.
- Density of sand and consistency of clay are estimates only.
- ** Specific conductivity = 1/10 uhos/cm.

UNITS

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

TOTAL DEPTH 157
PAGE 1 OF 2

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. <u>4465-35</u>	BORING I.D. <u>MW-18</u>
	DRILL RIG/METHOD <u>Gus Pech/HSA</u>	LOCATION <u>See Figure</u>
FIELD BORING LOG	OPERATOR <u>M. Allen/Keck</u>	DATE STARTED <u>1-20-86</u>
COMPANY <u>E.C. JORDAN</u>	INSPECTOR <u>T. Taylor/ECJ</u>	DATE COMPLETED <u>1-21-86</u>

COMPANY E.C. JORDAN

DRILL RIG/METHOD Gus Pech/HSA

INSPECTOR T. Taylor/ECJ

LOCATION See Figure

DATE COMPLETED 1-21-86

[illegible]

PPM	SU	**	°C
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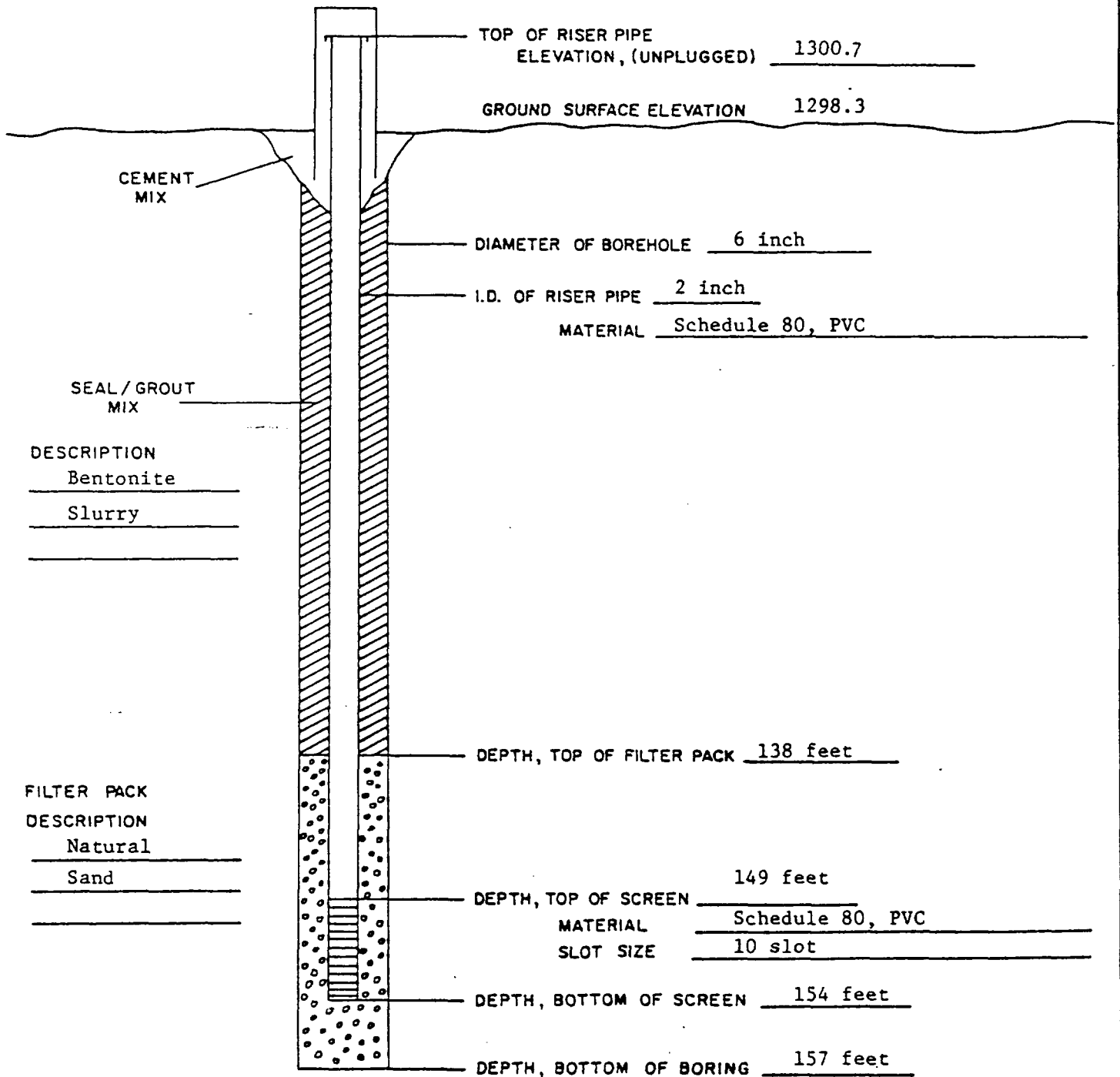
UNITS

PAGE 2 OF 2

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL MW-18



AQUITARD INFORMATION

DEPTH TO UPPER NA THICKNESS 0
DEPTH TO LOWER 155 feet

ECJORDANCO

NORTHERNAIRE GROUNDWATER STUDY						JOB NO. 4465-35.		BORING I.D. P-1								
FIELD BORING LOG						DRILL RIG/METHOD Gus Pech/HSA		LOCATION See Figure								
COMPANY E.C. JORDAN						OPERATOR M. Allen/Keck		DATE STARTED 1-17-86								
						INSPECTOR T. Taylor/ECJ		DATE COMPLETED 1-17-86								
SAMPLE DEPTH		TYPE OF SAMPLE	CLP/ECJ LD.	STRATA CHANGE	UNIFIED SOIL CLASSIFICATION: COLOR: GRADATION(PLASTICITY):DENSITY(CONSISTENCY): MOISTURE:STRUCTURE:GEOLOGIC NAME				FIELD PARAMETERS							
FROM	TO								HEAD SPACE	HEX CH.	pH	SPEC. COND.	TEMP.	OTHER		
1	10	cut			Med. to dark brown, med. to fine grained, med. dense, well sorted, quartz sand.											
10	32	cut			Med. to light brown, fine grained, med. dense, sand.											
32	33.5	cut			Brown, soft, clay lens.											
33.5	53	cut			Same sand as from 10-32 ft.											

GROUNDWATER ENCOUNTERED AT _____ FT.

NOTES:

No field screening done.

UNITS

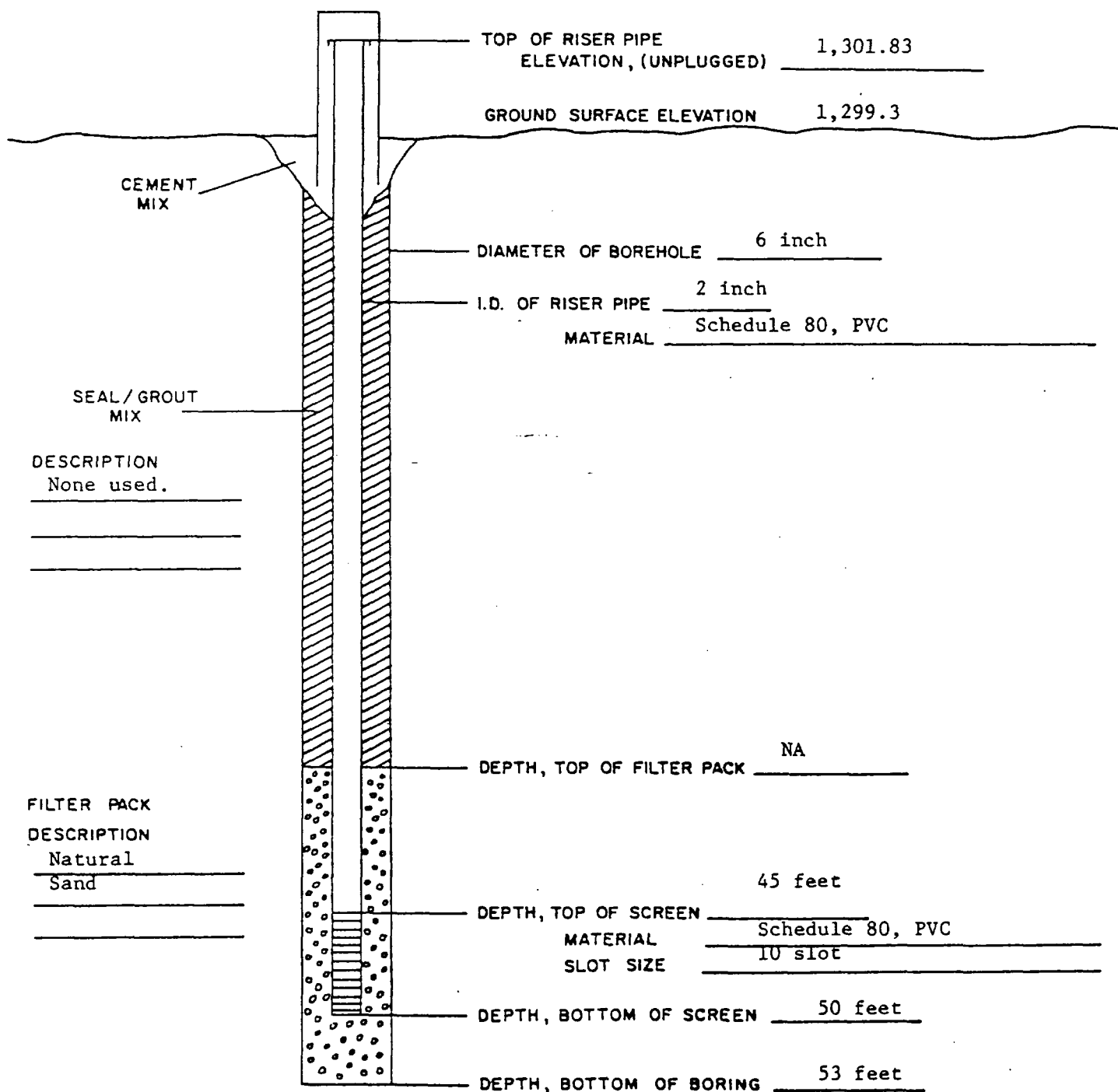
TOTAL DEPTH 53

PAGE OF

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL P-1



AQUITARD INFORMATION

DEPTH TO UPPER	<u>NA</u>	THICKNESS	<u>0 feet</u>
DEPTH TO LOWER	<u>NA</u>		

NORTHERNAIRE GROUNDWATER STUDY	JOB NO. <u>4465-35</u>	BORING I.D. <u>P-2</u>
	DRILL RIG/METHOD <u>Gus Pech/HSA</u>	LOCATION <u>See Figure</u>
FIELD BORING LOG COMPANY <u>E.C. JORDAN</u>	OPERATOR <u>M. Allen/Keck</u>	DATE STARTED <u>1-17-86</u>
	INSPECTOR <u>T. Taylor/ECJ</u>	DATE COMPLETED <u>1-17-86</u>

COMPANY E.C. JORDAN

INSPECTOR T. Taylor/ECJ

DATE COMPLETED 1-17-86

[illegible]

GROUNDWATER ENCOUNTERED AT _____ FT.

NOTES:

No field screening done.

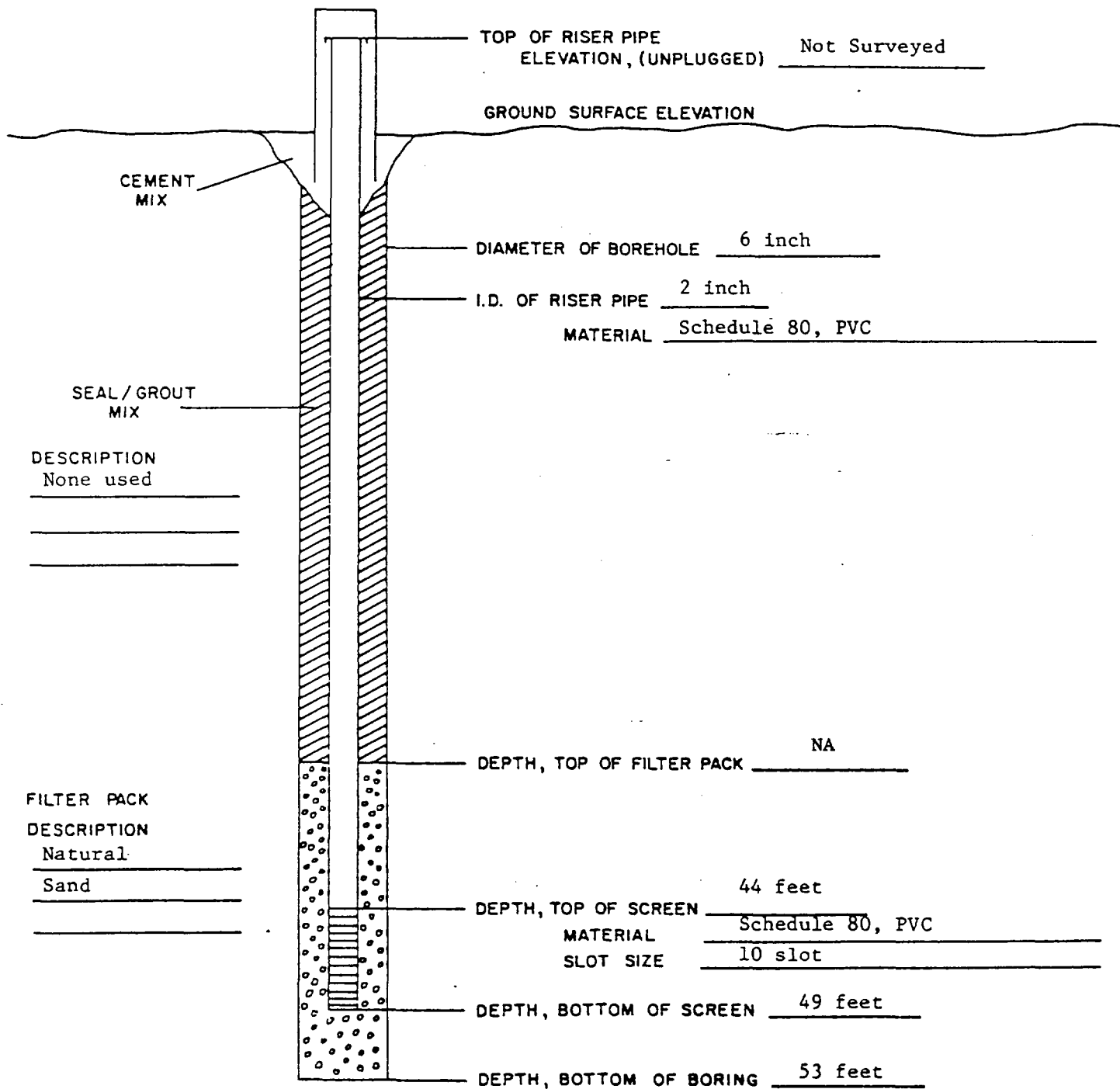
UNITS

TOTAL DEPTH		53
PAGE	1	OF 1

TRACE 0-10% LITTLE 10-20% SOME 20-35% AND 35-50%

MONITORING WELL INSTALLATION RECORD

MONITORING WELL P-2



AQUITARD INFORMATION

DEPTH TO UPPER NA THICKNESS 0 feet

DEPTH TO LOWER NA